

Topological Complexity and Related Invariants

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Eigenständigkeitserklärung

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbstständig und eigenhändig sowie ohne unerlaubte fremde Hilfe und ausschließlich unter Verwendung der aufgeführten Quellen und Hilfsmittel angefertigt habe.

Berlin, den

Sandro Erras

Abstract

This thesis is an introduction to the concept of topological complexity, which is a homotopy invariant of topological spaces established in [Far03] by M. Farber. This invariant can be interpreted as a lower bound of discontinuity for motion planning on a given topological space, where a motion planning algorithm can be viewed as a map $X \times X \rightarrow X^I$. Here X^I denotes the space of all continuous paths on X . Throughout this thesis, we study various connections of topological complexity to other topological invariants, such as Lusternik-Schnirelmann category, covering dimension and singular cohomology.

After that, we consider directed topological complexity, which was introduced in [GFS20] and which is a variant of topological complexity for directed spaces. A directed space X is a topological space together with a choice of directed paths, i.e. a subset $PX \subseteq X^I$, such that PX is closed under concatenation and reparametrization by non-decreasing continuous maps on the standard unit interval $I \rightarrow I$. Similar to the undirected case, directed complexity turns out to be invariant under simple dihomotopy equivalences, which resemble a stronger version of homotopy equivalence adapted to the setting of directed spaces.

Finally, a brief overview of geodesic complexity is given. This is a modification of topological complexity for metric spaces, where one considers motion planning algorithms that map to $GX \subseteq X^I$, the space of all geodesics on X . The notion was originally introduced in [RM21]. We will see that geodesic complexity is invariant under isometries. Furthermore, for a certain class of metric spaces, the geodesic complexity is completely determined by the total cut locus, which is the set of all pairs of points with more than one geodesic between them.

Kurzzusammenfassung

Diese Arbeit stellt eine Einführung in das Themengebiet der topologischen Komplexität dar. Topologische Komplexität ist eine Homotopie-Invariante topologischer Räume, die in [Far03] von M. Farber eingeführt wurde. Diese Invariante kann als ein Mindestmaß für die Unstetigkeit von Wegplanungsalgorithmen in einem gegebenen topologischen Raum interpretiert werden, wobei ein Wegplanungsalgorithmus als eine Abbildung $X \times X \rightarrow X^I$ aufgefasst wird. Dabei bezeichnet X^I den Raum aller stetigen Pfade auf X . In dieser Arbeit werden verschiedene Verbindungen der topologischen Komplexität zu anderen topologischen Invarianten, beispielsweise der Lusternik-Schnirelmann-Kategorie, der Überdeckungsdimension und der singulären Kohomologie thematisiert.

Danach betrachten wir die gerichtete topologische Komplexität, die in [GFS20] eingeführt wurde und welche als Variante der topologischen Komplexität für gerichtete Räume aufgefasst werden kann. Ein gerichteter Raum X ist ein topologischer Raum mit einer Auswahl an gerichteten Pfaden, d.h. einer Teilmenge $PX \subseteq X^I$, sodass PX unter Verkettung und Umparametrisierung durch monoton steigende stetige Abbildungen auf dem Standard-Einheitsintervall $I \rightarrow I$ abgeschlossen ist. Die gerichtete Komplexität erweist sich, ähnlich wie im ungerichteten Fall, als invariant unter einfachen Dihomotopie-Äquivalenzen, welche eine stärkere Version von Homotopie-Äquivalenzen für den Fall von gerichteten Räumen darstellen.

Abschließend wird ein kurzer Überblick über das Konzept der geodätischen Komplexität gegeben. Dabei handelt es sich um eine Abwandlung der topologischen Komplexität für metrische Räume, bei der man Wegplanungsalgorithmen $X \times X \rightarrow GX$ betrachtet, wobei $GX \subseteq X^I$ den Raum aller Geodäten auf X bezeichnet. Der Begriff wurde ursprünglich in [RM21] eingeführt. Wir werden sehen, dass die geodätische Komplexität invariant unter Isometrien ist. Außerdem stellt sich heraus, dass die geodätische Komplexität für eine bestimmte Klasse metrischer Räume vollständig durch den sogenannten totalen Schnittpunkt bestimmt wird, welcher die Menge aller Paare in $X \times X$ enthält, zwischen denen mehr als eine Geodäte verläuft.

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1 Introduction

In modern industry, more and more tasks are automated and handled by robots instead of being executed by hand. As part of this process of ever-increasing automation, software engineers have to come up with algorithms to efficiently guide those robots. More precisely, an automated mechanical system has to be told how to traverse from one state to another desired state, in a way determined by some motion planning algorithm.

In more mathematical terms, this can be described as follows: Suppose that X is the set of all possible states of our system, which we call the configuration space of the system ¹. Some examples, inspired by [Far03] and [Far04], are:

- i) All possible configurations of a single robot occupying a point in the plane. In this case we would have $X = \mathbb{R}^2$.
- ii) More generally, all possible configurations of n robots in some topological space Y , such that their positions do not overlap. In this case we would get a “classical” configuration space

$$X = \text{Conf}_n(Y)$$

where $\text{Conf}_n(Y)$ is given by all tuples $(y_1, \dots, y_n) \in Y^n$ with $y_i \neq y_j$ for $i \neq j$.

To dive even further into the real world, consider Y to be the floor of a factory with obstructions given by pillars or machines of some sort. We can think of this as a subspace of $\mathbb{R}^2$ where the obstructions give rise to holes. The colored dots in fig. 1 would then be an example of a configuration of three robots on the factory floor, i.e. a point in $\text{Conf}_3(Y)$.

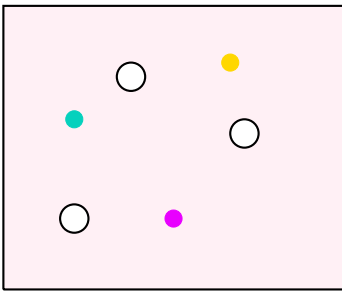


Figure 1: Sample configuration of three robots on a factory floor.

- iii) All possible configurations of a planar robot arm with n joints, where one joint fixed at the origin

¹Not to be confused with the notion of a configuration space in the topological sense. The term “configuration space of a mechanical/physical system” is often used in physics for the set of all parameters that describe all possible configurations of a system.

and fixed distances between them. In this case we have $X = (\mathbb{S}^1)^n$.

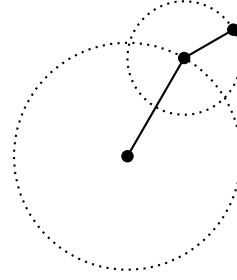


Figure 2: Sample configuration of a robot arm with two joints.

In all of those examples, the configuration space of interest carries at least a topological structure: In i), the structure is induced by the metric of $\mathbb{R}^2$, in ii) it is inherited from Y^n and in iii) it comes from $\mathbb{S}^1$. This means, that we are able to talk about continuity and other topological notions such as “closeness” in terms of neighborhoods and so on.

Indeed, the main topic in this thesis relies on analyzing motions of a given system with topological tools. This idea is brought up in [Far03] and the considerations in this introduction as well as in chapters 3.2 and 4 are heavily influenced by this paper. Thus, from now on we require that the configuration space X is a topological space.

Next, assume that the system is in some initial configuration $a \in X$ and should end up in some desired configuration $b \in X$. Given these configurations, a motion planning algorithm then should produce a way to transfer the system from state a to state b , possibly with many intermediate states. Hence, a motion planning algorithm can be regarded as a map which takes a pair $(a, b) \in X \times X$ of initial and desired configuration and produces a path γ in the configuration space X , which starts at a and ends at b . As most real-world examples don’t allow the system to just immediately “jump” from one state to another, it is reasonable to require the resulting path γ to be continuous with respect to the topology on X , which oftentimes is induced by a metric of some kind.

Before continuing further, let us introduce a couple of notions. Let I be the standard unit interval $[0, 1]$ with topology induced by the standard topology on $\mathbb{R}$. Denote by X^I the set of all continuous paths $\gamma : I \rightarrow X$ in X . Then by the previous discussion, a motion planning algorithm on X is just a map $s : X \times X \rightarrow X^I$ with the property that $s(a, b)(0) = a$ and $s(a, b)(1) = b$ for all $(a, b) \in X \times X$.

Now consider the following scenario: We are given

some configuration space X describing a system and are assigned to construct a motion planning algorithm for this space. First of all, it is natural to ask if a motion planning algorithm could exist in the first place. Of course, X being path-connected is sufficient, since in this case, every pair of start- and end-points can just be mapped to an arbitrary path between them. But even then, our goal should be to construct a “good” algorithm, whatever that might mean in practice. One quality feature to look out for could be that our algorithm should be “stable”² in some sense. To get a feeling what this could mean, consider the following example:

Let X be the configuration space of a robot arm with one joint, i.e. $X = \mathbb{S}^1$, and suppose we have an initial state $a \in X$ and some desired state $b \in X$ of the system.

As a first approach, we could construct a motion planning algorithm $s : X \times X \rightarrow X^I$ as follows: For $(a, b) \in X \times X$ with $a = b$, we choose the constant path at a as an output for our algorithm. Otherwise, there are two distinct arcs on $\mathbb{S}^1$ connecting the initial state and the desired state. We can then instruct our robot to just follow the shorter arc until the destination is reached. If both arcs are of equal length, which of course occurs if the destination is exactly antipodal to the initial state, the robot is instructed to follow the arc in the mathematically positive direction, which has to be defined beforehand in the program.

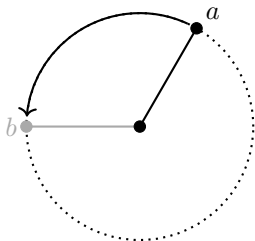


Figure 3: Possible movement in the mathematically positive direction to get from the initial state a to the desired state b .

Regarding stability, the reader may already have spotted a possible problem: Suppose the desired configuration is $b = -a$, i.e. the antipodal point of a . Our algorithm should then output the arc in the positive direction. But if there is even the smallest deviation a' of the input in the mathematically negative direction, the algorithm will produce a vastly different output, see fig. 4. This can be regarded as some kind of “discontinuity” of our motion planning algorithm $s : X \times X \rightarrow X^I$. So far, we can not really talk about continuity of s , since X^I does not yet carry the structure of a topological space. Later in chapter 3 this will be made mathematically precise by

equipping X^I with a suitable topology. In practice, such a deviation of the input could be caused by noise on the robot sensors, hence our algorithm could be regarded as not very stable.

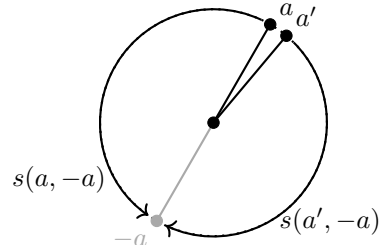


Figure 4: Two different outputs with slightly different inputs a and a' .

To fix this problem, we first should ask ourselves, whether it is possible to eliminate those kinds of discontinuities entirely with more sophisticated approaches when constructing an algorithm. Unfortunately, as we will see at the end of chapter 3.2, a continuous motion planning on the whole of $X \times X$ is only possible if and only if the configuration space X is contractible in the topological sense. The next best thing to aim for in terms of stability is to minimize the discontinuity of our motion planning. But in order to approach this goal, one first needs some kind of measurement of discontinuity for any motion planning algorithm on a given space X . At this point, topological complexity becomes a useful concept. By definition, the topological complexity $TC(X)$ of a space X is the smallest integer k , such that $X \times X$ may be covered by k open subsets $U_i \subseteq X \times X$, such that over each U_i there exists a continuous motion planning $s_i : U_i \rightarrow X^I$. As noted in [Far03], given such “local motion planning rules”, one may then define a global motion planning $s : X \times X \rightarrow X^I$ by choosing one local rule whenever there is overlap between one or more regions of the covering. Of course, if $TC(X) > 1$, a motion planning constructed in this way cannot be continuous.

What implications does this have in practice? Let us return to the example where $X = \mathbb{S}^1$. One can show that $TC(\mathbb{S}^1) = 2$ (See example 4.5), implying two things: First, it is not possible to construct a motion planning algorithm which is continuous at every pair $(a, b) \in X \times X$ of start- and end-points. Second, with regard to stability, the most optimal motion planning on X one could hope for consists of just two continuous local motion planning rules.

It turns out that topological complexity is a homotopy invariant, which means that the nature of discontinuities of motion planning algorithms is entirely dependent on the topological properties of the given configuration

²If you are concerned by the many quotation marks in this chapter, do not worry, as the considered concepts and definitions will be made mathematically precise throughout the thesis.

space X . We will also see that an upper bound of the complexity of a space can be stated in terms of its covering dimension, whereas a lower bound can be found via the structure of its cohomology ring. In some cases, those bounds can be used to compute the complexity of a space without constructing explicit motion planning algorithms, see section 4.7. Furthermore, topological complexity has connections with the Lusternik-Schnirelmann category, which is another homotopy invariant.

After that, we move to directed topological complexity, which is a variant of topological complexity for the setting of directed spaces. A directed space, or d-space for short, is a topological space X equipped with a d-structure PX , that encodes some sense of direction. This is done by choosing a subspace $PX \subseteq X^I$, which is closed under concatenation and reparametrization by non-decreasing maps $I \rightarrow I$. We call those paths directed paths, or d-paths. It turns out that directed complexity is invariant under simple dihomotopy equivalences, which is a stronger version of homotopy equivalence for the case of d-spaces.

Towards the end of the thesis, there will be a brief introduction to geodesic complexity, which is an adaption of topological complexity for metric spaces and which turns out to be invariant under isometries. We will see that the difference between topological complexity and geodesic complexity of a metric space (X, d) can get arbitrarily large, depending on the metric d . The reader will also be introduced to the concept of the total cut locus of a space and its connection to geodesic complexity.

2 Preliminaries

First, let us review some definitions and results, which will be needed throughout the thesis.

2.1 General Topology

The following definitions and results can be found in standard textbooks on (general) topology, for example [Arm83]. Note that the compact-open topology is not covered in this chapter, rather it is treated separately in chapter 3.

Definition 2.1. Let X be a topological space and $Y \subseteq X$ a subset. Then the *subspace topology* or *induced topology* on Y is obtained by defining $V \in Y$ to be open, if there is an open set U in X such that $V = U \cap Y$.

Example 2.2. The *standard unit interval* I is the subspace $[0, 1] \subseteq \mathbb{R}$ with respect to the standard topology on $\mathbb{R}$.

Proposition 2.3 (Glueing Lemma). *Let X, Y be subsets of a topological space and give each of X, Y and $X \cup Y$ the induced topology. Further let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be continuous maps, such that $f|_{X \cap Y} = g|_{X \cap Y}$. If X and Y are either both closed or both open in $X \cup Y$, then $f \cup g$ is continuous, with*

$$f \cup g : X \cup Y \rightarrow Z, \quad (f \cup g)(p) = \begin{cases} f(p), & p \in X \\ g(p), & p \in Y. \end{cases}$$

Corollary 2.4. *For a space X and for every $i = 1, \dots, n$, let $X_i \subseteq X$ be a subset, such that $X_i \cap X_j = \emptyset$ for $i \neq j$. Furthermore, consider the subspace $A = \bigcup_{i=1}^n X_i$ and suppose we have a continuous map $f_i : X_i \rightarrow Z$ for every i . If either every X_i is open in A or every X_i is closed in A , then the map $f : A \rightarrow Z$ defined by $f(x) = f_i(x)$ if $x \in X_i$ is continuous.*

Definition 2.5. Let X and Y be topological spaces and let $\mathcal{B}$ denote the family of all subsets $U \times V \subseteq X \times Y$ where U is open in X and V is open in Y . Then $\mathcal{B}$ is a base for a topology on $X \times Y$ which is called *product topology*. $X \times Y$ equipped with the product topology is called the *product space* of X and Y .

Proposition 2.6. *The product topology is the coarsest/smallest topology on $X \times Y$ for which both projections $X \times Y \rightarrow X$ and $X \times Y \rightarrow Y$ are continuous.*

Proposition 2.7. *Let X, Y, Z be topological spaces. A map $f : Z \rightarrow X \times Y$ is continuous if and only if both $p_1 \circ f$ and $p_2 \circ f$ are continuous.*

Definition 2.8. Given continuous maps $f : X \rightarrow Y$ and $g : X' \rightarrow Y'$, we denote by $f \times g$ the continuous map given by

$$X \times X' \rightarrow Y \times Y', \quad (x, x') \mapsto (f(x), g(x')).$$

Definition 2.9. Let (X, d) be a metric space. Then the *metric topology* on X is given by defining $U \subseteq X$ to be open if for any $x \in U$ there is some $r > 0$, such that

$$B_r(x) := \{x' \in X \mid d(x, x') < r\}$$

is contained in U . We call $B_r(x)$ an *open ball with radius r and center x* .

Note that any open ball is indeed open with respect to the metric topology. From now on, we always assume a metric space (X, d) to be equipped with its metric topology, unless otherwise noted.

Definition 2.10. Let $(X, d_X), (Y, d_Y)$ be metric spaces. Then we define the *product metric* on $X \times Y$ as

$$d((x, y), (x', y')) := \max(d_X(x, x'), d_Y(y, y')).$$

Proposition 2.11. Let (X, d_X) and (Y, d_Y) be metric spaces. Then the product topology on $X \times Y$ is the same as the metric topology induced by the product metric d .

Proof. Let $W \subseteq X \times Y$ be open with respect to the product topology. This means, that W is a union of sets of the form $\tilde{U} \times \tilde{V}$ where $\tilde{U}$ is open in X and $\tilde{V}$ is open in Y . Let $(x, y) \in W$, then there are open sets $U \subseteq X$ and $V \subseteq Y$ such that $(x, y) \in U \times V$. Since U and V are open in their respective spaces, there exist $r_x, r_y > 0$, such that

$$B_{r_x}(x) \subseteq U, \quad B_{r_y}(y) \subseteq V.$$

Choose $r := \min(r_x, r_y)$ and let d denote the product metric. Then

$$\begin{aligned} B_r(x, y) &= \{(x', y') \in X \times Y \mid d((x, y), (x', y')) < r\} \\ &= \{(x', y') \in X \times Y \mid d_X(x, x'), d_Y(y, y') < r\} \\ &= B_r(x) \times B_r(y) \\ &\subseteq B_{r_x}(x) \times B_{r_y}(y) \\ &\subseteq U \times V \\ &\subseteq W. \end{aligned}$$

Since $B_r(x, y)$ is an open ball with respect to the product metric, W is open with respect to the topology induced by the product metric.

On the other hand, let $A \subseteq X \times Y$ be open with respect to the topology induced by the product metric. This means, for $(x, y) \in A$, there exists $r_{xy} > 0$, such that $B_{r_{xy}}(x, y) \subseteq A$. As above, we have

$$B_{r_{xy}}(x, y) = B_{r_{xy}}(x) \times B_{r_{xy}}(y)$$

and $B_{r_{xy}}(x)$ is open in X , $B_{r_{xy}}(y)$ is open in Y , thus $B_{r_{xy}}(x, y)$ is a basic open set with respect to the product topology. Furthermore we have

$$A = \bigcup_{(x, y) \in A} \{(x, y)\} \subseteq \bigcup_{(x, y) \in A} B_{r_{xy}}(x, y) \subseteq A,$$

hence A is a union of sets which are open in the product topology. $\square$

Remark 2.12. There are many different choices for a product metric on $X \times Y$, which all induce the product topology. Another common choice is

$$d_p((x, y), (x', y')) := (|d_X(x, x')|^p + |d_Y(y, y')|^p)^{1/p}$$

for some $p \in [1, \infty)$.

Definition 2.13. A Hausdorff space X is called *paracompact*, if every open cover of X has a locally finite refinement.

The following results can be found in [Sin19], Theorem 9.1.7 and Theorem 9.1.18.

Proposition 2.14. Every metric space is paracompact.

Proposition 2.15. Every open cover of a paracompact space has a partition of unity subordinate to it.

2.2 Locally Compact Spaces

Definition 2.16. A topological space X is called *locally compact*, if for every $x \in X$, there exists an open neighborhood U of x such that its closure $\bar{U}$ is a compact subspace of X .

Definition 2.17. A subset $A \subseteq X$ is called *locally closed* if it is of the form $A = F \cap V$ where F is closed and V is open in X .

Note that any closed subset $A \subseteq X$ is locally closed, since one can write $A = A \cap X$. The same holds for any open subset of X .

The following result can be found in [Eng89], 3.3.10. Corollary.

Proposition 2.18. A subspace of a locally compact space is locally compact if and only if it is locally closed.

Example 2.19. The real line $\mathbb{R}$ equipped with the standard topology is locally compact and thus also the standard unit interval I is locally compact, since it is closed and thus locally closed.

Recall, that the product of two spaces $X \times Y$ is compact if and only if X and Y are compact, a closed subset of a compact space is compact, and a compact subset of a Hausdorff space is closed. See for example [Arm83], theorems (3.5), (3.6) and (3.15).

One then easily proves, that a product of two locally compact spaces is locally compact, and that the intersection of two locally compact subspaces of a Hausdorff space is also locally compact.

2.3 Euclidean Neighborhood Retracts

Definition 2.20. A subspace $A \subseteq Z$ is called a *neighborhood retract*, if it is a retract of some open neighborhood $A \subseteq U \subseteq Z$. A space X is called an *Euclidean Neighborhood Retract (ENR)*, if it is homeomorphic to some neighborhood retract $X' \subseteq \mathbb{R}^n$.

Recall that by the Tietze Extension Theorem (See for example [Die08] (1.1.2) Theorem), given a normal and Hausdorff space Y , a closed subset $A \subseteq Y$, and a continuous map $f : A \rightarrow \mathbb{R}^n$, there is a continuous extension $g : Y \rightarrow \mathbb{R}^n$ of f , i.e. $g|_A = f$.

The following two results can be found in [Dol95], chapter IV, 8.13 Exercises 1 & 2.

Proposition 2.21. *Given an ENR X , a normal space Y , a closed subset $A \subseteq Y$ and a continuous map $f : A \rightarrow X$, there exists an extension $\tilde{f} : V \rightarrow X$ for some open neighborhood V of A , i.e. $\tilde{f}|_A = f$.*

Proof. X being an ENR implies the existence of some subsets $X' \subseteq U \subseteq \mathbb{R}^n$, such that U is open in $\mathbb{R}^n$, there is a retraction $r : U \rightarrow X'$ and a homeomorphism $h : X \rightarrow X'$. By the Tietze Extension Theorem, there is a continuous extension $g : Y \rightarrow \mathbb{R}^n$ of $h \circ f : A \rightarrow X' \subseteq \mathbb{R}^n$. Consider $V := g^{-1}(U)$, which is an open neighborhood of A , since $g(A) = h(f(A)) \subseteq X' \subseteq U$. Define $\tilde{f} : V \rightarrow X$ as

$$\tilde{f} := h^{-1} \circ r \circ g|_V.$$

We know that r is a retraction, so $r|_{X'} = \text{id}_{X'}$. Thus we have $r \circ h \circ f = h \circ f$, implying

$$\begin{aligned} \tilde{f}|_A &= h^{-1} \circ r \circ g|_A \\ &= h^{-1} \circ r \circ h \circ f \\ &= h^{-1} \circ h \circ f \\ &= f. \end{aligned}$$

This is displayed in the following commutative diagram:

$$\begin{array}{ccccc} A & \xrightarrow{f} & X & \xleftarrow[h]{h^{-1}} & X' \\ \downarrow & \nearrow \tilde{f} & & & \downarrow r \\ V & \xrightarrow{g|_V} & U & & \\ \downarrow & & \downarrow & & \\ Y & \xrightarrow{g} & \mathbb{R}^n & & \end{array}$$

□

Proposition 2.22. *Let X be an ENR, Y a space such that $Y \times [0, 1]$ is normal, $A \subseteq Y$ a closed subset of Y and $f_0, f_1 : Y \rightarrow X$ such that $f_0|_A \simeq f_1|_A$. Then there exists a neighborhood $V \supseteq A$ such that $f_0|_V \simeq f_1|_V$.*

The following proposition can be found in [Dol95], chapter IV, 8.1 Proposition.

Proposition 2.23. *If $X \subseteq \mathbb{R}^n$ is a neighborhood retract, then X is locally closed.*

Corollary 2.24. *Any ENR is locally compact.*

Proof. Let X be an ENR, i.e. it is homeomorphic to some neighborhood retract $X' \subseteq \mathbb{R}^n$. Then X' is locally closed, hence locally compact by proposition 2.18. But then also X is locally compact, since local compactness is invariant under homeomorphism. □

We close this section with an additional result from [Die08], (18.4.6) Theorem.

Proposition 2.25. *Let X be a Hausdorff space that has an open cover $X = X_1 \cup \dots \cup X_n$ such that each X_i is an ENR. Then X is an ENR.*

3 Path Spaces and the Compact-open topology

As already discussed in the introduction, the concept of topological complexity is about motion planning algorithms on some space X , i.e. maps of the form $s : U \rightarrow X^I$, where U is some open subset of $X \times X$ and X^I is the space of all continuous paths in X . When talking about continuous maps, of course we have to specify which topologies are used for domain and codomain respectively. For the domain there is a canonical choice: Since X is a topological space, it induces a topology on $X \times X$ via the product topology. A common choice for X^I is the compact-open topology, whose properties we will discuss in the first part of this chapter. In the second part, we introduce path spaces and the path fibration of a space. With those tools at hand, we can formally introduce the notion of a continuous motion planning.

3.1 The Compact-open Topology

Definition 3.1. Let X and Y be topological spaces and denote

$$X^Y := \{f : X \rightarrow Y \mid f \text{ continuous}\}.$$

Recall, that a *subbasis* S of a topology T on X is a subset $S \subset T$, if the set of all finite intersections of elements in S together with the set X form a basis for T . In other words, every open set $U \in T$ can be written as a union of finite intersections of elements of S .

The set X^Y can be given the structure of a topological space via the compact-open topology, see for example [Hat02] Appendix.

Definition 3.2. For some compact subset $K \subseteq Y$ and an open subset $U \subseteq X$, define

$$V(K, U) := \{f \in X^Y \mid f(K) \subseteq U\}.$$

Then the collection of all such $V(K, U)$ is taken as a subbasis for the *compact-open topology* on X^Y . Furthermore, we refer to X^I equipped with the compact-open topology as the *(free) path space* of X , where I is the standard unit interval.

From now on, we always assume X^Y to be equipped with the compact-open topology, unless otherwise mentioned. The compact-open topology on X^Y can be regarded as a generalization of the topology of uniform convergence. This fact is reflected in the following proposition, which can be found in [Hat02], Proposition A.13.

Proposition 3.3. Let (X, d) be a metric space and Y be a compact topological space. Then the compact-open

topology on X^Y is the same as the metric topology induced by the metric

$$d(f, g) = \sup_{y \in Y} d(f(y), g(y)).$$

The next propositions, which are taken from [Hat02], Proposition A.14 and A.16, establish the main tools which we will use later when dealing with continuity of maps involving path spaces.

Proposition 3.4. Let X, Y, Z be topological spaces.

i) If Y is locally compact, then the evaluation map

$$e : X^Y \times Y \rightarrow X, \quad (f, y) \mapsto f(y)$$

is continuous.

ii) If $f : Y \times Z \rightarrow X$ is continuous, then so is the map

$$\hat{f} : Z \rightarrow X^Y, \quad \hat{f}(z)(y) = f(y, z).$$

iii) If Y is locally compact, then the converse to ii) holds.

Proposition 3.5. If Y is locally compact and Hausdorff and Z is Hausdorff, then the map

$$X^{Y \times Z} \rightarrow (X^Y)^Z, \quad f \mapsto \hat{f}$$

is a homeomorphism, where $\hat{f}$ is defined as in part ii) of the previous proposition.

As mentioned before, most of the time we are working with the path space X^I . Since I is locally compact and Hausdorff, we can state the following corollary as a combination of the two previous propositions for the special case $Y = I$.

Corollary 3.6. Let X, Z be topological spaces.

i) The evaluation map

$$e : X^I \times I \rightarrow X, \quad (\gamma, t) \mapsto \gamma(t)$$

is continuous.

ii) A map $f : I \times Z \rightarrow X$ is continuous if and only if the map

$$\hat{f} : Z \rightarrow X^I, \quad \hat{f}(z)(t) = f(t, z)$$

is continuous.

iii) If Z is Hausdorff, then

$$X^{I \times Z} \rightarrow (X^I)^Z, \quad f \mapsto \hat{f}$$

is a homeomorphism.

Remark 3.7. Part ii) of the corollary tells us, that the existence of a continuous map $h : Z \rightarrow X^I$ is nothing

else than the existence of a homotopy between the maps $h_0, h_1 : Z \rightarrow X$ given by $h_0 : z \mapsto h(z)(0)$ and $h_1 : z \mapsto h(z)(1)$.

As an application of this corollary, let us check continuity of some common path-operations. This will turn out useful when we deal with continuity of more complicated maps on path spaces.

Example 3.8. i) Let $Y \subseteq X^I \times X^I$ denote the subspace consisting of all pairs of paths for which the end-point of the first path is the starting point of the second path, i.e.

$$Y := \{(\alpha, \beta) \in X^I \times X^I \mid \alpha(1) = \beta(0)\}.$$

Then the concatenation map $*$: $Y \rightarrow X^I$, $(\alpha, \beta) \mapsto \alpha * \beta$ given by

$$(\alpha * \beta)(t) := \begin{cases} \alpha(2t), & 0 \leq t \leq \frac{1}{2} \\ \beta(2t - 1), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

is continuous:

By the previous corollary, it suffices to check continuity of the map

$$\tilde{*} : I \times Y \rightarrow X, \quad (t, \alpha, \beta) \mapsto (\alpha * \beta)(t).$$

On $[0, \frac{1}{2}] \times Y$, this map can be written as a composition of continuous maps

$$\begin{array}{ccc} [0, \frac{1}{2}] \times Y & \xrightarrow{(t, \alpha, \beta) \mapsto (t, \alpha)} & [0, \frac{1}{2}] \times X^I \\ & \searrow \tilde{*} & \downarrow (t, \alpha) \mapsto (2t, \alpha) \\ & & I \times X^I \\ & & \downarrow e \\ & & X \end{array}$$

where e denotes the evaluation map.

Similarly, on $[\frac{1}{2}, 1]$ this map can also be written as a composition of continuous maps.

Hence by the Glueing Lemma 2.3, we know that $\tilde{*}$ and thus also $*$ is continuous.

ii) Precomposing a path with any continuous map $I \rightarrow I$, i.e. the map

$$P : X^I \times I^I \rightarrow X^I, \quad (\alpha, \varphi) \mapsto \alpha \circ \varphi,$$

is continuous:

We denote by e_X and e_I the two evaluation maps $e_X : X^I \times I \rightarrow X$ and $e_I : I^I \times I \rightarrow I$, both of which are continuous. Continuity of P is equivalent to continuity of the map

$$\tilde{P} : I \times X^I \times I^I \rightarrow X, \quad (t, \alpha, \varphi) \mapsto \alpha(\varphi(t)).$$

We see that

$$\tilde{P}(t, \alpha, \varphi) = e_X(\alpha, \varphi(t)) = e_X(\alpha, e_I(\varphi, t))$$

and thus $\tilde{P}$ can be written as the composition

$$I \times X^I \times I^I \xrightarrow{(t, \alpha, \varphi) \mapsto (\alpha, e_I(\varphi, t))} X^I \times I \xrightarrow{e_X} X$$

of continuous maps.

iii) Suppose that ρ is the map that restricts a given path to $[t, t']$ in the sense that

$$\rho : I \times I \times X^I \rightarrow X^I, \quad (t, t', \alpha) \mapsto \alpha_{t, t'},$$

where $\alpha_{t, t'} : \tau \mapsto \alpha((t' - t) \cdot \tau + t)$.

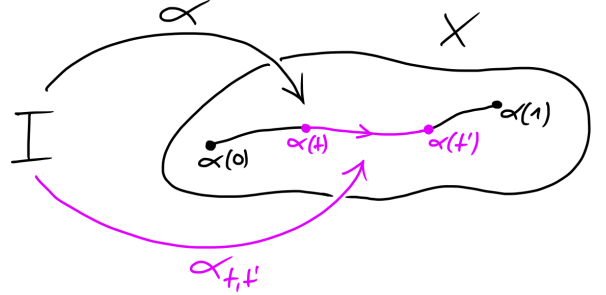


Figure 5: Restriction of a path.

Then ρ is continuous:

Consider for $(t, t') \in I \times I$ the map $\varphi_{t, t'} \in I^I$ given by $\varphi_{t, t'}(\tau) = (t' - t) \cdot \tau + t$. Then the map

$$I \times I \rightarrow I^I, \quad (t, t') \mapsto \varphi_{t, t'}$$

is continuous, since

$$I \times I \times I \rightarrow I, \quad (\tau, t, t') \mapsto \varphi_{t, t'}(\tau) = (t' - t) \cdot \tau + t$$

is continuous. Thus, ρ can be written as a composition of continuous maps

$$I \times I \times X^I \xrightarrow{(t, t', \alpha) \mapsto (\varphi_{t, t'}, \alpha)} X^I \times I^I \xrightarrow{P} X^I$$

where P denotes the precomposition map from ii).

With the help of the previous examples, we can now show that for a given space X , its corresponding path space X^I is homotopy-equivalent to the space itself.

Proposition 3.9. *Given a space X , we have $X \simeq X^I$.*

Proof. Consider the map

$$f : X \rightarrow X^I, \quad x \mapsto c_x$$

where c_x is the constant path at the point $x \in X$. Then f is continuous, since

$$I \times X \rightarrow X, \quad (t, x) \mapsto c_x(t) = x$$

is continuous.

As a possible homotopy inverse, consider

$$g : X^I \rightarrow X, \quad \alpha \mapsto \alpha(0).$$

This map is continuous: Take an open set $U \subseteq X$, then

$$g^{-1}(U) = \{\alpha \in X^I \mid \alpha(0) \in U\}$$

is a set of the form as given in definition 3.2, with the choice $K = \{0\}$. Hence it is open in X^I .

We immediately get $g \circ f = \text{id}_X$. In order to show $f \circ g \simeq \text{id}_{X^I}$, consider

$$H : I \times X^I \rightarrow X^I, \quad (t, \alpha) \mapsto H_t(\alpha) := \rho(0, t, \alpha),$$

where ρ is the restriction map introduced in part iii) of the previous examples. H is continuous, because the restriction map ρ is continuous. Furthermore we have

$$H_0(\alpha) = \rho(0, 0, \alpha) = c_{\alpha(0)} = (f \circ g)(\alpha)$$

and

$$H_1(\alpha) = \rho(0, 1, \alpha) = \alpha,$$

which finishes the proof. $\square$

3.2 Path Spaces and Continuous Motion Plannings

We are now equipped with enough tools to make our considerations about discontinuities of motion planning algorithms more precise. The definitions and considerations in this section are mainly based on [Far03] and [Far08], chapter 4.

Definition 3.10. For a topological space X , the *path fibration* π_X of X is given by

$$\pi_X : X^I \rightarrow X \times X, \quad \gamma \mapsto (\gamma(0), \gamma(1)).$$

A (*global*) *section* of π_X is a (possibly discontinuous) map $s_X : X \times X \rightarrow X^I$ such that $\pi_X \circ s_X = \text{id}_{X \times X}$, i.e.

$$s_X(a, b)(0) = a, \quad s_X(a, b)(1) = b.$$

As discussed in the introduction, a *motion planning algorithm* on X is just a section of π_X .

Note that most of the time we will omit the subscript X and just write π and s instead of π_X and s_X respectively.

Continuity of π immediately follows from corollary 3.6, since the evaluation map is continuous. Furthermore, it is indeed a fibration in the sense of algebraic

topology, as shown in the next chapter. This will be important later, when we define topological complexity as the sectional category of the path-fibration, see 4.1.

In the introduction we have discussed a motivating example of motion planning on $\mathbb{S}^1$ and noticed that for specific inputs, small deviations of the input can lead to vastly different outputs. Indeed, this intuitive view of “discontinuous” behaviour of a motion planning $s : X \times X \rightarrow X^I$ is coherent with our approach of making discontinuities mathematically precise by equipping X^I with the compact-open topology. To see this, consider the case where the configuration space of our system is some metric space (X, d) . In this case, the product $X \times X$ inherits the structure of a metric space via the product metric d from definition 2.10, such that the metric topology coincides with the product topology. Furthermore, in this case the compact-open topology on X^I is the same as the metric topology induced by the metric

$$\bar{d} : X^I \times X^I \rightarrow \mathbb{R}, \quad (\gamma, \gamma') \mapsto \max_{t \in I} d(\gamma(t), \gamma'(t)),$$

see proposition 3.3. Now suppose, we have a section $s : X \times X \rightarrow X^I$ which is discontinuous at some point $(a, b) \in X \times X$. By the ε/δ -criterion, there exists some $\varepsilon > 0$, such that for all $\delta > 0$ there exists another pair $(a', b') \in X \times X$ which satisfies $d((a, b), (a', b')) < \delta$ but

$$\max_{t \in I} d(s(a, b)(t), s(a', b')(t)) = \bar{d}(s(a, b), s(a', b')) > \varepsilon.$$

This means: There exist pairs (a', b') which are arbitrarily close to (a, b) with respect to the metric on $X \times X$, but for which the resulting paths $s(a, b)$ and $s(a', b')$ are not close in the sense that there will always be some time $t \in I$ where they differ by at least ε with respect to the metric in X .

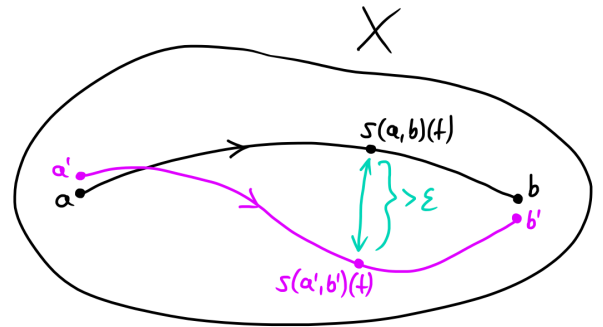


Figure 6: Discontinuous behaviour of a map $s : X \times X \rightarrow X^I$, where X is a metric space.

In this sense, a small error in the input (a, b) can lead to a large error in the resulting paths. On the other hand, if the section s is continuous, small input errors

result in small output errors.

More generally, translating our observations to the language of topology: Continuity of the section s ensures, that pairs (a, b) and (a', b') , which are “close” with respect to the topology on $X \times X$ result in paths $s(a, b)$ and $s(a', b')$ which are “close” with respect to the compact-open topology.

Let us have a look at two specific motion planning scenarios. The examples are inspired by [Far04] Examples 1.1 and 1.2.

Example 3.11. i) Consider $X = \mathbb{R}^n$ with the standard topology. Given a pair $(a, b) \in X \times X$, we can take the straight line between them given by

$$\gamma_{a,b} : I \rightarrow X, \quad \gamma_{a,b}(t) = (1-t)a + tb.$$

This defines a section of the path fibration via

$$s : X \times X \rightarrow X^I, \quad s(a, b) = \gamma_{a,b}.$$

Continuity of s follows by corollary 3.6, since the map

$$I \times X \times X \rightarrow X, \quad (t, a, b) \mapsto s(a, b)(t) = (1-t)a + tb$$

is continuous. This shows, that a continuous motion planning on $\mathbb{R}^n$ exists.

ii) Consider $\mathbb{R}^n$ with the origin removed, i.e. $X = \mathbb{R}^n \setminus \{0\}$. We will show that every motion planning, that is, a section of π , must be discontinuous. Suppose that a continuous section $s : X \times X \rightarrow X^I$ of the path fibration exists, i.e. $\pi \circ s = \text{id}_{X \times X}$. Pick some $a_0 \in X$ and consider the map

$$X \rightarrow X^I, \quad x \mapsto s(a_0, x),$$

which is continuous by assumption and thus by corollary 3.6 induces a continuous map

$$h : I \times X \rightarrow X, \quad (t, x) \mapsto h_t(x) := s(a_0, x)(t).$$

We obtain that h_0 is equal to the constant map $x \mapsto a_0$ and $h_1 = \text{id}_X$, implying that X is contractible. But this is a contradiction, since $X \simeq \mathbb{S}^{n-1}$ and $\mathbb{S}^{n-1}$ is not contractible. Hence a global continuous motion planning cannot exist in this case.

Remark 3.12. With the same reasoning as in i), one can show that a continuous motion planning exists for any convex subspace $X \in \mathbb{R}^n$.

In the second example, we modified the topology of the space from the first part by adding an obstruction, in this case a hole, which made it impossible for a continuous motion planning to exist. This serves as a motivating example for the main result of this chapter,

which was originally shown in [Far03] and which tells us that a continuous motion planning can only exist contractible spaces. Note that the same reasoning as in part ii) of the previous example will be used to prove that existence of a continuous section of π implies contractibility of the space itself.

Before moving to this result, we first introduce the notion of a local section and prove two more general results, the latter of which we will also revisit later when we consider relations between LS-category and topological complexity.

Definition 3.13. Given an open subset $U \subseteq X \times X$, a continuous map $s : U \rightarrow X^I$ with $\pi \circ s = \text{id}_U$ is called a *local section of π over U* .

As already mentioned in remark 3.7, a continuous map into the path-space X^I automatically defines a homotopy between specific maps, leading to the following result:

Proposition 3.14. *Let $U \subseteq X \times X$ be an open subset. Then a local section s of π over U exists if and only if the two projections $p_1, p_2 : U \rightarrow X$ onto the first and second factors are homotopic.*

Proof. The statement follows more or less directly from part ii) of corollary 3.6. First suppose that a local section $s : U \rightarrow X^I$ of π exists. Then the map $h : I \times U \rightarrow X$ defined by

$$h : (t, a, b) \mapsto h_t(a, b) := s(a, b)(t)$$

is continuous and thus defines a homotopy between the two projections

$$h_0 : (a, b) \mapsto s(a, b)(0) = a$$

and

$$h_1 : (a, b) \mapsto s(a, b)(1) = b.$$

On the other hand, suppose the two projections $p_1, p_2 : U \rightarrow X$ onto the first and second factors are homotopic, i.e. there is a homotopy

$$g : I \times U \rightarrow X, \quad (t, a, b) \mapsto g_t(a, b)$$

with $g_0 : (a, b) \mapsto a$ and $g_1 : (a, b) \mapsto b$. Then the map $\sigma : U \rightarrow X^I$ defined via

$$\sigma(a, b)(t) := g_t(a, b)$$

defines a continuous section of π , since

$$\pi \circ \sigma(a, b) = (\sigma(a, b)(0), \sigma(a, b)(1)) = (a, b). \quad \square$$

The next proposition is inspired by the proofs of Theorem 5 and Theorem 1 from [Far03].

Proposition 3.15. *Let X be a topological space.*

- i) If a local section of π exists over some open subset $U \subseteq X \times X$, then for some fixed $a_0 \in X$ the set

$$V := \{b \in X \mid (a_0, b) \in U\}$$

is open and contractible in X .

- ii) If $W \subseteq X$ is open and contractible in X , then there is a local section of π over $W \times W \subseteq X \times X$.

Proof. To prove i), let $U \subseteq X \times X$ be an open subset, such that a continuous section $s : U \rightarrow X^I$ of π exists. Fix some $a_0 \in X$. Consider the continuous map

$$g : X \rightarrow X \times X, \quad b \mapsto (a_0, b)$$

and define $V := g^{-1}(U) = \{b \in X \mid (a_0, b) \in U\}$. Then V is open because U is open. Furthermore, we can show that V is contractible in X by constructing a homotopy between the constant map $V \rightarrow \{a_0\}$ and the inclusion $V \hookrightarrow X$. Consider

$$h : I \times V \rightarrow X, \quad (t, b) \mapsto h_t(b) := s(a_0, b)(t).$$

This map is continuous, since it can be written as the concatenation

$$I \times V \xrightarrow{(t, b) \mapsto (t, a_0, b)} I \times \underbrace{\{a_0\} \times V}_{\subseteq I \times X \times X} \xrightarrow{(t, a, b) \mapsto s(a, b)(t)} X$$

where the latter map is continuous by corollary 3.6. Furthermore for all $b \in V$ we have $h_0(b) = s(a_0, b)(0) = a_0$ and $h_1(b) = s(a_0, b)(1) = b$, since s was assumed to be a local section of π over U . Hence V is contractible in X .

For ii), assume that $W \subseteq X$ is open and contractible in X . That is, there is a homotopy

$$\tilde{h} : I \times W \rightarrow X, \quad (t, b) \mapsto \tilde{h}_t(b)$$

such that for some point $w_0 \in W$ we have that $\tilde{h}_1$ is the constant map $W \rightarrow \{w_0\}$ and $\tilde{h}_0$ is the inclusion $W \hookrightarrow X$. For a given pair $(a, b) \in W \times W$, we can then define a path between those points by concatenating the path $t \mapsto \tilde{h}_t(a)$ with the inverse path of $t \mapsto \tilde{h}_t(b)$. Hence we can define a map $\tilde{s} : I \times W \times W \rightarrow X^I$ by

$$\tilde{s}(t, a, b) = \begin{cases} \tilde{h}_{2t}(a), & 0 \leq t \leq \frac{1}{2}, \\ \tilde{h}_{2(1-t)}(b), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

To see that $\tilde{s}$ is continuous, note that on $[0, \frac{1}{2}] \times W \times W$, the map is just the composition

$$[0, \frac{1}{2}] \times W \times W \xrightarrow{m} I \times W \times W \xrightarrow{p} I \times W \xrightarrow{h} X,$$

where

$$\begin{aligned} m : (t, a, b) &\mapsto (2t, a, b), \\ p : (t, a, b) &\mapsto (t, a), \end{aligned}$$

and is thus continuous. Similar we can show that $\tilde{s}$ is continuous on $[\frac{1}{2}, 1] \times W \times W$. Since those two subsets form a closed cover of $I \times W \times W$ and $\tilde{h}_1(a) = w_0 = \tilde{h}_1(b)$, we have that $\tilde{s}$ is continuous by the glueing lemma. Thus the map $s : W \times W \rightarrow X^I$ defined by

$$s(a, b)(t) := \tilde{s}(t, a, b)$$

is continuous by corollary 3.6. Furthermore we have $s(a, b)(0) = \tilde{h}_0(a) = a$ and $s(a, b)(1) = \tilde{h}_0(b) = b$, so s indeed a local section of π over $W \times W$. $\square$

Corollary 3.16. For a topological space X , a (global) continuous section of the path fibration exists if and only if X is contractible.

Proof. Let X be a topological space. First assume that a continuous section of π exists over $X \times X$. Then by proposition 3.15 i) we have that

$$X = \{b \in X \mid (a_0, b) \in X \times X\}$$

is contractible, where a_0 is some point in X .

On the other hand, suppose that X is contractible. Then proposition 3.15 ii) implies the existence of a continuous section of π over $X \times X$. $\square$

3.3 Fibrations, Cofibrations and the Path Fibration

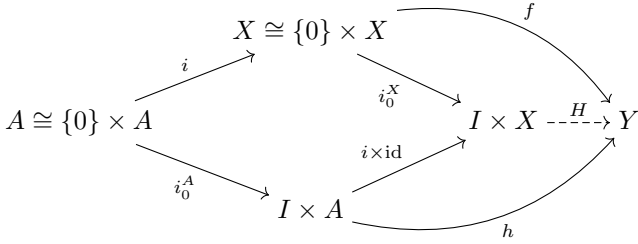
In this chapter, the concepts of fibrations and cofibrations are introduced and we will see that the path fibration is indeed a fibration.

Definition 3.17. Let E, B, X be topological spaces. A continuous map $p : E \rightarrow B$ has the *homotopy lifting property (HLP)* for the space X if the following holds: Let $h : I \times X \rightarrow B$ be a homotopy and $a : X \rightarrow E$ be a continuous map such that $p \circ a = h \circ i$, where $i : X \cong \{0\} \times X \hookrightarrow I \times X$ denotes the inclusion map. Then there exists a homotopy $H : I \times X \rightarrow E$ with $p \circ H = h$ and $H \circ i = a$. In this case, H is called a *lifting* of h with *initial condition* a .

$$\begin{array}{ccc} X \cong \{0\} \times X & \xrightarrow{a} & E \\ \downarrow i & \nearrow H & \downarrow p \\ I \times X & \xrightarrow{h} & B \end{array}$$

The map $p : E \rightarrow B$ is called a *fibration*, if it has the HLP for all spaces.

Definition 3.18. Let A, X, Y be topological spaces. A continuous map $i : A \rightarrow X$ has the *homotopy extension property (HEP)* for some space Y if the following holds: First, let $i_0^A : \{0\} \times A \hookrightarrow I \times A$ and $i_0^X : \{0\} \times X \hookrightarrow I \times X$ denote the inclusions. Furthermore, let $h : I \times A \rightarrow Y$ be a homotopy and $f : X \rightarrow Y$ be a continuous map such that $f \circ i = h \circ i_0^A$. Then there exists a homotopy $H : I \times X \rightarrow Y$ with $H \circ i_0^X = f$ and $H \circ (i \times \text{id}) = h$. In this case, we call H an *extension* of h with initial condition f .



The map $i : A \rightarrow X$ is called a *cofibration*, if it has the HEP for all spaces.

The following three results can be found in [Hat02], Proposition 4.61, [Bre93], chapter VII, 1.4. Corollary and [Spa66], chapter 2, §8, Theorem 2.

Proposition 3.19. *Given a fibration $p : E \rightarrow B$, the fibers $F_b = p^{-1}(b)$ over each path-component of B are all homotopy equivalent.*

Proposition 3.20. *If A is a subcomplex of a CW-complex X , then the inclusion $A \hookrightarrow X$ is a cofibration.*

Example 3.21. From the previous proposition we can conclude that $\partial I \hookrightarrow I$ is a cofibration.

Proposition 3.22. *Let X' and X be locally compact Hausdorff spaces, $f : X' \rightarrow X$ be a cofibration and Y be any space. Then*

$$p : Y^X \rightarrow Y^{X'}, \quad g \mapsto g \circ f$$

is a fibration.

The next result is based upon [Spa66], chapter 2, §8, Corollary 3.

Corollary 3.23. *For any topological space X , the path fibration*

$$\pi : X^I \rightarrow X \times X, \quad \gamma \mapsto (\gamma(0), \gamma(1))$$

is a fibration.

Proof. We have seen in example 3.21 that the inclusion $f : \partial I \hookrightarrow I$ is a cofibration. Let X be a topological space. Since I and ∂I are locally compact Hausdorff spaces, the map

$$p : X^I \rightarrow X^{\partial I}, \quad \gamma \mapsto \gamma \circ f$$

is a fibration. Furthermore we have $X^{\partial I} \cong X \times X$. To see this, consider the maps

$$\alpha : X^{\partial I} \rightarrow X \times X, \quad g \mapsto (g(0), g(1))$$

and

$$\beta : X \times X \rightarrow X^{\partial I}, \quad (a, b) \mapsto g_{a,b}$$

where $g_{a,b}$ is given by $g_{a,b}(0) = a, g_{a,b}(1) = b$. Those maps are obviously inverse to each other. Continuity of β is straightforward to check, using proposition 3.4 and the glueing lemma. Continuity of α follows by the same proposition, since we know that the evaluation map $e : X^{\partial I} \times \partial I \rightarrow X$ is continuous and $\alpha(g) = (e(g, 0), e(g, 1))$. Note that $\pi = \alpha \circ p$ and we know that p is a fibration, hence it has the HLP for all spaces. Let $h : I \times Z \rightarrow X \times X$ be a homotopy and $a : Z \rightarrow X^I$ be a continuous map, such that $\pi \circ a = h \circ i$ where i denotes the inclusion $Z \cong \{0\} \times Z \hookrightarrow I \times Z$. Then $\hat{h} := \beta \circ h$ is a homotopy $I \times Z \rightarrow X^{\partial I}$, so there is a homotopy $H : I \times Z \rightarrow X^I$ with $p \circ H = \hat{h}$ and $H \circ i = a$. Then we also have

$$\pi \circ H = \alpha \circ p \circ H = \alpha \circ \hat{h} = \alpha \circ \beta \circ h = h.$$

This shows that π also has the HLP with respect to any space, hence it is indeed a fibration. $\square$

Remark 3.24. Note that the same arguments as in the proof could be used to show that given spaces E, B_1, B_2 , a fibration $p_1 : E \rightarrow B_1$, a continuous map $p_2 : E \rightarrow B_2$ and a homeomorphism $f : B_1 \rightarrow B_2$ with $p_2 = f \circ p_1$, it follows that p_2 is a fibration.

4 Topological Complexity

The definitions, results and examples in this chapter are mainly based on [Far03] and [Far08], chapter 4.

4.1 Sectional Category and Definition of Topological Complexity

In order to define the topological complexity of a space, we use the following notation, which is adapted from [Cor+03], Definition 9.13. Note that our definition is 1 bigger than the definition in [Cor+03].

Definition 4.1. Let $p : E \rightarrow B$ be a fibration. The *sectional category* or *Schwarz genus* $\text{secat}(p)$ of p is the smallest integer k such that for $i = 1, \dots, k$ there exist open subsets $U_i \subseteq B$ and corresponding continuous maps $s_i : U_i \rightarrow E$ such that

$$B = U_1 \cup \dots \cup U_k,$$

and $p \circ s_i = \text{id}_{U_i}$ for each $i = 1, \dots, k$.

If no such number k exists, set $\text{secat}(p) := \infty$.

We denote the sets U_i given above as *domains of continuity* for p and refer to the maps s_i as *partial or local sections* of p .

Since the path fibration $\pi : X^I \rightarrow X \times X$ is indeed a fibration, see corollary 3.23, this allows us to define topological complexity of a space as the sectional category of its path fibration. The definition is due to [Far03].

Definition 4.2. Let X be a topological space. The *topological complexity* $\text{TC}(X)$ of X is defined as

$$\text{TC}(X) := \text{secat}(\pi_X),$$

where $\pi_X : X^I \rightarrow X \times X$ is the path fibration of X .

By definition, $\text{TC}(X)$ is the smallest integer k such that there exists a cover

$$X \times X = U_1 \cup \dots \cup U_k$$

by k open subsets of $X \times X$ and for each $i = 1, \dots, k$ there is a local section $s_i : U_i \rightarrow X^I$ of π_X , i.e. s_i is continuous and $\pi_X \circ s_i = \text{id}_{U_i}$. As in the definition of sectional category, we call the sets U_i *domains of continuity*.

Our first result about topological complexity is implied by proposition 3.14 and corollary 3.16.

Corollary 4.3. *Let X be a topological space. Then the following are equivalent:*

- i) $\text{TC}(X) = 1$.
- ii) X is contractible.
- iii) The projections $X \leftarrow X \times X \rightarrow X$ onto the first and second factors are homotopic to each other.

Remark 4.4. Note that the equivalence between ii) and iii) is a special case of a more general result which tells us that X is contractible if and only if for any space Y any two maps $f, g : Y \rightarrow X$ are homotopic.

Example 4.5. i) We have seen in example 3.11 that $\text{TC}(\mathbb{R}^n) = 1$ and $\text{TC}(\mathbb{R}^n \setminus \{0\}) > 1$.

- ii) This example is based upon one of the initial examples given in [Far03]. Consider the circle

$$\mathbb{S}^1 = \{x \in \mathbb{R}^2 \mid \|x\|_2 = 1\} \subseteq \mathbb{R}^2.$$

Since $\mathbb{S}^1$ is not contractible, we can conclude $\text{TC}(\mathbb{S}^1) \geq 2$. We will show that in fact $\text{TC}(\mathbb{S}^1) = 2$, by constructing an open cover with two domains of continuity for π , implying $\text{TC}(\mathbb{S}^1) \leq 2$.

Consider the two subsets

$$U_1 = \{(a, b) \in \mathbb{S}^1 \times \mathbb{S}^1 \mid a \neq -b\},$$

$$U_2 = \{(a, b) \in \mathbb{S}^1 \times \mathbb{S}^1 \mid a \neq b\}.$$

For $(a, b) \in U_1$, there is a unique shortest geodesic arc $\gamma_{a,b}$ from a to b . Hence, a continuous local section $s_1 : U_1 \rightarrow X^I$ over U_1 is given by $s_1 : (a, b) \mapsto \gamma_{a,b}$. While continuity of s_1 may intuitively be clear, it still has to be shown. Indeed, this can be done easily with tools from later in the thesis, see remark 6.21.

For a local section over U_2 , first fix an orientation of $\mathbb{S}^1$ by choosing a non-vanishing vector field on $\mathbb{S}^1$. Given $(a, b) \in U_2$, there exist exactly two geodesic arcs from a to b . Let $\tilde{\gamma}_{a,b}$ denote the geodesic, which is coherent with the chosen orientation, i.e. which moves a towards b with constant velocity in the positive direction. Then define

$$s_2 : U_2 \rightarrow X^I, \quad (a, b) \mapsto \tilde{\gamma}_{a,b}.$$

This section cannot be extended to $\mathbb{S}^1$, which can be seen as follows: For $(a, a) \in \mathbb{S}^1 \times \mathbb{S}^1$, choose sequences (a, b_n) and (a, c_n) where b_n converges to a from the positive direction and c_n converges to a from the negative direction. Then $s_2(a, b_n)$ converges to the constant path at a , while $s_2(a, c_n)$ converges to a path describing a full circle in the positive direction, see fig. 7.

As is often the case with topological invariants, they can be very hard to compute directly for some spaces and topological complexity is no different in that regard. Fortunately, there are some upper and lower bounds to assist in this matter, which will be explored in the upcoming chapters.

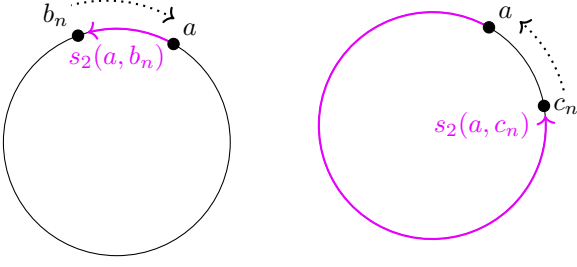


Figure 7: Discontinuous behaviour for any extension of s_2 to $\mathbb{S}^1$.

4.2 Topological Complexity as a Homotopy Invariant

As already mentioned in the introduction, topological complexity is a homotopy invariant, implying that the nature of discontinuities of motion planning algorithms depend entirely on the topological properties of the underlying configuration space. In order to show homotopy invariance of topological complexity, we first have a look at a more general result, from which homotopy invariance will follow. The proof is based upon [Far03], Theorem 3.

Proposition 4.6. *Let X, Y be path-connected topological spaces, such that X dominates Y , i.e. there exist maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$. Then $\text{TC}(Y) \leq \text{TC}(X)$.*

Proof. Let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$. First we will show that each domain of continuity for π_X will give us a domain of continuity for π_Y and after that we prove that those domains of continuity indeed cover the whole of $Y \times Y$.

Let $U \subseteq X \times X$ be an open subset such that there exists a continuous section $s : U \rightarrow X^I$ of π_X . Define

$$V := (g \times g)^{-1}(U)$$

which is open in $Y \times Y$. Let $h : I \times Y \rightarrow Y, (t, y) \mapsto h_t(y)$ be a homotopy between id_Y and $f \circ g$, i.e. $h_0 = \text{id}_Y$ and $h_1 = f \circ g$. Then define a map $\tilde{r} : I \times V \rightarrow Y$ by

$$\tilde{r}(t, a, b) = \begin{cases} h_{3t}(a), & 0 \leq t \leq \frac{1}{3}, \\ f(s(g(a), g(b))(3t-1)) & \frac{1}{3} \leq t \leq \frac{2}{3}, \\ h_{3(1-t)}(b), & \frac{2}{3} \leq t \leq 1. \end{cases}$$

Note that every restriction of $\tilde{r}$ to a subset $[\frac{i-1}{3}, \frac{i}{3}] \times V$, where $i = 1, \dots, 3$, coincides with the composition of continuous maps and is therefore continuous. Also note that on the overlaps of those subsets, the individual maps coincide, since

$$\begin{aligned} f(s(g(a), g(b))(0)) &= f(g(a)) = h_1(a), \\ f(s(g(a), g(b))(1)) &= f(g(b)) = h_1(b). \end{aligned}$$

Thus, by the glueing lemma 2.3 the whole map $\tilde{r}$ is continuous. But then by remark 3.6 also the map $r : V \rightarrow Y^I$ given by $r(a, b)(t) := \tilde{r}(t, a, b)$ is continuous.

Furthermore we have $r(a, b)(0) = h_0(a) = a$ and $r(a, b)(1) = h_0(b) = b$, so r is indeed a local section of π_Y over V .

Now assume that $k := \text{TC}(X) \neq \infty$, otherwise the statement is obviously true. This means there is an open cover

$$X \times X = U_1 \cup \dots \cup U_k$$

with a corresponding local section s_i of π_X over each U_i . Then the construction above yields open sets of the form $V_i = (g \times g)^{-1}(U_i)$ and corresponding local sections r_i of π_Y for each $i = 1, \dots, k$. We have

$$\begin{aligned} Y \times Y &= (g \times g)^{-1}(X \times X) \\ &= (g \times g)^{-1} \left(\bigcup_{i=1}^k U_i \right) \\ &= \bigcup_{i=1}^k (g \times g)^{-1}(U_i) \\ &= \bigcup_{i=1}^k V_i, \end{aligned}$$

hence the sets V_i form an open cover of $Y \times Y$, yielding $\text{TC}(Y) \leq k = \text{TC}(X)$. $\square$

Corollary 4.7 (Homotopy invariance). *If $X \simeq Y$, then $\text{TC}(X) = \text{TC}(Y)$.*

Note that this automatically implies $\text{TC}(Y) = \text{TC}(X)$ if Y is a deformation retract of X . However, there is also a connection between $\text{TC}(Y)$ and $\text{TC}(X)$ for the case where Y is just a retract of X , as shown in the next proposition. The proof is adapted from [Far08], Lemma 4.25, and is based upon the idea that a path in X can be “projected” to Y via the retraction map.

Proposition 4.8. *Let $Y \subseteq X$ be a retract. Then $\text{TC}(Y) \leq \text{TC}(X)$.*

Proof. Assume $k := \text{TC}(X)$ is finite, otherwise this is obviously true. Then there is an open cover

$$X \times X = U_1 \cup \dots \cup U_k$$

with corresponding continuous sections $s_i : U_i \rightarrow X^I$ of π_X . Let $r : X \rightarrow Y$ be a retraction, i.e. $r \circ \iota = \text{id}_Y$ where $\iota : Y \hookrightarrow X$ denotes the inclusion. Define

$$V_i := U_i \cap (Y \times Y),$$

which are open sets in $Y \times Y$. We have

$$\begin{aligned} Y \times Y &= (X \times X) \cap (Y \times Y) \\ &= \left(\bigcup_{i=1}^k U_i \right) \cap (Y \times Y) \\ &= \bigcup_{i=1}^k (U_i \cap (Y \times Y)) \\ &= \bigcup_{i=1}^k V_i. \end{aligned}$$

Furthermore, for each i consider the map

$$\tilde{\sigma}_i : I \times V_i \rightarrow Y, \quad \tilde{\sigma}_i(t, a, b) = r(s_i(a, b)(t)).$$

By corollary 3.6, this map is continuous, since r and s_i are continuous, and thus also $\sigma_i : V_i \rightarrow Y^I$ defined by

$$\sigma_i(a, b)(t) := \tilde{\sigma}_i(t, a, b)$$

is continuous. This is indeed a local section of π_Y over V_i , since

$$\sigma_i(a, b)(0) = r(s_i(a, b)(0)) = r(a) = a$$

and likewise $\sigma_i(a, b)(1) = b$.

This concludes $\text{TC}(Y) \leq k = \text{TC}(X)$. $\square$

4.3 Equivalent Characterisations of Topological Complexity

In [Far08] it is mentioned at the beginning of chapter 4.2 that in many practical applications, the configuration space X of interest is given by a semi-algebraic set and is thus homeomorphic to a polyhedron. It is then reasonable for practical purposes to consider the special case of X being an ENR, since the class of ENR spaces includes finite-dimensional cell complexes as well as compact manifolds (see proposition 2.25), which also oftentimes occur in real-world applications. Indeed, as proven with proposition 4.12 in [Far08], there are several equivalent characterisations of $\text{TC}(X)$ when X is an ENR. However, before we get to that result, we need some facts about normal spaces. Recall that a space is called *normal*, if any two disjoint nonempty closed neighborhoods have disjoint open neighborhoods.

The following two results can be found in [Mun14], Theorem 32.2 and [Eng89], 1.5.18 Theorem, respectively.

Proposition 4.9. *Every metrizable space is normal.*

Note that any ENR is metrizable and thus normal. Further, note that the product of two metrizable spaces is again metrizable, where one can use the product metric from definition 2.10.

Proposition 4.10 (Shrinking Lemma). *Let X be a normal space and let $\{U_i\}_{i \in J}$ be a point-finite open cover of X , i.e. for every $x \in X$ the set $\{i \in J \mid x \in U_i\}$ is finite. Then there exists an open cover $\{V_i\}_{i \in J}$ of X , such that $\overline{V_i} \subseteq U_i$ for all $i \in J$.*

In particular this holds true for any finite open cover.

The proof of the following result is based upon [Far08], Proposition 4.12.

Proposition 4.11. *Let X be an ENR.*

- i) *Let k denote the smallest integer such that there exists a (not necessarily continuous) section $\sigma : X \times X \rightarrow X^I$ of π and an increasing sequence of k open subsets*

$$\emptyset = V_0 \subseteq V_1 \subseteq \dots \subseteq V_k = X \times X,$$

with the property that for any $i = 0, 1, \dots, k-1$ the restriction $\sigma|_{V_{i+1} \setminus V_i}$ is continuous. Then $\text{TC}(X) = k$.

- ii) *Let ℓ denote the smallest integer such that there exists a (not necessarily continuous) section $\rho : X \times X \rightarrow X^I$ of π and an increasing sequence of ℓ closed subsets*

$$\emptyset = W_0 \subseteq W_1 \subseteq \dots \subseteq W_\ell = X \times X,$$

with the property that for any $i = 0, 1, \dots, \ell-1$ the restriction $\rho|_{W_{i+1} \setminus W_i}$ is continuous. Then $\text{TC}(X) = \ell$.

- iii) *Let r denote the smallest integer such that there exists a (not necessarily continuous) section $\tau : X \times X \rightarrow X^I$ of π and a partition*

$$X \times X = F_1 \cup \dots \cup F_r, \quad F_i \cap F_j = \emptyset, i \neq j,$$

of $X \times X$ into r locally compact subsets F_i with the property that for any $i = 1, \dots, r$ the restriction $\tau|_{F_i}$ is continuous. Then $\text{TC}(X) = r$.

Proof. For the rest of the proof, let $N := \text{TC}(X)$, meaning there is an open cover

$$X \times X = U_1 \cup \dots \cup U_N$$

together with local continuous sections $s_i : U_i \rightarrow X^I$ of π .

We will prove $N = k = \ell = r$ by showing $N \geq k \geq r \geq N$ and $N \geq \ell \geq r \geq N$. Note that when proving the inequality $r \geq N$, it becomes clear, that $r = \infty$ if $N = \infty$, since otherwise N would be finite. In the same way, the inequalities $k \geq r$ and $\ell \geq r$ then imply $k = \ell = \infty$ if $\text{TC}(X) = \infty$.

$N \geq k :$

Set $V_0 := \emptyset$ and $V_i := U_1 \cup \dots \cup U_i$ for $i = 1, \dots, N$. Then

every V_i is open and we have

$$V_0 \subseteq V_1 \subseteq \cdots \subseteq V_N = X \times X.$$

Now for every $(x, y) \in X \times X$ let

$$j_{x,y} := \min\{i \in \{1, \dots, N\} \mid (x, y) \in U_i\}$$

be the smallest index such that $(x, y) \in U_{j_{x,y}}$. Then we can define a section of π via

$$\sigma : X \times X \rightarrow X^I, \quad (x, y) \mapsto s_{j_{x,y}}(x, y).$$

Note that

$$V_{i+1} \setminus V_i = U_{i+1} \setminus (U_1 \cup \cdots \cup U_i).$$

Thus for every $(x, y) \in V_{i+1} \setminus V_i$ we have $j_{x,y} = i + 1$ and therefore $\sigma|_{V_{i+1} \setminus V_i} = s_{i+1}$ is continuous.

$N \geq \ell$:

First note that, as an ENR, X is metrizable. As noted before, the product of two metrizable spaces is again metrizable and thus normal, hence we can apply the shrinking lemma 4.10 to $X \times X$ in order to obtain closed subsets $\tilde{U}_i \subseteq U_i$ such that

$$X \times X = \tilde{U}_1 \cup \cdots \cup \tilde{U}_N.$$

Note that every restriction $s_i|_{\tilde{U}_i}$ is continuous. Now if we choose $W_0 := \emptyset$ and $W_i := \tilde{U}_1 \cup \cdots \cup \tilde{U}_i$, then each W_i is closed, we have

$$W_0 \subseteq W_1 \subseteq \cdots \subseteq W_N = X \times X$$

and we can continue similar to before to first obtain a section $\rho : X \times X \rightarrow X^I$ and then show that each restriction $\rho|_{W_{i+1} \setminus W_i}$ is continuous.

$k \geq r$:

Suppose there is a section $\sigma : X \times X \rightarrow X^I$ of π together with an increasing sequence of k open subsets

$$\emptyset = V_0 \subseteq V_1 \subseteq \cdots \subseteq V_k = X \times X,$$

with the property that for any $i = 0, 1, \dots, k-1$ the restriction $\sigma|_{V_{i+1} \setminus V_i}$ is continuous.

For $i = 1, \dots, k$ define $F_i := V_i \setminus V_{i-1}$. Then

$$\bigcup_{i=1}^k F_i = \bigcup_{i=1}^k V_i \setminus V_{i-1} = \bigcup_{i=1}^k V_i = X \times X.$$

Each F_i is locally closed in the sense of definition 2.17, since $F_i = V_i \cap ((X \times X) \setminus V_{i-1})$ and V_i is open in $X \times X$, whereas $(X \times X) \setminus V_{i-1}$ is closed in $X \times X$. Since X is an ENR, also $X \times X$ is an ENR, so in particular it is locally compact. Then by 2.18 also F_i is locally compact as a subspace of $X \times X$.

$\ell \geq r$:

This can be shown similar to the proof of $k \geq r$.

$r \geq N$:

Suppose there is a section $\tau : X \times X \rightarrow X^I$ of π and a partition

$$X \times X = F_1 \cup \cdots \cup F_r, \quad F_i \cap F_j = \emptyset, i \neq j,$$

such that F_i is locally compact and $\tau|_{F_i}$ is continuous for each i . From now on let $i \in \{1, \dots, r\}$. By proposition 3.14, continuity of $\tau|_{F_i}$ is equivalent to the existence of a homotopy

$$h : I \times F_i \rightarrow X, \quad (t, a, b) \mapsto h_t(a, b)$$

such that $h_0 = p_1 : F_i \rightarrow X$ is the projection onto the first factor and $h_1 = p_2 : F_i \rightarrow X$ is the projection onto the second factor. Since F_i is locally compact, we can write it as $F_i = \overline{F_i} \cap W_i$ for some open $W_i \subseteq X \times X$ by proposition 2.18. This means that F_i is closed in W_i with respect to the subspace topology on W_i . Furthermore, F_i is metrizable as a subspace of $X \times X$, so $F_i \times I$ is also metrizable, thus normal. We can now apply proposition 2.22 to the maps $p_1, p_2 : W_i \rightarrow X$ to obtain a neighborhood $A_i \supseteq F_i$ with the property that $p_1|_{A_i} \simeq p_2|_{A_i}$. But by proposition 3.14, we know that this implies the existence of a continuous section $\alpha_i : A_i \rightarrow X^I$.

Furthermore, those sets A_i form a cover of $X \times X$ since

$$X \times X = \bigcup_{i=1}^r F_i \subseteq \bigcup_{i=1}^r A_i.$$

Thus $\text{TC}(X) \leq r$. □

As an application of this result, one can show that for the suspension $X = \Sigma Y$ of an ENR Y the inequality

$$\text{TC}(X) \leq 3$$

holds. For details see [Far08], Example 4.17.

Let us close this section with an additional remark on practical applications for topological complexity. As noted in the introduction, $\text{TC}(X)$ can be regarded as some kind of lower bound for discontinuities of any motion planning algorithm on X and in particular has connections to instabilities of motion planning algorithms. More explicitly, this is reflected in some results from [Far04], which we will roughly outline here.

Let X be a path-connected ENR and assume $\text{TC}(X)$ is finite. A motion planning algorithm $s : X \times X \rightarrow X^I$ is called *tame*, if there is a partition of

$$X \times X = F_1 \cup \cdots \cup F_k$$

into k pairwise disjoint subsets F_i , such that each F_i is

an ENR and each restriction $s|_{F_i}$ is continuous. We refer to the maps $s|_{F_i}$ as *local rules* of the motion planning algorithm. Of course, by proposition 4.11 iii), we know that k must be greater or equal than $\text{TC}(X)$.

Consider a tame motion planning algorithm $s : X \times X \rightarrow X^I$. Then the *order of instability* $\text{ord}_{a,b}(s)$ at a point $(a, b) \in X \times X$ of s is defined as the largest number n , such that any neighborhood of (a, b) has nonempty intersection with n distinct sets of the partition $\{F_i\}_i$. Furthermore, the *order of instability* $\text{ord}(s)$ of s is defined as

$$\text{ord}(s) := \max_{(a,b) \in X \times X} \text{ord}_{a,b}(s).$$

In other words: If $\text{ord}(s) = m$, then for any input (a, b) , small deviations from (a, b) may lead to at most m essentially distinct motions, in the sense that it may lead to m different motions given by the local motion planning rules $s|_{F_i}$.

Theorem 6.2 from [Far04] then tells us, that if X is a connected smooth manifold, there exists a motion planning algorithm on X , which has both order of instability and number of local rules equal to $n = \text{TC}(X)$. Moreover, this is the best one can hope for, in the sense that for any other motion planning algorithm on X , the order of instability and number of local rules have to be at least $n = \text{TC}(X)$.

While this is an interesting connection between topology and real-world applications, we will not go further in this direction. In this thesis, the focus is rather on viewing topological complexity as an invariant of topological spaces.

4.4 Connections with Lusternik-Schnirelmann Category and Dimensional Upper Bound

In this chapter, we will explore connections between topological complexity and Lusternik-Schnirelmann category, which is a well-established topological invariant. The LS-category $\text{cat } M$ of a smooth manifold M , among other things, provides lower bounds for the number of critical points for any smooth function on M (See Theorem 1.15 of [Cor+03]) and thus establishes a connection between topology and analysis on manifolds. For us, however, the Lusternik-Schnirelmann category will prove useful in order to obtain upper and lower bounds for topological complexity.

As mentioned in the introduction of chapter 1.2 in [Cor+03], the idea behind LS-category is decomposing a topological space into “simple” objects and then analyzing how they fit together to form the space, as is a common theme in mathematics.

For the LS-category $\text{cat}(X)$, the simple objects of interest are open subsets which are contractible in X and we relate them to each other by requiring that they form an

open cover of X . The following definition is adapted from [Cor+03] Definition 1.1.

Definition 4.12. Let X be a topological space. The *Lusternik-Schnirelmann category* or *LS-category* $\text{cat}(X)$ of X is the smallest integer k , such that there exists a cover

$$X = U_1 \cup \dots \cup U_k$$

by k open subsets, where each U_i is contractible in X .

If no such number k exists, set $\text{cat}(X) := \infty$.

We denote the sets U_i given above as *categorical sets* and refer to a cover given by such sets as *categorical cover*.

Note that, as before when defining sectional category, we use a definition for $\text{cat}(X)$, which is 1 bigger than the original definition from the authors of [Cor+03]. As remarked in their book, there seems to be no consensus in the literature as to which notation is better or worse, as some formulas appear nicer with one version, while others become more complicated. We use the version as given above, since it fits better to the other invariants in this thesis. Note that this version is also used by the authors of [Jam78].

We have $\text{cat}(X) = 1$ if and only if X is contractible.

The following propositions can be found in [Jam78], Proposition (1.1) and (2.3).

Proposition 4.13. *If X dominates Y , then $\text{cat}(Y) \leq \text{cat}(X)$.*

Corollary 4.14 (Homotopy Invariance). *If $X \simeq Y$, then $\text{cat}(X) = \text{cat}(Y)$.*

Proposition 4.15. *If X and Y are path-connected and paracompact, then*

$$\text{cat}(X \times Y) \leq \text{cat}(X) + \text{cat}(Y) - 1.$$

There is also a connection between the sectional category of a fibration and the LS-category of the corresponding base space. The proof can be found in [Cor+03], Proposition 9.14.

Proposition 4.16. *Let $p : E \rightarrow B$ be a fibration. Then we have $\text{secat}(p) \leq \text{cat}(B)$.*

For certain spaces, the LS-category can be estimated by the covering dimension, which is defined as follows:

Definition 4.17. Let X be a paracompact space.

- i) Let $\mathcal{U} = \{U_i\}_{i \in \mathcal{I}}$ be an open cover of X . The *order* of $\mathcal{U}$ is defined as the smallest integer k , such that there exist $k + 1$ sets of $\mathcal{U}$ with non-trivial intersection, but the intersection of any $k + 2$ sets of $\mathcal{U}$ is trivial.

- ii) The *covering dimension* $\dim(X)$ of X is the smallest integer ℓ , such that any open cover of X has a refinement of order ℓ . Set $\dim(X) := \infty$ if there is no such integer.

The proof of the next result can be found in [Cor+03] Theorem 1.7.

Proposition 4.18. *If X is path-connected, locally contractible and paracompact, then*

$$\text{cat}(X) \leq \dim(X) + 1.$$

The following proposition establishes the connection between $\text{TC}(X)$ and $\text{cat}(X)$ as well as $\text{cat}(X \times X)$ and is based upon [Far03] Theorem 5. It shows, that the topological complexity of X can be regarded as some kind of interpolation between $\text{cat}(X)$ and $\text{cat}(X \times X)$.

Proposition 4.19. *Let X be a path-connected space. Then*

$$\text{cat}(X) \leq \text{TC}(X) \leq \text{cat}(X \times X).$$

If furthermore X is path-connected and paracompact, we have

$$\text{cat}(X) \leq \text{TC}(X) \leq 2 \cdot \text{cat}(X) - 1.$$

Proof. For the first inequality, let $k := \text{TC}(X)$ be finite, otherwise the inequality is obviously true. This means there exists a cover

$$X \times X = U_1 \cup \dots \cup U_k$$

by k open sets together with local sections $s_i : U_i \rightarrow X^I$ of π for each $i = 1, \dots, k$. By proposition 3.15 i) we know that for some fixed $a_0 \in X$ and for each $i = 1, \dots, k$ we get an open set

$$V_i := g^{-1}(U_i)$$

which is contractible in X , where

$$g : X \rightarrow X \times X, \quad b \mapsto (a_0, b).$$

Furthermore we have

$$\begin{aligned} \bigcup_{i=1}^k V_i &= \bigcup_{i=1}^k g^{-1}(U_i) \\ &= g^{-1} \left(\bigcup_{i=1}^k U_i \right) \\ &= g^{-1}(X \times X) \\ &= X. \end{aligned}$$

Thus, the sets $\{V_i\}_{i=1}^k$ form a categorical open cover of X , implying $\text{cat}(X) \leq \text{TC}(X)$.

The second inequality follows from the fact that $\pi : X^I \rightarrow X \times X$ is a fibration, so $\text{TC}(X) = \text{secat}(\pi) \leq \text{cat}(X \times X)$ by proposition

4.16.

Now if additionally X is path-connected and paracompact, proposition 4.15 immediately implies

$$\text{cat}(X) \leq \text{TC}(X) \leq \text{cat}(X \times X) \leq 2 \cdot \text{cat}(X) - 1.$$

□

Remark 4.20. Note that contrary to what one could think, part ii) of proposition 3.15 does not imply $\text{TC}(X) \leq \text{cat}(X)$. To see this, take a categorical cover $\{U_i\}_{i=1}^k$ of X . Then the proposition guarantees a local section of π over each $U_i \times U_i$. But in general $\{U_i \times U_i\}_{i=1}^k$ does not form a cover of $X \times X$. Think of the case where $X = I$, $U_1 = [0, \frac{2}{3}]$ and $U_2 = (\frac{1}{3}, 1]$. Then

$$(0, 1) \notin (U_1 \times U_1) \cup (U_2 \times U_2).$$

As a direct consequence of propositions 4.19 and 4.18 we obtain the following result, which was originally stated in [Far03] Theorem 4.

Corollary 4.21. *If X is a path-connected, locally contractible and paracompact space, we have*

$$\text{TC}(X) \leq 2 \cdot \dim(X) + 1.$$

4.5 Product Inequality

We have seen in proposition 4.15 that a product inequality for the LS-category holds for the case of path-connected and paracompact spaces. In fact, the same kind of product inequality can also be shown for topological complexity.

Note that by propositions 4.15 and 4.16 we already have

$$\begin{aligned} \text{TC}(X \times Y) &= \text{secat}(\pi_{X \times Y}) \\ &\leq \text{cat}(X \times Y) \\ &\leq \text{cat}(X) + \text{cat}(Y) - 1, \end{aligned}$$

but the following product inequality gives an even better upper bound. The proof is based upon [Far03], Theorem 11, and uses similar arguments than the proof of proposition 4.15.

Proposition 4.22. *If X and Y are path-connected metric spaces, then*

$$\text{TC}(X \times Y) \leq \text{TC}(X) + \text{TC}(Y) - 1.$$

Proof. We assume $\text{TC}(X), \text{TC}(Y) < \infty$, otherwise the inequality is obviously true. Let $k := \text{TC}(X), \ell := \text{TC}(Y)$. This means there are open covers

$$X \times X = U_1 \cup \dots \cup U_k, \quad Y \times Y = V_1 \cup \dots \cup V_\ell,$$

together with continuous sections $s_i : U_i \rightarrow X^I$ of π_X for $i = 1, \dots, k$ and $r_j : V_j \rightarrow Y^I$ of π_Y for $j = 1, \dots, \ell$.

Since both spaces are metric, also $X \times X$ and $Y \times Y$ are metric spaces, hence by propositions 2.14 and 2.15, there are continuous partitions of unity $f_i : X \times X \rightarrow \mathbb{R}$ subordinate to the cover $\{U_i\}_{i=1,\dots,k}$ and $g_i : Y \times Y \rightarrow \mathbb{R}$ subordinate to $\{V_j\}_{j=1,\dots,\ell}$ respectively.

For any two nonempty subsets $S \subseteq \{1, \dots, k\}$ and $T \subseteq \{1, \dots, \ell\}$ we define

$$W(S, T) \subseteq (X \times Y) \times (X \times Y)$$

as follows: If $S \times T \neq \{1, \dots, k\} \times \{1, \dots, \ell\}$ then $(a, b, c, d) \in W(S, T)$ if and only if for any $(i, j) \in S \times T$ and for any $(i', j') \notin S \times T$ we have

$$f_i(a, c) \cdot g_j(b, d) > f_{i'}(a, c) \cdot g_{j'}(b, d).$$

If $S \times T = \{1, \dots, k\} \times \{1, \dots, \ell\}$, then $(a, b, c, d) \in W(S, T)$, if and only if

$$f_i(a, c) \cdot g_j(b, d) > 0$$

for all $(i, j) \in S \times T$. Note that

$$W(\{1, \dots, k\}, \{1, \dots, \ell\}) \subseteq \bigcap_{\substack{i \in \{1, \dots, k\} \\ j \in \{1, \dots, \ell\}}} U_i \times V_j.$$

The case $S \times T \neq \{1, \dots, k\} \times \{1, \dots, \ell\}$ can also be written in terms of preimages of continuous maps: For $(i, j) \in S \times T$ and $(i', j') \notin S \times T$ consider the continuous map

$$\begin{aligned} h_{i,j,i',j'} : (X \times Y) \times (X \times Y) &\rightarrow \mathbb{R}, \\ (a, b, c, d) &\mapsto f_i(a, c) \cdot g_j(b, d) - f_{i'}(a, c) \cdot g_{j'}(b, d). \end{aligned}$$

Then $(a, b, c, d) \in W(S, T)$ is equivalent to the condition that

$$(a, b, c, d) \in (h_{i,j,i',j'})^{-1}(\mathbb{R}_{>0})$$

for all $(i, j) \in S \times T$ and $(i', j') \notin S \times T$, or in other words

$$W(S, T) = \bigcap_{\substack{(i,j) \in S \times T \\ (i',j') \notin S \times T}} (h_{i,j,i',j'})^{-1}(\mathbb{R}_{>0}).$$

From this we can immediately deduce that each $W(S, T)$ is open as a finite intersection of open sets, since $\mathbb{R}_{>0}$ is open and the considered maps are continuous.

We will now show that the sets $W(S, T)$ form an open cover of $(X \times Y) \times (X \times Y)$ and admit local sections of $\pi_{X \times Y}$. We then use this knowledge to construct an open cover by $k + \ell - 1$ subsets together with local sections of $\pi_{X \times Y}$.

First we show that $W(S, T) \subseteq U_i \times V_j$ whenever $(i, j) \in$

$S \times T$. This then implies the existence of a local section of $\pi_{X \times Y}$ over $W(S, T)$, since for $(i, j) \in S \times T$ the map

$$\sigma_{S,T} : W(S, T) \rightarrow (X \times Y)^I, (a, b, c, d) \mapsto (s(a, c), r(b, d))$$

is well-defined. Here we implicitly used that the map $(a, b, c, d) \mapsto ((a, c), (b, d))$ is continuous.

Note that the maps f_i, g_j for some $(i, j) \in \{1, \dots, k\} \times \{1, \dots, \ell\}$ come from partitions of unity, hence they are non-negative and $f_i(a, c) \cdot g_j(b, d) > 0$ implies $(a, c) \in U_i$ and $(b, d) \in V_j$.

The inclusion is then immediate for $S \times T = \{1, \dots, k\} \times \{1, \dots, \ell\}$. For the case $S \times T \neq \{1, \dots, k\} \times \{1, \dots, \ell\}$, let $(i, j) \in S \times T$, $(a, b, c, d) \in W(S, T)$ and choose some $(i', j') \notin S \times T$. Then we have

$$f_i(a, c) \cdot g_j(b, d) > f_{i'}(a, c) \cdot g_{j'}(b, d) \geq 0,$$

implying $(a, c) \in U_i$ and $(b, d) \in V_j$.

Now we check that those sets indeed form a cover of $(X \times Y) \times (X \times Y)$. Let $(a, b, c, d) \in (X \times Y) \times (X \times Y)$. Consider

$$\begin{aligned} S &:= \{i \in \{1, \dots, k\} \mid f_i(a, c) = \max_{m=1,\dots,k} f_m(a, c)\}, \\ T &:= \{j \in \{1, \dots, \ell\} \mid g_j(b, d) = \max_{n=1,\dots,\ell} g_n(b, d)\}. \end{aligned}$$

Then $(a, b, c, d) \in W(S, T)$. Hence we have

$$(X \times Y) \times (X \times Y) \subseteq \bigcup W(S, T),$$

where the union is taken over all sets S, T with the properties described above.

Finally, we construct an open cover by $k + \ell - 1$ subsets together with local sections.

For $n = 2, \dots, k + \ell$ consider

$$W_n := \bigcup_{|S|+|T|=n} W(S, T).$$

Then

$$(X \times Y) \times (X \times Y) = W_2 \cup \dots \cup W_{k+\ell},$$

since the sets $W(S, T)$ cover $(X \times Y) \times (X \times Y)$.

Note that $W(S, T) \cap W(S', T') = \emptyset$ whenever $S \times T \not\subseteq S' \times T'$ and $S' \times T' \not\subseteq S \times T$. To see this, assume that the condition holds and let $(a, b, c, d) \in W(S, T) \cap W(S', T')$. Then there exists $(i, j) \in (S \times T) \setminus (S' \times T')$ and $(i', j') \in (S' \times T') \setminus (S \times T)$ for which we obtain

$$f_i(a, c) \cdot g_j(b, d) > f_{i'}(a, c) \cdot g_{j'}(b, d) > f_i(a, c) \cdot g_j(b, d),$$

a contradiction.

Now assume that $|S| + |T| = |S'| + |T'| = n$. Then either

$S = S'$ and $T = T'$ which implies $W(S, T) = W(S', T')$ or we have $S \times T \not\subseteq S' \times T'$ and $S' \times T' \not\subseteq S \times T$. In the latter case it follows that $W(S, T) \cap W(S', T') = \emptyset$ by the discussion above.

Thus, each W_n consists of either a single set $W(S, T)$ or is actually a union of disjoint sets

$$W_n = \bigsqcup_{|S|+|T|=n} W(S, T).$$

In the first case, a local section over W_n is given by $\sigma_{S, T}$ as defined above and in the second case we can just take the disjoint union of maps

$$\bigsqcup_{|S|+|T|=n} \sigma_{S, T} : W_n \rightarrow (X \times Y)^I,$$

which is continuous.

This completes the proof that

$$\text{TC}(X \times Y) \leq k + \ell - 1 = \text{TC}(X) + \text{TC}(Y) - 1.$$

□

Corollary 4.23. *If X is path-connected and metric, then*

$$\text{TC}(X^n) \leq n \cdot (\text{TC}(X) - 1) + 1.$$

4.6 Cohomological Lower Bounds

In this chapter, we consider a lower bound for $\text{TC}(X)$ given by the structure of the cohomology of X . Before we do this, let us first review some definitions and results about the singular cohomology of a space.

Throughout this chapter, let $\mathbb{F}$ be a field. Furthermore, we denote by $C_n(X)$ the singular n -chains of X with coefficients in $\mathbb{Z}$, by $C_n(X; \mathbb{F})$ the singular n -chains with coefficients in $\mathbb{F}$, and by $C^n(X; \mathbb{F}) = \text{hom}(C_n(X); \mathbb{F})$ the singular n -cochains with coefficients in $\mathbb{F}$. Likewise, denote the n -th singular homology group by $H_n(X)$, the n -th singular homology group with coefficients in $\mathbb{F}$ by $H_n(X; \mathbb{F})$, the n -th singular cohomology group with coefficients in $\mathbb{F}$ by $H^n(X; \mathbb{F})$, and likewise the n -th reduced cohomology group as $\tilde{H}^n(X; \mathbb{F})$.

The definitions and results in this chapter are taken from [Hat02], unless otherwise noted. It should be noted that some of those results also hold for the case where we take coefficients in a ring R rather than a field $\mathbb{F}$.

Proposition 4.24. *Let X be a topological space and $\mathbb{F}$ be a field. Then $H^n(X; \mathbb{F})$ is the dual vector space of $H_n(X; \mathbb{F})$, i.e.*

$$H^n(X; \mathbb{F}) \cong \text{hom}_{\mathbb{F}}(H_n(X; \mathbb{F}), \mathbb{F}).$$

Furthermore, if $H_n(X; \mathbb{F})$ is finitely generated, then $H^n(X; \mathbb{F}) \cong H_n(X; \mathbb{F})$.

Definition 4.25. The *cup product* is the bilinear map given by

$$\smile : C^k(X; \mathbb{F}) \times C^\ell(X; \mathbb{F}) \rightarrow C^{k+\ell}(X; \mathbb{F}), (\varphi, \psi) \mapsto \varphi \smile \psi$$

where $\varphi \smile \psi$ is the cochain whose value on a singular simplex $\sigma : \Delta^{k+\ell} \rightarrow X$ is given by

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma|_{[v_0, \dots, v_k]})\psi(\sigma|_{[v_k, \dots, v_{k+\ell}]}).$$

Proposition 4.26. *For $\varphi \in C^k(X; \mathbb{F})$ and $\psi \in C^\ell(X; \mathbb{F})$ we have*

$$\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^k \varphi \smile \delta\psi.$$

Thus there is an induced cup product

$$\smile : H^k(X; \mathbb{F}) \times H^\ell(X; \mathbb{F}) \rightarrow H^{k+\ell}(X; \mathbb{F}).$$

Proposition 4.27. *For a continuous map $f : X \rightarrow Y$, the induced maps $f^* : H^n(Y; \mathbb{F}) \rightarrow H^n(X; \mathbb{F})$ satisfy*

$$f^*(\varphi \smile \psi) = f^*(\varphi) \smile f^*(\psi).$$

Proposition 4.28. *For $\varphi \in H^k(X; \mathbb{F})$ and $\psi \in H^\ell(X; \mathbb{F})$ we have*

$$\varphi \smile \psi = (-1)^{k\ell} \psi \smile \varphi.$$

Definition 4.29. For some space X , define $H^*(X; \mathbb{F}) := \bigoplus_{i \in \mathbb{N}} H^i(X; \mathbb{F})$, i.e. elements of $H^*(X; \mathbb{F})$ are finite sums $\sum_i \varphi_i$ with $\varphi_i \in H^i(X; \mathbb{F})$. This has the structure of a graded $\mathbb{F}$ -algebra with (graded-commutative) multiplication given by the cup product, i.e.

$$\left(\sum_i \varphi_i \right) \cdot \left(\sum_j \psi_j \right) = \sum_{i,j} \varphi_i \smile \psi_j.$$

Furthermore, we write $|\varphi| = i$ if $\varphi \in H^i(X; \mathbb{F})$.

Recall that for the reduced cohomology groups we have $\tilde{H}^n(X; \mathbb{F}) = H^n(X; \mathbb{F})$ for all $n \geq 1$, since they are obtained by dualizing the augmented singular chain complex of X . In dimension 0, we have that $\tilde{H}^0(X; \mathbb{F})$ consists of all maps $X \rightarrow \mathbb{F}$ that are constant on path-components modulo the maps that are constant on all of X , as discussed in [Hat02], p. 199. Thus, if X is path-connected, we have $\tilde{H}^0(X; \mathbb{F}) \cong 0$.

Definition 4.30. The *cup-length* $\text{cup}(X; \mathbb{F})$ of X with coefficients in $\mathbb{F}$ is defined as the smallest integer k , such that we have

$$\varphi_1 \smile \dots \smile \varphi_{k+1} = 0 \in \tilde{H}^*(X; \mathbb{F})$$

for all $\varphi_i \in \tilde{H}^*(X; \mathbb{F})$.

Set $\text{cup}(X; \mathbb{F}) := \infty$ if no such k exists.

We have the following connection between LS-category and cup-length, which can be found in [Cor+03], Proposition 1.5. As remarked before, our definition of LS-category is 1 bigger than in the referenced book. Keep this in mind, when comparing the statement here with the result from the book.

Proposition 4.31. *Let X be a topological space, then $\text{cup}(X; \mathbb{F}) < \text{cat}(X)$.*

Together with proposition 4.19 this immediately establishes a connection between topological complexity of a space X and its cohomology.

Corollary 4.32. *For any space X we have $\text{TC}(X) > \text{cup}(X; \mathbb{F})$.*

Consider the tensor product of $\mathbb{F}$ -algebras $H^*(X; \mathbb{F}) \otimes H^*(X; \mathbb{F})$, which is also a $\mathbb{F}$ -algebra with multiplication given by

$$(\varphi \otimes \psi) \cdot (\varphi' \otimes \psi') = (-1)^{|\psi| \cdot |\varphi'|} (\varphi \smile \varphi') \otimes (\psi \smile \psi').$$

The cup product induces an algebra homomorphism

$$\smile: H^*(X; \mathbb{F}) \otimes H^*(X; \mathbb{F}) \rightarrow H^*(X; \mathbb{F}).$$

The following definition is due to [Far03].

Definition 4.33. The kernel of the cup product

$$\smile: H^*(X; \mathbb{F}) \otimes H^*(X; \mathbb{F}) \rightarrow H^*(X; \mathbb{F})$$

is called the *ideal of zero-divisors of $H^*(X; \mathbb{F})$* . The *zero-divisors-cup-length $\text{zcl}(X; \mathbb{F})$ of $H^*(X; \mathbb{F})$* is the length of the longest nontrivial product in the ideals of zero-divisors of $H^*(X; \mathbb{F})$.

We now arrive at the main proposition of this chapter. Unfortunately, a proof of this result is outside the scope of this thesis, rather it can be found in [Far17], Theorem 3.3.

Proposition 4.34. *Given a space X , we have*

$$\text{TC}(X) > \text{zcl}(X; \mathbb{F}).$$

In the upcoming section, we will see how this lower bound can aid in computing the topological complexity of some spaces.

4.7 Topological Complexity of Certain Spaces

As noted before, the results of the previous chapters are useful in order to compute the topological complexity of some spaces.

First we have a look at complexity of graphs, which can be regarded as 1-dimensional CW-complexes. Recall that a CW-complex is called finite, if it has only finitely many cells and is said to be n -dimensional, if there are no cells with dimension greater than n .

For computing the complexity of a graph, we need some knowledge about Betti numbers.

Definition 4.35. If $H_k(X)$ is finitely generated, then the k -th Betti number $b_k(X)$ of X is defined as the rank of the abelian group $H_k(X)$.

The following definitions and results can be found in [Bre93], chapter III, 7.11. Definition, 7.13. Lemma and 7.14. Theorem.

Definition 4.36. For a finite connected graph G , its *Euler characteristic* $\chi(G)$ is defined as $\chi(G) := V - E$, where V denotes the number of vertices and E denotes the number of edges of G .

Proposition 4.37. *If G is a finite connected graph, then G is homotopy-equivalent to the one-point union of n circles, where $n = 1 - \chi(G)$.*

Proposition 4.38. *If G is a finite connected graph, then $\pi_1(G)$ is a free group on $1 - \chi(G)$ generators.*

Corollary 4.39. *Let G be a finite connected graph. Then*

$$b_k(G) = \begin{cases} 1, & k = 0, \\ E - V + 1, & k = 1, \\ 0, & k > 1, \end{cases}$$

where V denotes the number of vertices and E denotes the number of edges of G . In particular we have $\chi(G) = 1 - b_1(G)$.

Proof. Since G is connected, we have $H_0(G) \cong \mathbb{Z}$. By proposition 4.38, we have that $\pi_1(G)$ is the free group on $1 - \chi(G) = E - V + 1$ generators. Furthermore, by the Hurewicz isomorphism, we know that $H_1(G)$ is isomorphic to the abelianization of $\pi_1(G)$, which is $\mathbb{Z}^{E-V+1}$. Lastly, because G is a finite 1-dimensional CW-complex, we have $H_k(G) \cong 0$ for $k > 1$. $\square$

Now we are ready to state a result which follows from corollary 4.21 and proposition 4.34. The proof is based upon [Far04], Theorem 7.3.

Proposition 4.40. *If G is a connected finite graph, then*

$$\text{TC}(G) = \min(b_1(G), 2) + 1 = \begin{cases} 1, & b_1(G) = 0, \\ 2, & b_1(G) = 1, \\ 3, & b_1(G) > 1, \end{cases}$$

where $b_1(G)$ is the first Betti number of G .

Proof. If $b_1(G) = 0$, then by proposition 4.37, we know that G is homotopy-equivalent to the wedge of $1 - \chi(G) = 1 - (1 - b_1(G)) = 0$ circles, i.e. G is contractible. In this case we have $\text{TC}(G) = 1$.

If $b_1(G) = 1$, then $G \simeq \mathbb{S}^1$ by the same proposition. Since topological complexity is a homotopy invariant, this implies $\text{TC}(G) = \text{TC}(\mathbb{S}^1) = 2$, see example 4.5.

Lastly, if $b_1(G) > 1$, then

$$G \simeq \bigvee_{i=1}^n \mathbb{S}^1,$$

where $n = 1 - \chi(G) = b_1(G) > 1$. Using the Mayer-Vietoris sequence for pushouts, one can show that

$$\tilde{H}^k(G) = \tilde{H}^k\left(\bigvee_{i=1}^n \mathbb{S}^1; \mathbb{R}\right) \cong \bigoplus_{i=1}^n \tilde{H}^k(\mathbb{S}^1; \mathbb{R})$$

for all $k \in \mathbb{N}$. For $k > 0$, this also holds for unreduced cohomology groups, since reduced and unreduced groups only differ in dimension 0. In particular, by using

$$\tilde{H}^k(\mathbb{S}^1; \mathbb{R}) \cong \begin{cases} \mathbb{R}, & k = 1, \\ 0, & k \neq 1, \end{cases}$$

we obtain

$$H^1(G; \mathbb{R}) \cong \mathbb{R}^n \quad \text{and} \quad H^2(G; \mathbb{R}) \cong 0.$$

Since $n > 1$, there are two linearly independent cohomology classes $u_1, u_2 \in H^1(G; \mathbb{R})$, which correspond to basis elements of $\mathbb{R}^n$. Consider for $i = 1, 2$

$$\bar{u}_i := 1 \otimes u_i - u_i \otimes 1 \in H^*(G; \mathbb{R}) \otimes H^*(G; \mathbb{R}),$$

with $1 \in H^0(G; \mathbb{R})$. We have

$$\begin{aligned} \smile(\bar{u}_i) &= \smile(1 \otimes u_i - u_i \otimes 1) \\ &= 1 \smile u_i - u_i \smile 1 \\ &= 1 \smile u_i - 1 \smile u_i \\ &= 0, \end{aligned}$$

where we made use of proposition 4.28. Note that $u_1 \smile u_2 = 0$, since $H^2(G; \mathbb{R}) \cong 0$. Hence we obtain

$$\begin{aligned} &(1 \otimes u_1 - u_1 \otimes 1) \cdot (1 \otimes u_2 - u_2 \otimes 1) \\ &= (1 \otimes u_1) \cdot (1 \otimes u_2) - (u_1 \otimes 1) \cdot (1 \otimes u_2) \\ &\quad - (1 \otimes u_1) \cdot (u_2 \otimes 1) + (u_1 \otimes 1) \cdot (u_2 \otimes 1) \\ &= (1 \smile 1) \otimes (u_1 \smile u_2) - (u_1 \smile 1) \otimes (1 \smile u_2) \\ &\quad + (1 \smile u_2) \otimes (u_1 \smile 1) + (u_1 \smile u_2) \otimes (1 \smile 1) \\ &= 1 \otimes (u_1 \smile u_2) - u_1 \otimes u_2 + u_2 \otimes u_1 + (u_1 \smile u_2) \otimes 1 \\ &= u_2 \otimes u_1 - u_1 \otimes u_2 \\ &\neq 0, \end{aligned}$$

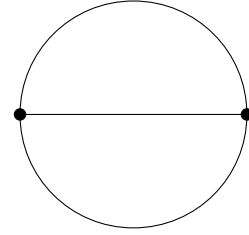
since $u_1 \otimes u_2$ and $u_2 \otimes u_1$ are basis elements of $H^1(G; \mathbb{R}) \otimes H^1(G; \mathbb{R}) \cong \mathbb{R}^n \otimes \mathbb{R}^n$. This implies that we have two elements in the ideal of zero-divisors of $H^*(G; \mathbb{R})$ such that their product is nontrivial, i.e. $\text{zcl}(G; \mathbb{R}) \geq 2$.

Proposition 4.34 then yields $\text{TC}(G) \geq 3$. Additionally, by corollary 4.21 we know

$$\text{TC}(G) \leq 2 \cdot \dim(G) + 1 = 2 \cdot 1 + 1 = 3$$

and thus $\text{TC}(G) = 3$. $\square$

Example 4.41. Consider the following graph G :



We have $b_1(G) = 2$, hence $\text{TC}(G) = 3$ by proposition 4.40.

In the introduction to this thesis, an example for a configuration space was that of a planar robot arm with n joints, which led us to considering the space $(\mathbb{S}^1)^n$. Similarly, the configuration space of a spatial robot arm with n joints can be considered as $(\mathbb{S}^2)^n$. The complexity of those spaces can be deduced from a more general result about the complexity of $\text{TC}((\mathbb{S}^m)^n)$, see proposition 4.43. Before we get to that, we need a result about the complexity of spheres. The proof is based upon [Far08], Proposition 4.41.

Proposition 4.42. *We have*

$$\text{TC}(\mathbb{S}^m) = \begin{cases} 2, & m \text{ odd}, \\ 3, & m \text{ even}. \end{cases}$$

Proof. First note that $H^m(\mathbb{S}^m; \mathbb{R})$ is the dual space of $H_m(\mathbb{S}^m; \mathbb{R}) \cong \mathbb{R}$ by proposition 4.24. Hence there is some element $u \in H^m(\mathbb{S}^m; \mathbb{R})$ with $u(1) = 1$. Note that u is then a generator of $H^m(\mathbb{S}^m; \mathbb{R})$. Consider

$$\bar{u} := 1 \otimes u - u \otimes 1 \in H^m(\mathbb{S}^m; \mathbb{R}) \otimes H^m(\mathbb{S}^m; \mathbb{R}),$$

where $1 \in H^0(\mathbb{S}^m; \mathbb{R})$. Then for any m -simplex σ we have

$$\smile(\bar{u})(\sigma) = (1 \smile u - u \smile 1)(\sigma) = u(\sigma) - u(\sigma) = 0.$$

Hence $\bar{u}$ is an element of $\ker(\smile)$, implying $\text{zcl}(\mathbb{S}^m; \mathbb{R}) \geq 1$.

For the case of m even, consider

$$\begin{aligned} \bar{u}^2 &= (1 \otimes u)^2 - (1 \otimes u)(u \otimes 1) - (u \otimes 1)(1 \otimes u) + (u \otimes 1)^2 \\ &= -(-1)^{m^2} (1 \smile u) \otimes (u \smile 1) - (u \smile 1) \otimes (1 \smile u) \\ &= ((-1)^{m^2+1} - 1)u \otimes u, \end{aligned}$$

where we used that $u \smile u \in H^{2m}(\mathbb{S}^m; \mathbb{R}) \cong 0$ and $u \smile 1 = u = 1 \smile u$. Since u is a generator of $H^m(\mathbb{S}^m; \mathbb{R})$, we know that $u \otimes u$ is a generator of $H^m(\mathbb{S}^m; \mathbb{R}) \otimes H^m(\mathbb{S}^m; \mathbb{R})$, so $u \otimes u \neq 0$. Now if m is even, then also m^2 is even and thus this simplifies to

$$\bar{u}^2 = -2u \otimes u \neq 0.$$

But also $\bar{u}^2 \in \ker(\smile)$, since $u \smile u = 0$. Hence for m even we obtain $\text{zcl}(\mathbb{S}^m; \mathbb{R}) \geq 2$.

Using proposition 4.34 we obtain

$$\text{TC}(\mathbb{S}^m) \geq \begin{cases} 2, & m \text{ odd}, \\ 3, & m \text{ even}. \end{cases}$$

The reverse inequality can be found in [Far08], Example 4.8, where a cover of $\mathbb{S}^m$ in the sense of proposition 4.11 iii) is constructed. This cover consists of 3 sets for m even, whereas for m odd, a cover consisting of 2 sets can be constructed with the help of a non-vanishing vector field. $\square$

With this information, the product inequality 4.23 implies

$$\text{TC}((\mathbb{S}^m)^n) \leq n \cdot (\text{TC}(\mathbb{S}^m) - 1) + 1 = \begin{cases} n + 1, & m \text{ odd}, \\ 2n + 1, & m \text{ even}. \end{cases}$$

The following observation is due to [Far03] Theorem 13. Similar to the calculation in proposition 4.42, one can find a cohomology class $u \in H^m(\mathbb{S}^m, \mathbb{R})$ with $u(1) = 1$ and then define $a_i := p_i^*(u) \in H^m((\mathbb{S}^m)^n; \mathbb{R})$ for $i = 1, \dots, n$, where $p_i : (\mathbb{S}^m)^n \rightarrow \mathbb{S}^m$ is the projection onto the i -th factor. By defining

$$\bar{a}_i := 1 \otimes a_i - a_i \otimes 1,$$

one then obtains

$$\bar{a}_1 \cdot \dots \cdot \bar{a}_n \neq 0$$

and additionally

$$\bar{a}_1^2 \cdot \dots \cdot \bar{a}_n^2 \neq 0$$

for m even. Proposition 4.34 then implies

$$\text{TC}((\mathbb{S}^m)^n) \geq \begin{cases} n + 1, & m \text{ odd}, \\ 2n + 1, & m \text{ even}. \end{cases}$$

This leads to the following result.

Proposition 4.43. *We have*

$$\text{TC}((\mathbb{S}^m)^n) = \begin{cases} n + 1, & m \text{ odd}, \\ 2n + 1, & m \text{ even}. \end{cases}$$

Returning to our introductory examples regarding motion planning of planar or spatial robot arms

with n joints, it follows that $\text{TC}((\mathbb{S}^1)^n) = n + 1$ and $\text{TC}((\mathbb{S}^2)^n) = 2n + 1$.

Similar techniques can be used to compute topological complexity for more types of spaces. One can show that

$$\text{TC}(\Sigma_g) = \begin{cases} 3, & g \leq 1 \\ 5, & g > 1 \end{cases}$$

for a compact orientable surface of genus g . The computation for $g \leq 1$ follows immediately from $\Sigma_0 \cong \mathbb{S}^2$ and $\Sigma_1 \cong (\mathbb{S}^1)^2$. For $g > 1$, one can again find suitable cohomology classes and then use proposition 4.34 in conjunction with corollary 4.21. See [Far03], Theorem 9 for details.

Furthermore, using the cohomological lower bound, it can be shown that $\text{TC}(\mathbb{CP}^n) \geq 2n + 1$, see [Far03], Theorem 10.

5 Directed Topological Complexity

After having established the notion of topological complexity and reviewing some of its properties, we transition to the setting of directed spaces and directed topological complexity. The definitions, results and examples in this chapter are mainly based on [GFS20].

5.1 Directed Algebraic Topology

Definition 5.1. A *directed topological space*, or *d-space* for short, is a pair $\vec{X} := (X, PX)$ consisting of a topological space X together with a set PX of continuous maps $\gamma : I \rightarrow X$, called *directed paths* or *d-paths*, with the following properties:

- i) Every constant map $I \rightarrow X$ is directed.
- ii) PX is closed under composition with continuous non-decreasing maps from $I \rightarrow I$.
- iii) PX is closed under concatenation.

For $(a, b) \in X \times X$ we define the subspace

$$PX(a, b) := \{\gamma \in PX \mid \gamma(0) = a, \gamma(1) = b\} \subseteq PX,$$

i.e. $PX(a, b)$ consists of all directed paths which start at a and end at b .

Recall that we considered X^I equipped with the compact-open topology. Since the space of directed paths PX is a subset of X^I , it inherits a topology induced by the compact-open topology on X^I .

Remark 5.2. i) A product $X \times Y$ of two d-spaces $\vec{X} = (X, PX)$ and $\vec{Y} = (Y, PY)$ has a natural d-space structure: A path $\gamma : I \rightarrow X \times Y$ consists of two components $\gamma_X : I \rightarrow X, \gamma_Y : I \rightarrow Y$ such that $\gamma = (\gamma_X, \gamma_Y)$. Then we define the d-spaces structure on $X \times Y$ by saying $P(X \times Y)$ consists of all those paths $\gamma = (\gamma_X, \gamma_Y)$ where $\gamma_X \in PX$ and $\gamma_Y \in PY$. We denote this as

$$\vec{X} \times \vec{Y} := (X \times Y, P(X \times Y)).$$

- ii) Given a d-space $\vec{X} = (X, PX)$ and an equivalence relation $\sim$ on X , there is an induced d-paths structure on $X/\sim$ given by

$$PX/\sim := \{p \circ \gamma \mid \gamma \in PX\},$$

where p is the projection $X \rightarrow X/\sim$. We denote this as

$$\vec{X}/\sim := (X/\sim, PX/\sim).$$

- iii) The disjoint union $X \sqcup Y$ of two directed spaces (X, PX) and (Y, PY) also comes with an obvious d-structure, since a continuous path in $X \sqcup Y$ lies either in X or Y . Hence $P(X \sqcup Y) = PX \sqcup PY$.

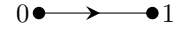
The following examples are based on [GFS20], Examples 1, 2 and 4.

Example 5.3. i) The only possible d-space structure for a one-point space $\{*\}$ consists of the single constant map $I \rightarrow \{*\}$

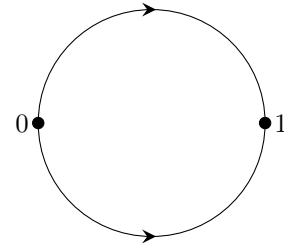
- ii) The standard d-space structure on the unit interval I is given by

$$PI = \{\gamma \in I^I \mid \gamma \text{ non-decreasing}\}.$$

We denote the *standard directed unit interval* as $\vec{I} := (I, PI)$.



- iii) Another example of a d-space is given by the directed circle $\vec{\mathbb{S}}^1$.



We construct this d-space as follows: First consider two copies of the directed unit interval, i.e. $I_+ = \vec{I} \times \{1\}$ and $I_- = \vec{I} \times \{0\}$. Then we glue them together via the equivalence relation $\sim$ induced by $(0, 0) \sim (0, 1)$ and $(1, 0) \sim (1, 1)$, i.e. we define

$$\vec{\mathbb{S}}^1 := (I_+ \sqcup I_-) / \sim.$$

Further, define $0, 1 \in \vec{\mathbb{S}}^1$ as the equivalence classes

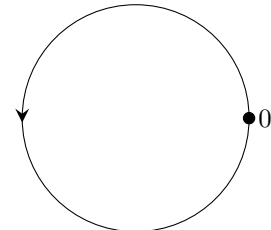
$$0 := [(0, 0)]_\sim, \quad 1 := [(1, 0)]_\sim.$$

The d-paths $P\vec{\mathbb{S}}^1$ then consist of the paths coming from I_+ and those coming from I_- which we denote as PX_+ and PX_- respectively, so

$$P\vec{\mathbb{S}}^1 = PX_+ \cup PX_-.$$

We also have that $PX_+ \cap PX_-$ consists only of the two constant paths at 0 or 1 respectively.

- iv) A further example is the directed loop $\vec{\mathbb{O}}^1$.



Consider

$$X = \{z \in \mathbb{C} \mid |z| = 1\} \cong \mathbb{S}^1.$$

Any path $\gamma : I \rightarrow X$ is given by

$$\gamma(t) = e^{i\varphi(t)}$$

for some continuous function $\varphi : I \rightarrow \mathbb{R}$. We then define the d-paths PX to be consisting of only those paths where φ is non-decreasing. We denote the directed loop as $\tilde{\mathbb{O}}^1 := (X, PX)$.

Definition 5.4. Let $(X, PX), (Y, PY)$ be d-spaces. A map $f : X \rightarrow Y$ is called a *d-map*, if it is continuous and for any directed path $\gamma \in PX$, we have $f \circ \gamma \in PY$.

Thus, a d-map $f : X \rightarrow Y$ induces a map

$$Pf : PX \rightarrow PY, \quad \gamma \mapsto f \circ \gamma.$$

This map is indeed continuous: By corollary 3.6 we have that $Pf : PX \rightarrow PY \subseteq Y^I$ is continuous, if and only if the map

$$F : I \times PX \rightarrow Y, \quad (t, \gamma) \mapsto Pf(\gamma)(t)$$

is continuous. Consider the evaluation map

$$e : I \times X^I \rightarrow X, \quad e(t, \gamma) = \gamma(t),$$

which is continuous by the same corollary. We have

$$Pf(\gamma)(t) = f(\gamma(t)) = f(e(t, \gamma)),$$

so F can be written as a composition of continuous maps

$$\begin{array}{ccc} I \times PX & \xrightarrow{e} & X \xrightarrow{f} Y \\ & \searrow F & \nearrow \\ & & \end{array}$$

and is therefore continuous.

Furthermore for some continuous maps $f : X \rightarrow Y, g : Y \rightarrow Z$ and any path $\gamma \in PX$ we have

$$P(\text{id}_X)(\gamma) = \text{id}_X \circ \gamma = \gamma = \text{id}_{PX}(\gamma)$$

and

$$P(g \circ f)(\gamma) = g \circ f \circ \gamma = Pg(f \circ \gamma) = Pg(Pf(\gamma)),$$

i.e. $P(\text{id}_X) = \text{id}_{PX}$ and $P(g \circ f) = Pg \circ Pf$.

Remark 5.5. Summarizing these results, we obtain a functor P from the category DIR of d-spaces together with d-maps to the category TOP of topological spaces with continuous maps, that assigns

$$\begin{aligned} (X, PX) &\mapsto PX, \\ (f : (X, PX) \rightarrow (Y, PY)) &\mapsto (Pf : PX \rightarrow PY). \end{aligned}$$

5.2 The Concept of Directed Topological Complexity

After having established the notion of directed spaces, one can consider the setting of directed motion planning, i.e. studying maps of the form $X \times X \rightarrow PX$ for some d-space (X, PX) . This can be regarded as a special case of motion planning, where some constraints only allow for paths following a certain direction which is prescribed by the d-structure PX .

In the previous chapter we have seen some examples of d-spaces, and while some of them may resemble some directed motion planning problem occurring in practice, they are mainly constructed for the purpose of introducing the reader to directed topology. However, it should be noted that there are indeed naturally arising d-spaces. Some examples are given in [GFS20] and include d-structures appearing in the context of control theory or concurrency and distributed systems theory.

In order to define the notion of directed topological complexity, we use a variation of the path space fibration which was used to define (undirected) topological complexity.

Definition 5.6. Let $\vec{X} = (X, PX)$ be a d-space. We define the *d-paths map* of X as

$$\vec{\pi}_X : PX \rightarrow X \times X, \quad \gamma \mapsto (\gamma(0), \gamma(1)).$$

We sometimes omit the subscript $\vec{X}$ if it doesn't cause any confusion.

One should note that the d-paths map of some space X is just the path fibration restricted to the directed paths of X , i.e. $\vec{\pi}_X = \pi_X|_{PX}$. In particular we have $\vec{\pi}_X = \pi_X$ if $PX = X^I$. However, a consequence of this restriction is that $\vec{\pi}$ is in general not a fibration. This is discussed later in more detail, see example 5.12.

Contrary to the undirected setting where $\pi(X^I) = X \times X$ if X is path-connected, for a given pair $(a, b) \in X \times X$ there does not necessarily exist a directed path starting at a and ending at b . For example consider the directed unit interval $\vec{I}$ from example 5.3. There is no directed path going from 1 to 0.

Given a d-space $\vec{X} = (X, PX)$, we denote the image of the d-paths map as

$$\Gamma_X := \Gamma_{\vec{X}} := \vec{\pi}(PX) \subseteq X \times X.$$

This is the set of all pairs $(a, b) \in X \times X$ where there exists a directed path in X starting at a and ending at b . This leads us to the following definition:

Definition 5.7. A d-space $\vec{X} = (X, PX)$ is called *strongly-connected*, if $\Gamma_{\vec{X}} = X \times X$.

Note that a strongly-connected space is always path-connected.

As before, the following examples are based upon [GFS20], Examples 1,2 and 4.

Example 5.8. i) We have $\Gamma_{\tilde{I}} = \{(a, b) \in I^2 \mid b \geq a\}$. Note that $\Gamma_{\tilde{I}}$ is homeomorphic to the 2-disk, hence contractible and path-connected. Furthermore, for every $(x, x') \in \Gamma_{\tilde{I}}$, we have that

$$(\tilde{\pi})^{-1}(x, x') = PI(x, x')$$

is contractible and thus also path-connected. To see this, consider

$$h : I \times PI(x, x') \rightarrow PI(x, x') \quad (t, \gamma) \mapsto h_t(\gamma)$$

defined by

$$h_t(\gamma)(\tau) := (1 - t)\gamma(\tau) + t \cdot \gamma_{x, x'}(\tau),$$

where $\gamma_{x, x'} \in PI$ denotes the linear interpolation

$$\gamma_{x, x'} : \tau \mapsto (1 - \tau)x + \tau x'$$

between x and x' . Denote by $e : PI(x, x') \times I \rightarrow I$ the evaluation map, which is continuous by corollary 3.6. We have

$$h_t(\gamma)(\tau) = (1 - t)e(\gamma, \tau) + t \cdot e(\gamma_{x, x'}, \tau),$$

which implies continuity of h by the same corollary. Furthermore, we have $h_0 = \text{id}_{PI(x, x')}$ and h_1 is the constant map $\gamma \mapsto \gamma_{x, x'}$.

ii) Consider the directed circle $\tilde{\mathbb{S}}^1$. Here we have

$$\Gamma_{\tilde{\mathbb{S}}^1} = \tilde{\pi}(PX_+) \cup \tilde{\pi}(PX_-)$$

and

$$\tilde{\pi}(PX_+) \cap \tilde{\pi}(PX_-) = \{(0, 0), (0, 1), (1, 1)\}.$$

Note that $\tilde{\pi}(PX_+)$ and $\tilde{\pi}(PX_-)$ can each be identified with $\Gamma_{\tilde{I}}$. Thus, $\Gamma_{\tilde{\mathbb{S}}^1}$ consists of two overlapping path-connected sets, so it is path-connected itself.

iii) Consider the directed loop $\tilde{\mathbb{O}}^1$. In this case, for any pair of points there is a directed path between them, hence

$$\Gamma_{\tilde{\mathbb{O}}^1} = \mathbb{S}^1 \times \mathbb{S}^1,$$

i.e. $\tilde{\mathbb{O}}^1$ is strongly-connected.

Similar to before with Pf , a d-map $f : X \rightarrow Y$ induces a continuous map

$$\Gamma f : \Gamma_X \rightarrow \Gamma_Y, \quad (a, b) \mapsto (f(a), f(b)).$$

The image of Γf indeed lies in Γ_Y , since for some path $\gamma \in PX(a, b)$ the path $Pf(\gamma)$ is an element of $PY(f(a), f(b))$.

Note that $\Gamma(\text{id}_X) = \text{id}_{\Gamma_X}$ and for some d-maps $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ we have

$$\Gamma(g \circ f) = \Gamma g \circ \Gamma f.$$

Remark 5.9. As before with remark 5.5, this means we obtain a functor

$$\begin{aligned} \Gamma : \text{DIR} &\rightarrow \text{TOP}, \\ (X, PX) &\mapsto \Gamma_X, \\ (f : (X, PX) \rightarrow (Y, PY)) &\mapsto (\Gamma f : \Gamma_X \rightarrow \Gamma_Y). \end{aligned}$$

Let $f : X \rightarrow Y$ be a d-map. For some directed path $\gamma \in PX$ we have

$$\begin{aligned} \tilde{\pi}_Y(Pf(\gamma)) &= \tilde{\pi}_Y(f \circ \gamma) \\ &= (f(\gamma(0)), f(\gamma(1))) \\ &= \Gamma f(\gamma(0), \gamma(1)) \\ &= \Gamma f(\tilde{\pi}_X(\gamma)), \end{aligned}$$

that is, the following diagram commutes:

$$\begin{array}{ccc} PX & \xrightarrow{Pf} & PY \\ \tilde{\pi}_X \downarrow & & \downarrow \tilde{\pi}_Y \\ \Gamma_X & \xrightarrow{\Gamma f} & \Gamma_Y \end{array}$$

Remark 5.10. Together with remarks 5.5 and 5.9 we obtain that $\tilde{\pi}$ resembles a natural transformation between the functors P and Γ .

We now arrive at the main concept of this chapter, which was originally introduced in [Gou17].

Definition 5.11. Let (X, PX) be a d-space where X is an ENR. The *directed topological complexity* $\overrightarrow{\text{TC}}(X, PX)$ is the minimum number k , such that there exists a (possibly discontinuous) map $s : \Gamma_X \rightarrow PX$ and a partition of Γ_X into k subsets

$$\Gamma_X = U_1 \cup \dots \cup U_k, \quad U_i \cap F_j = \emptyset, i \neq j,$$

such that:

- i) Each U_i is an ENR.
- ii) s is a section of $\tilde{\pi}_{\tilde{X}} : PX \rightarrow X$, i.e. $\tilde{\pi}_{\tilde{X}} \circ s = \text{id}_{\Gamma_X}$.
- iii) $s|_{U_i} : U_i \rightarrow PX$ is continuous for all $i = 1, \dots, k$.

If no such number k exists, set $\overrightarrow{\text{TC}}(X, PX) := \infty$.

We denote the sets U_i given above as *domains of continuity* for $\tilde{\pi}_{\tilde{X}}$ and refer to the maps $s|_{U_i}$ as *local sections of $\tilde{\pi}_{\tilde{X}}$* .

Again, the ideas used in the following examples are based on [GFS20], Example 1 and 2.

Example 5.12. i) Consider the directed unit interval $\vec{I}$. The map $s : \Gamma_I \rightarrow PI$ obtained by defining

$$s(a, b)(t) := (1 - t)a + tb,$$

for $(a, b) \in \Gamma_I$ and $t \in I$, is a global continuous section of $\vec{\pi}$, since $\vec{\pi}(s(a, b)) = (a, b)$ and continuity follows directly from corollary 3.6. Thus we have a single domain of continuity, implying $\overrightarrow{TC}(\vec{I}) = 1$.

ii) Consider the directed circle $\vec{\mathbb{S}}^1$. First we will show that there is no global section of $\vec{\pi} : P\vec{\mathbb{S}}^1 \rightarrow \Gamma_{\vec{\mathbb{S}}^1}$. We will do this by showing that $\vec{\pi}$ admits no continuous section over any neighborhood of $(0, 1) \in \Gamma_{\vec{\mathbb{S}}^1}$. Let U be a neighborhood of $(0, 1) \in \Gamma_{\vec{\mathbb{S}}^1}$. As discussed in example 5.3, we have

$$P\vec{\mathbb{S}}^1 = PX_+ \cup PX_- \quad \text{and} \quad PX_+ \cap PX_- = \{\gamma_0, \gamma_1\},$$

where γ_0 and γ_1 denote the constant paths at 0 or 1 respectively. This means that the preimage $\vec{\pi}^{-1}(0, 1) = PI(0, 1)$ consists of those paths coming from $PX_+(0, 1)$ and $PX_-(0, 1)$ both of which are path-connected (see example 5.8), hence also connected. This shows that $\vec{\pi}^{-1}(0, 1)$ consists of two disjoint connected components, because $\gamma_0, \gamma_1 \notin \vec{\pi}^{-1}(0, 1)$.

Now assume that there is a continuous section s of $\vec{\pi}$ over U . We can find two sequences $(x_n^+, y_n^+)_{n \in \mathbb{N}}$ in $(\vec{\pi}(PX_+) \cap U) \setminus \{(0, 1)\}$ and $(x_n^-, y_n^-)_{n \in \mathbb{N}}$ in $(\vec{\pi}(PX_-) \cap U) \setminus \{(0, 1)\}$ both converging to $(0, 1)$. The limits of the images under s of those two sequences would need to land in different connected components of $\vec{\pi}^{-1}(0, 1)$, thus s cannot be (sequentially) continuous, showing $\overrightarrow{TC}(\vec{\mathbb{S}}^1) \geq 2$.

On the other hand, there is a partition

$$\Gamma_{\vec{\mathbb{S}}^1} = U_1 \cup U_2$$

with $U_1 = \vec{\pi}(PX_+)$ and $U_2 = \vec{\pi}(PX_-) \setminus \{(0, 0), (0, 1), (1, 1)\}$ and from i) we know that over each of those sets there exists a continuous section of $\vec{\pi}$. Thus we conclude $\overrightarrow{TC}(\vec{\mathbb{S}}^1) = 2$.

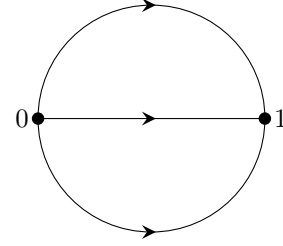
Furthermore, the map $\vec{\pi}$ is not a fibration in this case. As discussed above, the fiber $\vec{\pi}^{-1}(0, 1)$ consists of two disjoint connected components. If we take another pair $(a, b) \in \vec{\pi}(PX_+) \setminus \vec{\pi}(PX_-)$ then the fiber $\vec{\pi}^{-1}(a, b)$ would be homeomorphic to $PI(a, b)$ which is path-connected as shown earlier. From example 5.8 we also know that $\Gamma_{\vec{\mathbb{S}}^1}$ is path-connected. Hence if $\vec{\pi}$ would be a fibration, the different fibers would have to be homotopy-equivalent by proposition 3.19. But this is not the case, as we have seen that one fiber is path-connected and the other is not.

5.3 Relations to Undirected Topological Complexity

In this section we discuss relationships between undirected and directed topological complexity in more detail.

Contrary to what one might expect, the next example shows that the directed complexity of a d-space can in some cases be smaller than the undirected complexity of the underlying topological space.

Example 5.13. Consider the following directed space from [GFS20], Example 5:



Similar to $\vec{\mathbb{S}}^1$, this d-space can be constructed by glueing together three copies of $\vec{I}$, i.e. take $\vec{I}_i := \vec{I} \times \{i\}$ for $i = 1, \dots, 3$ and then define

$$\vec{X} := (\vec{I}_1 \sqcup \vec{I}_2 \sqcup \vec{I}_3) / \sim,$$

where $\sim$ is induced by $(0, 1) \sim (0, 2) \sim (0, 3)$ and $(1, 1) \sim (1, 2) \sim (1, 3)$.

Similar to the observations we made for $\vec{\mathbb{S}}^1$, there exists no global section for $\vec{\pi}$ in this case, so $\overrightarrow{TC}(\vec{X}) \geq 1$.

In order to construct local sections of $\vec{\pi}$, consider the sets

$$U_1 := \{(0, 1)\}, \quad U_2 := \Gamma_{\vec{X}} \setminus \{(0, 1)\}.$$

A continuous section of $\vec{\pi}$ over U_1 is just given by a choice of any d-path between 0 and 1, since every map from a one-point space into another space is continuous. Regarding U_2 , note that U_2 is a union of disjoint sets $\Gamma_1 \sqcup \Gamma_2 \sqcup \Gamma_3$ where each Γ_i is a copy of $\Gamma_{\vec{I}}$ without its boundary. We have seen before, that there exists a continuous section of $\vec{\pi}_{\vec{I}}$ over $\Gamma_{\vec{I}}$, so there is a continuous section s_i of $\vec{\pi}_{\vec{X}}$ over each Γ_i . Then we obtain a continuous section s of $\vec{\pi}_{\vec{X}}$ over U_2 by defining

$$s := s_1 \sqcup s_2 \sqcup s_3.$$

This shows that $\overrightarrow{TC}(\vec{X}) = 2$, whereas $TC(X) = 3$, see example 4.41. Here, X denotes the underlying space of $\vec{X}$.

Directed Complexity sometimes being smaller than undirected complexity can happen due to the fact that, while the space of possible paths PX is smaller than X^I , also the space of considered start- and end-points Γ_X is smaller than in the undirected case, since in general we have $\Gamma_X \neq X \times X$.

However, in the case of a strongly-connected d-space X , the directed complexity will always be at least as big as the topological complexity. The proof is based upon [GFS20], Proposition 2.

Proposition 5.14. *Let $\vec{X} = (X, PX)$ be a strongly connected d-space, where X is an ENR. Then*

$$\mathrm{TC}(X) \leq \overrightarrow{\mathrm{TC}}(\vec{X}).$$

Proof. Let $k = \overrightarrow{\mathrm{TC}}(\vec{X})$. Since $\vec{X}$ is strongly connected, we have $\Gamma_{\vec{X}} = X \times X$. This means there is a partition

$$X \times X = U_1 \cup \dots \cup U_k$$

in the sense of definition 5.11. Note that $\vec{\pi}$ is just a restriction of the path space fibration π , and any ENR is locally compact, see corollary 2.24. Thus, the same partition can be used for $\mathrm{TC}(X)$ in the sense of proposition 4.11 iii), implying $\mathrm{TC}(X) \leq \overrightarrow{\mathrm{TC}}(\vec{X})$. $\square$

Later in chapter 5.5 we will see that the two notions of complexity coincide for strongly connected directed graphs.

Proposition 5.14 can be useful for computing the directed complexity of a space, as seen in the next example which is based upon [GFS20] Example 4.

Example 5.15. Consider the directed loop $(X, PX) = \vec{\mathbb{O}}^1$ from example 5.3. This space is strongly connected, thus $\overrightarrow{\mathrm{TC}}(\vec{\mathbb{O}}^1) \geq \mathrm{TC}(\mathbb{S}^1) = 2$.

We show that also $\overrightarrow{\mathrm{TC}}(\vec{\mathbb{O}}^1) \leq 2$. Consider

$$\begin{aligned} U_1 &= \{(a, b) \in \mathbb{S}^1 \times \mathbb{S}^1 \mid a = b\}, \\ U_2 &= \{(a, b) \in \mathbb{S}^1 \times \mathbb{S}^1 \mid a \neq b\}. \end{aligned}$$

A continuous section $s_1 : U_1 \rightarrow PX$ of $\vec{\pi}$ over $U_1 = \{(a, a) \mid a \in \mathbb{S}^1\}$ is obtained by defining

$$s_1(a, a) := \gamma_a,$$

where γ_a is the constant path at a .

On the other hand, for $(a, b) \in U_2$ assign (a, b) to the path $\tilde{\gamma}_{a,b}$ obtained by moving a in the positive direction with constant speed towards b , similar to example 4.5 ii). Hence $\overrightarrow{\mathrm{TC}}(\vec{\mathbb{O}}^1) = 2$.

5.4 Regular Directed Spaces and Product Inequality

As in the undirected case, a product inequality can be shown for a certain class of directed spaces. The following definition is due to [GFS20].

Definition 5.16. A d-space (X, PX) is called *regular*, if there is a partition

$$\Gamma_X = U_1 \cup \dots \cup U_n,$$

where $n := \overrightarrow{\mathrm{TC}}(X)$, such that the following conditions are satisfied:

- i) Each U_i is an ENR;
- ii) $\vec{\pi}$ admits a continuous section over each U_i ;
- iii) The set $\bigcup_{i=1}^r U_i$ is closed for any $r = 1, \dots, n$.

Remark 5.17. As remarked in [GFS20], a partition as defined in 5.16 has the property that

$$\overline{U_i} \cap U_{i'} = \emptyset$$

for $i < i'$.

One can prove this by induction over i :

By property iii) we know that U_1 is closed and thus $\overline{U_1} \cap U_{i'} = U_1 \cap U_{i'} = \emptyset$ for all $i' > 1$, since the sets form a partition of Γ_X .

Now let $i \in \{1, \dots, n-1\}$ and assume that $\overline{U_i} \cap U_{i'} = \emptyset$ holds for all $i' > i$. Furthermore let $i'' > i+1$. Then we have

$$\overline{\bigcup_{j=1}^i U_j} \cap U_{i''} = \bigcup_{j=1}^i (\overline{U_j} \cap U_{i''}) = \emptyset.$$

By property iii) we know that $\bigcup_{j=1}^{i+1} U_j = \bigcup_{j=1}^{i+1} \overline{U_j}$. Hence we obtain

$$\begin{aligned} \overline{U_{i+1}} \cap U_{i''} &= (\overline{U_{i+1}} \cap U_{i''}) \cup \left(\overline{\bigcup_{j=1}^i U_j} \cap U_{i''} \right) \\ &= \overline{\bigcup_{j=1}^{i+1} U_j} \cap U_{i''} \\ &= \left(\bigcup_{j=1}^{i+1} U_j \right) \cap U_{i''} \\ &= \bigcup_{j=1}^{i+1} (U_j \cap U_{i''}) \\ &= \emptyset, \end{aligned}$$

again using that the sets $\{U_i\}$ form a partition of Γ_X .

The proof of the following product inequality is based upon [GFS20] Proposition 1.

Proposition 5.18. *For regular d-spaces $(X, PX), (Y, PY)$, we have*

$$\overrightarrow{\mathrm{TC}}(X \times Y, P(X \times Y)) \leq \overrightarrow{\mathrm{TC}}(X, PX) + \overrightarrow{\mathrm{TC}}(Y, PY) - 1.$$

Proof. First note that

$$\Gamma_{X \times Y} = \Gamma_X \times \Gamma_Y.$$

This holds, since $P(X \times Y)$ consists only of paths $\gamma = (\gamma_X, \gamma_Y)$ with $\gamma_X \in PX$ and $\gamma_Y \in PY$, hence the

existence of a directed path from $(x, y) \in X \times Y$ to $(x', y') \in X \times Y$ is equivalent to the existence of directed paths from x to x' in X and from y to y' in Y .

Let

$$k := \overrightarrow{\text{TC}}(X, PX) \quad \text{and} \quad \ell := \overrightarrow{\text{TC}}(Y, PY)$$

both be finite, because otherwise the inequality is clear. Since we required the spaces to be regular, there are partitions

$$\Gamma_X = U_1 \cup \dots \cup U_k, \quad \Gamma_Y = V_1 \cup \dots \cup V_\ell,$$

such that each U_i and each V_j is an ENR, π_X admits a continuous section s_i over each U_i , π_Y admits a continuous section r_j over each V_j , and the sets $\bigcup_{i=1}^n U_i, \bigcup_{j=1}^m V_j$ are closed for any $n = 1, \dots, k, m = 1, \dots, \ell$.

For $(i, j) \in \{1, \dots, k\} \times \{1, \dots, \ell\}$, define $W_{ij} := U_i \times V_j$. Each W_{ij} is an ENR and they form a partition of $X \times Y$. For each (i, j) , we obtain a continuous section σ_{ij} of $\pi_{X \times Y}$ over W_{ij} by defining

$$\sigma_{ij} := s_i \times r_j.$$

Now, for $p = 2, \dots, k + \ell$, define

$$G_p := \bigcup_{i+j=p} W_{ij}.$$

Those sets also form a partition of $X \times Y$.

We know that $W_{ij} \cap W_{i'j'} = \emptyset$ for $(i, j) \neq (i', j')$, so G_p is in fact a union of disjoint sets, which allows us to easily define a section $\tilde{\sigma}_p$ of $\pi_{X \times Y}$ over G_p by setting

$$\tilde{\sigma}_p := \bigsqcup_{i+j=p} \sigma_{ij}.$$

We still have to check continuity of those sections. If we know that each W_{ij} is open in G_p , then $\tilde{\sigma}_p$ is continuous by corollary 2.4, implying

$$\overrightarrow{\text{TC}}(X \times Y) \leq k + \ell - 1 = \overrightarrow{\text{TC}}(X) + \overrightarrow{\text{TC}}(Y) - 1.$$

All that is left to show is that the sets W_{ij} are open in the respective G_p .

Let $p \in \{2, \dots, k + \ell\}$ and (i, j) such that $i + j = p$. We show that $G_p \setminus W_{ij}$ is closed in G_p , then W_{ij} is open in G_p .

Consider the set

$$\overline{G_p \setminus W_{ij}} = \bigcup_{\substack{a+b=p \\ (a,b) \neq (i,j)}} \overline{W_{ab}} = \bigcup_{\substack{a+b=p \\ (a,b) \neq (i,j)}} \overline{U_a \times V_b}$$

which is closed in $X \times Y$. We have

$$G_p \cap \overline{(G_p \setminus W_{ij})} = \bigcup_{\substack{i'+j'=p \\ a+b=p \\ (a,b) \neq (i,j)}} (U_{i'} \times V_{j'}) \cap (\overline{U_a \times V_b}).$$

Let $i' + j' = p$ and $a + b = p$ with $(a, b) \neq (i, j)$. Without loss of generality, assume $a < i$, then $b > j$. Since the spaces X and Y are regular, remark 5.17 holds. If $(i', j') = (i, j)$, then

$$(U_{i'} \times V_{j'}) \cap (\overline{U_a \times V_b}) = \emptyset,$$

since $\overline{U_a} \cap U_i = \emptyset$. On the other hand, if $(i', j') \neq (i, j)$, there are two cases: Either $(i', j') \neq (a, b)$ or $(i', j') = (a, b)$. In the first case, we can assume $i' < a, j' > b$ without loss of generality. But then again

$$(U_{i'} \times V_{j'}) \cap (\overline{U_a \times V_b}) = \emptyset,$$

since $\overline{V_b} \cap V_{j'} = \emptyset$. For the second case, we obtain

$$(U_{i'} \times V_{j'}) \cap (\overline{U_a \times V_b}) = (U_{i'} \times V_{j'}) = W_{i'j'} = W_{ab}.$$

Summarizing those results, we obtain

$$G_p \cap \overline{(G_p \setminus W_{ij})} = \bigcup_{\substack{a+b=p \\ (a,b) \neq (i,j)}} W_{ab} = G_p \setminus W_{ij}.$$

This implies that $G_p \setminus W_{ij}$ is closed in G_p , since $\overline{G_p \setminus W_{ij}}$ is of course closed in $X \times Y$.

Lastly, note that each G_p is an ENR by proposition 2.25.

This concludes the proof. $\square$

Remark 5.19. As noted in [GFS20], for the case of undirected topological complexity, each ENR X with finite $\text{TC}(X) =: n$ could be considered regular, where of course the definition of regularity would have to be adapted for undirected complexity. This is due to proposition 4.11 ii), since given a sequence

$$\emptyset = W_0 \subseteq W_1 \subseteq \dots \subseteq W_n = X \times X$$

as in the proposition, one obtains a cover in the sense of definition 5.16 by defining $U_i := W_i \setminus W_{i-1}$. Remark 5.17 then also applies to those sets, hence similar arguments as in the previous proof could be used to show the product inequality for path-connected ENRs in the undirected case, but this requires further investigation.

With the help of the product inequality and our results about strongly connected d-spaces 5.14, we can compute the directed complexity of the directed n -torus $\vec{T}^n := (\vec{\mathbb{S}}^1)^n$. The proof is based upon [GFS20], Corollary 2.

Proposition 5.20. *We have $\overrightarrow{\text{TC}}(\vec{T}^n) = n + 1$.*

Proof. Note that the directed loop is regular and we have $\overrightarrow{\text{TC}}(\vec{\mathbb{S}}^1) = 2$, see example 5.15. The product inequality then implies $\overrightarrow{\text{TC}}(\vec{T}^n) \leq n + 1$. But $\vec{T}^n$ is also strongly connected and thus $\overrightarrow{\text{TC}}(\vec{T}^n) \geq \text{TC}((\mathbb{S}^1)^n) = n + 1$, see proposition 4.43. $\square$

5.5 Directed Graphs

The following observations are based upon [GFS20], section 5.

In section 4.7, we analyzed the topological complexity of graphs, where a graph is regarded as a 1-dimensional CW-complex, i.e. a collection of copies of the standard unit interval, glued together along their boundaries via attaching maps. For a directed graph G , every edge is given some orientation. Hence, similar to the undirected case, a directed graph can be regarded as a collection of copies of the standard directed interval $\vec{I}$ glued together along their end-points in some way. This induces a d-structure on G , as discussed in remark 5.2, which we denote as PG . We write $\vec{G} := (G, PG)$ for the associated d-space. After this glueing process, each edge of $\vec{G}$ can be identified with either $\vec{I}$ or $\vec{O}^1$.

Definition 5.21. Given a directed graph $\vec{G}$, a *small directed path* is a directed path which lies on only a single edge. Given some edge e of $\vec{G}$, we denote the interior of e as $\text{Int}(e)$.

Small directed paths are exactly the paths induced by the d-structures of $\vec{I}$ and $\vec{S}^1$. Furthermore, directed paths in $\vec{G}$ are concatenations of small directed paths. Note that for some edge e of $\vec{G}$ and for $x \in \text{Int}(e), y \in G \setminus e$ with $(x, y) \in \Gamma_G$ we have $(x', y) \in \Gamma_G$ for all $x' \in \text{Int}(e)$.

The following result tells us that the directed complexity of a directed graph is at most 3. This coincides with the theory of (undirected) topological complexity, see proposition 4.40. Note that “connected” in this context means, that the underlying undirected graph is connected, which has to be distinguished from the notion of strongly connectedness. The proof is based upon [GFS20], Proposition 3.

Proposition 5.22. *Given a finite connected directed graph $\vec{G} = (G, PG)$, we have $\overrightarrow{\text{TC}}(\vec{G}) \leq 3$.*

Proof. In order to find a suitable partition of $\Gamma_{\vec{G}}$, we consider the following subsets:

U_1 consists of all pairs of vertices $(x, y) \in \Gamma_{\vec{G}}$. U_2 consists of all pairs of points $(x, y) \in \Gamma_{\vec{G}}$, such that one of the points is a vertex and the other lies in the interior of an edge. U_3 consists of all pairs of points $(x, y) \in \Gamma_{\vec{G}}$, such that each point is an interior point of some edge. We now construct continuous sections of $\vec{\pi}$ over each U_i .

Let $(v_i, v_j) \in U_1$ be a pair of vertices. Since $U_1 \subseteq \Gamma_{\vec{G}}$ by definition, there is some directed path γ_{ij} between them. This defines a section of $\vec{\pi}$ via

$$s_1 : U_1 \rightarrow PG, \quad (v_i, v_j) \mapsto \gamma_{ij}.$$

This section is continuous, since U_1 is a discrete subspace of $\Gamma_{\vec{G}}$, and every map from a discrete space to some

other space is continuous.

Consider now a pair $(x, v_j) \in U_2$ such that $x \in \text{Int}(e)$ for some directed edge e and v_j is a vertex. Denote the endpoint of e as v_i . Similarly to before, there is a directed path γ_{ij} joining v_i and v_j . Furthermore, there is a constant velocity directed path γ_{x, v_i} joining x and v_i . (Think of e as a copy of I , then $v_i \in e$ corresponds to $1 \in I$. Then consider the straight path $t \mapsto (1 - t)x + t$ in I .)

This gives us a continuous section of $\vec{\pi}$ over $\text{Int}(e) \times \{v_j\}$ via $(x, v_j) \mapsto \gamma_{x, v_i} * \gamma_{ij}$. Continuity follows by corollary 3.6, since both $(x, 1) \mapsto (1 - t)x + t$ and $(v_i, v_j) \mapsto \gamma_{ij}$ are continuous as discussed before, and concatenation is a continuous operation.

Similarly, we can construct a continuous section over $\{v_j\} \times \text{Int}(e)$ for some vertex v_j .

Note that U_2 is a finite disjoint union of sets of the form $\text{Int}(e) \times \{v_j\}$ or $\{v_j\} \times \text{Int}(e)$ for some directed edges e and vertices v_j , so by taking the disjoint union of the corresponding sections constructed above, we obtain a continuous section s_2 of $\vec{\pi}$ over U_2 .

Finally, consider a pair $(x, y) \in U_3$, i.e. $x \in \text{Int}(e)$ and $y \in \text{Int}(e')$ for some directed edges e and e' .

First assume $e = e'$. In this case, identify e with $\vec{I}$, then x and y can be considered as points lying in the interior of $\vec{I}$. A continuous section of $\vec{\pi}$ over $\text{Int}(e) \times \text{Int}(e)$ is thus given by $(x, y) \mapsto \gamma_{xy}$ where γ_{xy} is the constant velocity path between x and y along $e \cong I$.

If $e \neq e'$, we proceed similarly to the construction for U_2 . We denote the end point of e by v_i and the initial point of e' by v_j . Then a continuous section of $\vec{\pi}$ over $\text{Int}(e) \times \text{Int}(e')$ is given by

$$(x, y) \mapsto \gamma_{x, v_i} * \gamma_{ij} * \gamma_{v_j, y},$$

where γ_{x, v_i} denotes the constant velocity path from x to v_i along e , $\gamma_{v_j, y}$ denotes the constant velocity path from v_j to y and γ_{ij} is again some directed path between the vertices v_i and v_j .

Similar to before, U_3 is a (finite) disjoint union of sets of the form $\text{Int}(e) \times \text{Int}(e')$, so with this construction we obtain a continuous section over U_3 . $\square$

Recall, that in example 5.13 we saw that there are directed graphs $\vec{G} = (G, PG)$ where $\overrightarrow{\text{TC}}(\vec{G}) < \text{TC}(G)$. Thus, although we know that both directed and undirected complexity of a graph must be smaller or equal than 3, in general we know nothing about the relationship of $\text{TC}(G)$ and $\overrightarrow{\text{TC}}(\vec{G})$. However, the next proposition shows that directed and undirected complexity coincide for strongly connected directed graphs. The proof is based upon [GFS20], Proposition 4.

Proposition 5.23. *Given a finite strongly connected di-*

rected graph $\vec{G} = (G, PG)$, we have

$$\overrightarrow{TC}(\vec{G}) = TC(G) = \min(b_1(G), 2) + 1.$$

Proof. We already know that

$$\overrightarrow{TC}(\vec{G}) \geq TC(G) = \begin{cases} 1, & b_1(G) = 0, \\ 2, & b_1(G) = 1, \\ 3, & b_1(G) > 1, \end{cases}$$

by propositions 4.40 and 5.14.

In order to show the inverse inequality, first note that G is homotopy-equivalent to the wedge of $b_1(G)$ circles, see proposition 4.37 and corollary 4.39.

If $b_1(G) = 0$, then G is contractible. We show that then $\vec{G}$ must be single point. Assume the contrary, i.e. $\vec{G}$ has at least one directed edge e , which can be identified with $\vec{I}$. Let $x, y \in \text{Int}(e)$, such that $y < x$ (with respect to the order on $\vec{I}$). Since $\vec{G}$ is strongly connected, there must be a sequence of other edges $e_1, \dots, e_n$ together with a d-path along those edges that connects x and y . But then there is a loop in G , which contradicts the contractibility of G . Hence G is a single point and $\overrightarrow{TC}(\vec{G}) = 1$.

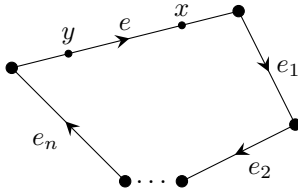


Figure 8: Contradiction for the case $b_1(G) = 0$.

If $b_1(G) = 1$, then we know that G is homotopy-equivalent to a circle. But then $\vec{G}$ must have n vertices $v_1, \dots, v_n$ and n oriented edges $e_1, \dots, e_n$ such that e_i goes from v_i to v_{i+1} and e_n connects v_n and v_1 , since otherwise the directed graph would not be strongly connected.

Then one can proceed similar to example 5.15 to obtain $\overrightarrow{TC}(\vec{G}) \leq 2$, by concatenating small directed paths along the edges in positive direction whenever start- and end-point are not equal.

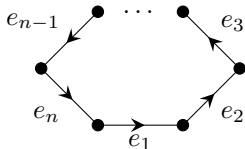


Figure 9: The case $b_1(G) = 1$.

If $b_1(G) > 1$, we already know $\overrightarrow{TC}(\vec{G}) \geq TC(G) = 3$ but also $\overrightarrow{TC}(\vec{G}) \leq 3$ by proposition 5.22. $\square$

5.6 Dihomotopy Equivalence and Invariance

We have seen in chapter 4.2 that (undirected) topological complexity is a homotopy invariant. For directed spaces, however, the underlying topological spaces being homotopy equivalent is not sufficient for their directed topological complexity to be the same.

Example 5.24. Consider the following directed graphs:

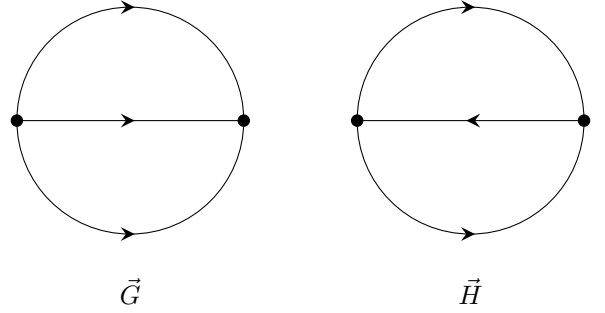


Figure 10: Two directed graphs with the same underlying undirected graphs

The underlying spaces G and H are obviously homotopic to each other, but $\overrightarrow{TC}(\vec{G}) = 2$ as seen in example 5.13 and $\overrightarrow{TC}(\vec{H}) = TC(H) = 3$ by example 4.41 and proposition 5.23, since $\vec{H}$ is strongly connected.

Thus we need a stronger notion of homotopy between d-spaces, which takes the d-structures of those spaces into consideration. This leads us to the notion of simple dihomotopy equivalence, which will turn out to indeed preserve directed topological complexity.

In order to define what a simple dihomotopy equivalence is, we need the notion of continuously graded maps. But first, we introduce two notations

Definition 5.25. Let $f : (X, PX) \rightarrow (Y, PY)$ be a d-map. With PY^f we denote the subspace of PY which consists of all d-paths that go from $f(x)$ to $f(x')$ for some $(x, x') \in \Gamma_X$, i.e.

$$PY^f = (\pi_Y)^{-1}(f(\Gamma_X)) \subseteq PY.$$

Furthermore, we define the subspace

$$\begin{aligned} T^f &:= \{(\gamma, x, x') \in PY^f \times \Gamma_X \mid \gamma \in PY(f(x), f(x'))\} \\ &= \{(\gamma, x, x') \in PY^f \times \Gamma_X \mid \pi(\gamma) = (f(x), f(x'))\}. \end{aligned}$$

The following definition is based upon [Gou17] and [GFS20].

Definition 5.26. Let $f : (X, PX) \rightarrow (Y, PY)$ be a d-map. Suppose that for any $(x, x') \in \Gamma_X$ there is a continuous map

$$F_{x, x'} : PY(f(x), f(x')) \rightarrow PX(x, x'),$$

with the following properties:

- i) Let $(f(x), f(x')) \in (f \times f)(\Gamma_X)$ and consider the preimage $W := (f \times f)^{-1}(f(x), f(x'))$. Then the map

$$W \rightarrow PX^{PY(f(a), f(a'))}, \quad (a, a') \mapsto F_{a, a'}$$

is continuous, where $PX^{PY(f(a), f(a'))}$ is considered as a subspace of PX^{PY} .

- ii) The map

$$F : T^f \rightarrow PX, \quad (\gamma, x, x') \mapsto F_{x, x'}(\gamma)$$

is continuous.

In this case, we say that this is a *map* $F : PY \multimap PX$ *continuously graded by* $f : X \rightarrow Y$.

Suppose that $F : PY \multimap PX$ is continuously graded by f and there is a map $h : \Gamma_X \rightarrow PY^f$ with $h(x, x') \in PY(f(x), f(x'))$. Then property ii) implies that the map

$$\Gamma_X \rightarrow PX, \quad (x, x') \mapsto F_{x, x'}(h(x, x'))$$

is continuous, since $F_{x, x'}(h(x, x')) = F(h(x, x'), x, x')$ and this can be written as the composition

$$\Gamma_X \xrightarrow{\Delta} \Gamma_X \times \Gamma_X \xrightarrow{h \times \text{id}} T^f \xrightarrow{F} PX$$

of continuous maps, where $\Delta : (x, x') \mapsto ((x, x'), (x, x'))$ is the diagonal map.

Recall, that a d-map $f : (X, PX) \rightarrow (Y, PY)$ induces a continuous map

$$Pf : PX \rightarrow PY, \quad \gamma \mapsto f \circ \gamma,$$

and that we defined $PX(x, x')$ as the subspace of PX consisting of all d-paths which start at x and end at x' . Also note that

$$Pf|_{PX(x, x')} : PX(x, x') \rightarrow PY(f(x), f(x')).$$

The following example is based on a remark in [Gou17].

Example 5.27. Let (X, PX) and (Y, PY) be two d-spaces which are isomorphic, meaning there are d-maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ with $f \circ g = \text{id}_Y$ and $g \circ f = \text{id}_X$. We show that both Pf and Pg induce continuously graded maps.

For $(x, x') \in \Gamma_X$, define $F_{x, x'} := Pg|_{PY(f(x), f(x'))}$ which is a map

$$PY(f(x), f(x')) \rightarrow PX(g(f(x)), g(f(x'))) = PX(x, x').$$

The first property of a continuous graded map is automatically satisfied in this case. For the second property, note that given $(\gamma, x, x') \in T_X^f$, we have $\gamma \in$

$PY(f(x), f(x'))$. Thus, for $F : T^f \rightarrow PX$ defined by

$$(\gamma, x, x') \mapsto F_{x, x'}(\gamma) = Pg|_{PY(f(x), f(x'))}(\gamma) = Pg(\gamma),$$

we obtain that F is the composition

$$T^f \xrightarrow{(\gamma, x, x') \mapsto \gamma} PY \xrightarrow{Pg} PX,$$

implying continuity of F . Hence $F : PY \multimap PX$ is continuously graded by f . Similarly, Pf induces a continuously graded map $G : PX \multimap PY$ by $G_{y, y'} := Pf|_{PX(g(y), g(y'))}$.

Now we can define the notion of a simple dihomotopy equivalence, see [GFS20] Definition 10.

Definition 5.28. Let (X, PX) and (Y, PY) be d-spaces. A *simple dihomotopy equivalence* between X and Y is a d-map $f : X \rightarrow Y$, such that the following conditions are satisfied:

- i) f is a homotopy equivalence between X and Y such that the homotopy inverse $g : Y \rightarrow X$ is also a d-map.
- ii) There is a map $F : PY \multimap PX$ continuously graded by f , such that for any $(x, x') \in \Gamma_X$ the map $Pf|_{PX(x, x')}$ is a homotopy equivalence between $PX(x, x')$ and $PY(f(x), f(x'))$ with homotopy inverse $F_{x, x'}$.
- iii) There is a map $G : PX \multimap PY$ continuously graded by g , such that for any $(y, y') \in \Gamma_Y$ the map $Pg|_{PY(y, y')}$ is a homotopy equivalence between $PY(y, y')$ and $PX(g(y), g(y'))$ with homotopy inverse $G_{y, y'}$.

Remark 5.29. It should be noted at this point that different notions of directed homotopy equivalences can be found in the literature. See for example [Rau21], where a different kind of directed homotopy equivalence is introduced, which also preserves directed complexity.

The following examples are based on [GFS20] Example 8.

Example 5.30. i) Let (X, PX) and (Y, PY) be two d-spaces which are isomorphic, meaning there are d-maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ with $f \circ g = \text{id}_Y$ and $g \circ f = \text{id}_X$. Then X and Y are simply dihomotopically equivalent.

The first property is obviously satisfied, since f is a homeomorphism with inverse g .

To verify the second property, let $F := Pg$. As we already know from example 5.27, this is a graded map where the grading is given by

$$F_{x, x'} = Pg|_{PY(f(x), f(x'))}$$

for any $(x, x') \in \Gamma_X$. Furthermore we obtain

$$\begin{aligned} F_{x,x'} \circ Pf|_{PX(x,x')} &= Pg|_{PY(f(x),f(x'))} \circ Pf|_{PX(x,x')} \\ &= (Pg \circ Pf)|_{PX(x,x')} \\ &= P(g \circ f)|_{PX(x,x')} \\ &= \text{id}_{PX(x,x')} \end{aligned}$$

and similarly

$$Pf|_{PX(x,x')} \circ F_{x,x'} = \text{id}_{PY(f(x),f(x'))}.$$

Likewise by setting $G := Pf$ the third property is satisfied.

- ii) We show that $\vec{I} = (I, PI)$ is simply dihomotopically equivalent to the one-point space $Y := \{*\}$, where the d-space structure on $\{*\}$ is given by the set $PY = \{c_* : I \rightarrow *\}$ consisting of only the constant path. Note that $PI(0, 0)$ consists only of the constant map $c_0 : I \rightarrow \{0\} \subseteq I$. Consider the maps

$$f : I \rightarrow Y, x \mapsto *, \quad g : Y \rightarrow I, * \mapsto 0.$$

It is well-known that these two maps form a (classical) homotopy equivalence between I and Y . Furthermore, they are obviously d-maps, so the first property is satisfied.

Note that the induced maps on path-spaces are given by $Pf : PI \rightarrow PY, \gamma \mapsto c_*$ and $Pg : PY \rightarrow PI, c_* \mapsto c_0$.

To obtain $F : PY \rightrightarrows PI$, first define for $(x, x') \in \Gamma_I$ the linear interpolation $\gamma_{x,x'} \in PI$ between them by

$$\gamma_{x,x'}(t) := (1-t)x + tx'.$$

Then define

$$F_{x,x'} : PY(*, *) \rightarrow PI(x, x'), \quad c_* \mapsto \gamma_{x,x'}.$$

We have $PY^f = PY$ and $T^f = PY \times \Gamma_I$, thus the map

$$F : T^f \rightarrow PI, \quad (c_*, x, x') \mapsto F_{x,x'}(c_*) = \gamma_{x,x'}$$

is continuous, since the assignment $(x, x') \mapsto \gamma_{x,x'}$ is continuous by corollary 3.6.

Furthermore, we have $Pf|_{PI(x,x')} \circ F_{x,x'} = \text{id}_{PY(*,*)}$ and

$$F_{x,x'} \circ Pf|_{PI(x,x')}(\gamma) = \gamma_{x,x'}.$$

In example 5.8 we have seen that each $PI(x, x')$ is contractible, hence any two maps from some space to $PI(x, x')$ are homotopic to each other. In particular, this implies

$$F_{x,x'} \circ Pf|_{PI(x,x')} \simeq \text{id}_{PI(x,x')}.$$

Hence, the maps $F_{x,x'}$ and $Pf|_{PI(x,x')}$ form indeed a homotopy equivalence for any $(x, x') \in \Gamma_X$.

For the third property, define a graded map $G : PI \rightrightarrows PY$ by the constant map

$$G_{*,*} : PI(g(*), g(*)) = PI(0, 0) \rightarrow PY(*, *).$$

Then obviously $Pg|_{PY(*,*)} \circ G_{*,*} = \text{id}_{PX(0,0)}$ and $G_{*,*} \circ Pg|_{PY(*,*)} = \text{id}_{PY(*,*)}$.

This concludes the example.

We now arrive at the main result of this chapter. The proof is based upon [GFS20], Proposition 7.

Proposition 5.31. *Let X and Y be two simply dihomotopically equivalent d-spaces. Then $\vec{\text{TC}}(X) = \vec{\text{TC}}(Y)$.*

Proof. Let (X, PX) and (Y, PY) be simply dihomotopically equivalent. This means we have d-maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that g is the homotopy inverse of f . Furthermore, there are maps

$$F : PY \rightrightarrows PX, \quad G : PX \rightrightarrows PY,$$

continuously graded by f and g respectively.

We first show $\vec{\text{TC}}(Y) \leq \vec{\text{TC}}(X)$. Suppose $k := \vec{\text{TC}}(X)$. This means we have a partition of Γ_X into pairwise disjoint subsets

$$\Gamma_X = U_1 \cup \dots \cup U_k,$$

such that each U_i is an ENR and for each i there is a continuous section s_i of $\vec{\pi}_X$ over U_i . Define

$$V_i := (g \times g)^{-1}(U_i) \cap \Gamma_Y.$$

We show that these sets form a partition of Γ_Y . They are pairwise disjoint, since preimages of pairwise disjoint sets remain pairwise disjoint. Note that

$$(g \times g)(\Gamma_Y) = \Gamma g(\vec{\pi}_Y(PY)) = \vec{\pi}_X(Pg(PY))$$

by remark 5.10 and thus $(g \times g)(\Gamma_Y) \subseteq \vec{\pi}_X(PX) = \Gamma_X$. Hence we know that $\Gamma_Y \subseteq (g \times g)^{-1}(\Gamma_X)$ and obtain

$$\begin{aligned} \bigcup_{i=1}^k V_i &= \bigcup_{i=1}^k ((g \times g)^{-1}(U_i) \cap \Gamma_Y) \\ &= (g \times g)^{-1} \left(\bigcup_{i=1}^k U_i \right) \cap \Gamma_Y \\ &= (g \times g)^{-1}(\Gamma_X) \cap \Gamma_Y \\ &= \Gamma_Y. \end{aligned}$$

Continuous sections t_i of $\vec{\pi}_Y$ over V_i can then be defined via concatenations

$$\begin{array}{ccc} V_i & \xrightarrow{(y,y') \mapsto (s_i(g(y), g(y')), y, y')} Tg & \xrightarrow{G} PY. \\ & \searrow t_i & \nearrow \end{array}$$

The individual maps are continuous, since $g \times g$, s_i and G are continuous. Furthermore, note that for $(y, y') \in V_i$ we have $s_i(g(y), g(y')) \in PX(g(y), g(y'))$, since s_i is a section of $\vec{\pi}$. Also G is graded by g as mentioned in the beginning, hence

$$G_{y,y'} : PX(g(y), g(y')) \rightarrow PY(y, y'),$$

which implies

$$G(s(g(y), g(y')), y, y') = G_{y,y'}(s(g(y), g(y'))) \in PY(y, y').$$

This shows that the maps t_i are indeed local sections of $\vec{\pi}_Y$.

Similarly, one can prove $\overrightarrow{\text{TC}}(X) \leq \overrightarrow{\text{TC}}(Y)$.

The obtained inequalities also show that both $\overrightarrow{\text{TC}}(X)$ and $\overrightarrow{\text{TC}}(Y)$ must be infinite if either of them is, since otherwise it would lead to a contradiction. $\square$

The next results are concerned with dcontractible spaces.

Definition 5.32. A d-space X is called *dcontractible*, if it is dihomotopically equivalent to a point.

The previous proposition immediately implies a result similar to corollary 3.16 for the undirected case. Note, however, that without any additional work, we only get an implication instead of an equivalence.

Corollary 5.33. If X is a dcontractible d-space, then $\overrightarrow{\text{TC}}(X) = 1$.

Example 5.30 then implies $\overrightarrow{\text{TC}}(\vec{I}) = 1$, which coincides with example 5.12 i).

In order to obtain an equivalence, the underlying topological space is required to be contractible. The proof is based upon [GFS20], Theorem 1.

Proposition 5.34. Let (X, PX) be a d-space such that X is contractible. Then (X, PX) is dcontractible if and only if $\overrightarrow{\text{TC}}(X, PX) = 1$.

Proof. The implication that $\overrightarrow{\text{TC}}(X) = 1$ if X is dcontractible is already given by the previous corollary.

Assume that $\overrightarrow{\text{TC}}(X) = 1$. We denote the one-point space as $Y := \{*\}$ and its unique d-structure as PY . X being contractible implies the existence of maps $f : X \rightarrow Y$, $x \mapsto *$ and $g : Y \rightarrow X$, $y \mapsto x_0$, where x_0 is some point in X , such that f and g are homotopy inverses to each other. Those maps are obviously d-maps.

Furthermore, denote by c_* and c_0 the constant paths at

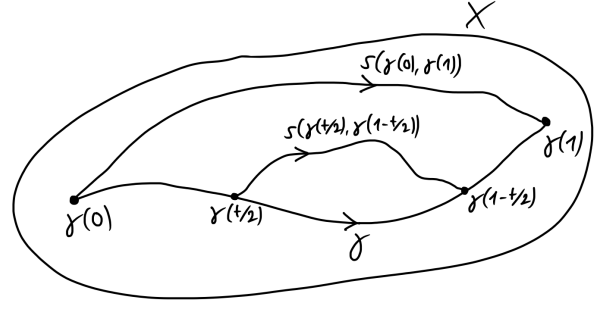


Figure 11: Visualization of the path $\alpha_{\gamma,t} : \gamma$ to $\gamma(t/2)$, then follow $s(\gamma(t/2), \gamma(1-t/2))$ from start to end, then follow γ again to $\gamma(1)$.

$*$ and x_0 respectively. We have

$$\begin{aligned} PY &= \{c_*\}, \\ Pf : PX &\rightarrow PY, \quad \gamma \mapsto c_*, \\ Pg : PY &\rightarrow PX, \quad c_* \mapsto c_0, \\ T^f &= PY \times \Gamma_X, \\ T^g &= PX(x_0, x_0) \times \Gamma_Y. \end{aligned}$$

Since $\overrightarrow{\text{TC}}(X) = 1$, there exists a continuous section $s : \Gamma_X \rightarrow PX$ of $\vec{\pi}$. For $\gamma \in PX$ and $t \in I$, define $\alpha_{\gamma,t} \in PX$ by

$$\alpha_{\gamma,t}(\tau) := \begin{cases} \gamma(\tau), & \text{if } 0 \leq \tau \leq \frac{t}{2}, \\ s(\gamma(\frac{t}{2}), \gamma(1 - \frac{t}{2}))(\frac{\tau - \frac{t}{2}}{1 - t}), & \text{if } \frac{t}{2} \leq \tau \leq 1 - \frac{t}{2}, \\ \gamma(\tau), & \text{if } 1 - \frac{t}{2} \leq \tau \leq 1, \end{cases}$$

see fig. 11. Note that for some fixed (γ, t) , the map $\alpha_{\gamma,t}$ is continuous by the Glueing Lemma and remark 3.6. Furthermore define

$$H : I \times PX \rightarrow PX, \quad (t, \gamma) \mapsto H_t(\gamma) := \alpha_{\gamma,t}.$$

This map is continuous, for details see lemma 8.1 in the appendix.

We have $H_1 = \text{id}_{PX}$ and H_0 coincides with the map $\gamma \mapsto s(\gamma(0), \gamma(1))$. Furthermore, for $(x, x') \in \Gamma_X$ the map H restricts to

$$H_{x,x'} : I \times PX(x, x') \rightarrow PX(x, x')$$

and we have $(H_{x,x'})_1 = \text{id}_{PX(x,x')}$ and $(H_{x,x'})_0$ is the constant map

$$PX(x, x') \rightarrow PX(x, x'), \quad \gamma \mapsto s(x, x').$$

In other words, $H_{x,x'}$ is a homotopy between those two maps.

Now define for $(x, x') \in \Gamma_X$ the map

$$F_{x,x'} : PY(*, *) \rightarrow PX(x, x'), \quad c_* \mapsto s(x, x').$$

Since $T_f = PY \times \Gamma_X$, PY consists of only a single element and $F_{x,x'}$ is obviously continuous, this defines a continuous map

$$F : T^f \rightarrow PX, \quad (c_*, x, x') \mapsto F_{x,x'}(c_*) = s(x, x').$$

In other words, the map $F : PY \multimap PX$ is graded by f . Also we obtain

$$F_{x,x'} \circ Pf|_{PX(x,x')} = (H_{x,x'})_0 \simeq \text{id}_{PX(x,x')}$$

and $Pf|_{PX(x,x')} \circ F_{x,x'} = \text{id}_{PY(*,*)}$, implying that the second property of dihomotopy equivalence between X and Y is satisfied.

Next, consider the continuous map

$$G_{*,*} : \underbrace{PX(x_0, x_0)}_{=PX(g(*), g(*))} \rightarrow PY(*, *), \quad \gamma \mapsto c_*.$$

Since $T^g = PX(x_0, x_0) \times \Gamma_Y$, this induces a continuous map

$$G : T^g \rightarrow PY, \quad (\gamma, *, *) \mapsto G_{*,*}(\gamma) = c_*,$$

i.e. gives rise to a map $G : PX \multimap PY$ graded by g . Furthermore note that $G_{*,*} \circ Pg|_{PY(*,*)} = \text{id}_{PY(*,*)}$ and $Pg|_{PY(*,*)} \circ G_{*,*}$ is the constant map

$$PX(x_0, x_0) \rightarrow PX(x_0, x_0), \quad \gamma \mapsto c_0.$$

All that is left to show is that the latter map is homotopic to $\text{id}_{PX(x_0, x_0)}$.

Note that $\gamma \in PX(x_0, x_0)$ implies that γ is a loop in X based at x_0 . Now X is contractible, hence there is a homotopy with fixed end-points between γ and the constant path c_0 . In particular we have $s(x_0, x_0) \in PX(x_0, x_0)$, thus there is a homotopy

$$h : I \times I \rightarrow X, \quad (t, \tau) \mapsto h_t(\tau)$$

such that $h_0 = s(x_0, x_0)$, $h_1 = c_0$ and $h_t(0) = h_t(1) = x_0$ for all $t \in I$. But this induces a homotopy

$$\tilde{h} : I \times PX(x_0, x_0) \rightarrow PX(x_0, x_0), \quad (t, \gamma) \mapsto \tilde{h}_t(\gamma) := h_t,$$

with $\tilde{h}_0(\gamma) = s(x_0, x_0)$ and $\tilde{h}_1(\gamma) = c_0$, i.e. a homotopy between the two constant maps $\gamma \mapsto c_0$ and $\gamma \mapsto s(x_0, x_0)$. Note that the assignment $t \mapsto h_t \in X^I$ is continuous by corollary 3.6.

We also know from before, that H_{x_0, x_0} is a homotopy between $\text{id}_{PX(x_0, x_0)}$ and the constant map $\gamma \mapsto s(x_0, x_0)$. By combining those two homotopies, one obtains a homotopy between $\text{id}_{PX(x_0, x_0)}$ and the constant map $\gamma \mapsto c_0$. Altogether, this shows dicontractibility of (X, PX) . $\square$

In [GFS20], there is an interesting remark after this result: In proposition 5.34, the equivalence only holds if the underlying topological space X is contractible. This is maybe sometimes not easy to show, depending on the space in question. But for some cases there is a nice solution: Suppose $\overrightarrow{\text{TC}}(X) = 1$, that is, there is a continuous section $s : \Gamma_X \rightarrow PX \subseteq X^I$ and additionally assume there is a point $a_0 \in X$, from which every point is reachable, i.e. $(a_0, a) \in \Gamma_X$ for all $a \in X$. Then we can define a section of the (undirected) path fibration π by

$$\sigma : X \times X \rightarrow X^I, \quad (a, b) \mapsto (s(a_0, a))^{-1} * s(a_0, b),$$

where $(s(a_0, a))^{-1}$ denotes the inverse path. This map is continuous, since concatenation of paths is continuous (see example 3.8) and s was assumed to be continuous. By corollary 4.3 this implies that X is contractible and thus (X, PX) is dicontractible.

This also holds, if there is a point $b_0 \in X$ which can be reached from any other point, i.e. $(a, b_0) \in \Gamma_X$ for all $a \in X$. In this case, a continuous section of π is obtained by

$$(a, b) \mapsto s(a, b_0) * (s(b, b_0))^{-1}.$$

6 A Brief Introduction to Geodesic Complexity

Concluding this thesis, we now come to geodesic complexity, which is an adaption of topological complexity for the setting of metric spaces. The definitions and considerations in this chapter are mainly based on [RM21].

6.1 Metric Spaces and Geodesics

Definition 6.1. Let (X, d) be a metric space. The length of a path $\gamma : I \rightarrow X$ is given by

$$\ell(\gamma) = \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \sum_{i=1}^N d(\gamma(t_{i-1}), \gamma(t_i))$$

where the supremum is taken over all finite partitions of I .

Remark 6.2. Note that the length of a path is independant of its parametrization. Suppose γ' is a reparametrization of some path γ , i.e. there is a continuous non-decreasing surjective map $\varphi : I \rightarrow I$ with $\gamma = \gamma' \circ \varphi$ and suppose we are given a finite partition

$$0 = t_0 \leq t_1 \leq \dots \leq t_N = 1$$

of I . Then, because φ is non-decreasing and surjective, we have $\varphi(t_N) = 1$ and

$$0 = \varphi(t_0) \leq \varphi(t_1) \leq \dots \leq \varphi(t_N) = 1$$

is just another possible finite partition of I . Since in the definition of the length of a curve we take the supremum over all possible finite partitions of I , we obtain

$$\begin{aligned} \ell(\gamma) &= \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \sum_{i=1}^N d(\gamma(t_{i-1}), \gamma(t_i)) \\ &= \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \sum_{i=1}^N d(\gamma'(\varphi(t_{i-1})), \gamma'(\varphi(t_i))) \\ &= \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \sum_{i=1}^N d(\gamma'(t_{i-1}), \gamma'(t_i)) \\ &= \ell(\gamma'). \end{aligned}$$

We can now consider specific paths in a metric space, which will turn out to be length-minimizing.

Definition 6.3. Let (X, d) be a metric space. A path $\gamma : I \rightarrow X$ is called a *geodesic*, if there exists a number $\lambda \in \mathbb{R}$ such that

$$d(\gamma(t), \gamma(t')) = \lambda \cdot |t - t'|$$

for all $0 \leq t < t' \leq 1$.

Geodesics can be thought of as some sort of generalization of straight line segments, as we will see in the next example.

Example 6.4. A straight line segment in $(\mathbb{R}^n, d)$, where d is the standard metric $d(a, b) = \|a - b\|_2$, is a geodesic: For $(a, b) \in \mathbb{R}^n \times \mathbb{R}^n$, define

$$\gamma : I \rightarrow \mathbb{R}^n, \quad \gamma(t) := (1 - t)a + tb.$$

Then for $0 \leq t \leq t' \leq 1$ we have

$$\begin{aligned} d(\gamma(t), \gamma(t')) &= \|(1 - t)a + tb - (1 - t')a - t'b\|_2 \\ &= \|(1 - t - 1 + t')a + (t - t')b\|_2 \\ &= \|(t - t')b - (t - t')a\|_2 \\ &= d(a, b) \cdot |t - t'|. \end{aligned}$$

Thus γ is a geodesic in the sense of definition 6.3 if we choose $\lambda = d(a, b)$.

Given a geodesic γ , with the notation as in definition 6.3, we have

$$\begin{aligned} \ell(\gamma) &= \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \sum_{i=1}^N \lambda |t_{i-1} - t_i| \\ &= \lambda \cdot \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \underbrace{\sum_{i=1}^N |t_{i-1} - t_i|}_{=1} \\ &= \lambda \\ &= \lambda \cdot |0 - 1| \\ &= d(\gamma(0), \gamma(1)). \end{aligned}$$

For every other path $\tilde{\gamma}$ with $\tilde{\gamma}(0) = \gamma(0)$, $\tilde{\gamma}(1) = \gamma(1)$ we have

$$\ell(\tilde{\gamma}) \geq d(\tilde{\gamma}(1), \tilde{\gamma}(0)) = d(\gamma(1), \gamma(0)) = \ell(\gamma).$$

Thus, a geodesic is indeed the shortest possible path between its end-points.

The converse does not always hold, i.e. a shortest path between two points does not necessarily need to be a geodesic. To see this, consider the path $\gamma : I \rightarrow \mathbb{R}$, $\gamma(t) = t$. As we have seen in example 6.4, this is a geodesic in $\mathbb{R}$ (equipped with the standard metric) in the sense of definition 6.3, where $\lambda = 1$. Consider a non-trivial reparametrization of γ , for example $\tilde{\gamma} = \gamma \circ \varphi$ with $\varphi : t \mapsto t^2$. Assume $\tilde{\gamma}$ is also a geodesic, so there is some constant $\tilde{\lambda} \in \mathbb{R}$ with

$$d(\tilde{\gamma}(t), \tilde{\gamma}(t')) = \tilde{\lambda} \cdot |t - t'|$$

for all $0 \leq t < t' \leq 1$. Then

$$\frac{1}{4} = d(0, 1/4) = d(\tilde{\gamma}(0), \tilde{\gamma}(1/2)) = \tilde{\lambda} \cdot \frac{1}{2},$$

implying $\tilde{\lambda} = \frac{1}{2}$. But on the other hand we have

$$d(\tilde{\gamma}(0), \tilde{\gamma}(1/3)) = d(0, 1/9) = \frac{1}{9} \neq \frac{1}{6} = \tilde{\lambda} \cdot \frac{1}{3},$$

so $\tilde{\gamma}$ cannot be a geodesic.

But $\ell(\gamma) = \ell(\tilde{\gamma})$, as seen in remark 6.2, so $\tilde{\gamma}$ is also a path of shortest possible length, but is no geodesic.

However, any shortest path between two points can be reparametrized to be a geodesic, see [RM21], section 7 for details.

Similar to the setting of directed complexity, even for a path-connected metric space X there does not necessarily exist a geodesic between two given points $a, b \in X$. This motivates the following definition.

Definition 6.5. A metric space (X, d) is called *geodesic*, if any two points $a, b \in X$ can be connected by a geodesic. It is called *uniquely geodesic*, if this connecting geodesic is unique.

We close this chapter with some additional definitions and a result from [BH99].

Definition 6.6. Let (X, d) be a metric space.

- i) A curve is called *rectifiable*, if it has finite length.
- ii) A subspace $K \subseteq X$ is called *convex*, if for any $x, y \in K$, every geodesic γ from x to y is entirely contained in K , i.e. $\gamma(t) \in K$ for every $t \in I$.
- iii) X is called *proper*, if all closed balls in X are compact.
- iv) X is called a *length space*, if for every pair $(x, y) \in X \times X$, the distance $d(x, y)$ is equal to the infimum of the length of rectifiable curves between them.

Proposition 6.7 (Hopf-Rinow Theorem). *Let (X, d) be a complete and locally compact length space. Then X is a geodesic space and every closed bounded subset of X is compact.*

6.2 Riemannian Manifolds as Metric Spaces

Recall that a Riemannian manifold (M, g) is a smooth manifold together with a Riemannian metric, i.e. a choice of some scalar product g_p on $T_p M$ for each $p \in M$, such that g_p is smoothly varying depending on p . For some piecewise smooth curve $\gamma : [a, b] \rightarrow M$, one can then define its *Riemannian length* $\ell_R(\gamma)$ as

$$\ell_R(\gamma) := \int_a^b \sqrt{g(\dot{\gamma}, \dot{\gamma})} dt$$

If M is connected, it can then be given the structure of a metric space (M, d) by defining

$$d(p, q) = \inf_{\gamma} \ell_R(\gamma),$$

where the infimum is taken over all piecewise smooth curves between p and q , see [BH99], chapter I, 3.18 Proposition. Furthermore, the topology on M coincides with the metric topology induced by d , and (M, d) is a length space. We say that d is induced by g .

Note that for any piecewise smooth curve $\gamma : I \rightarrow M$ we have

$$\begin{aligned} \ell(\gamma) &= \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \sum_{i=1}^N d(\gamma(t_{i-1}), \gamma(t_i)) \\ &\leq \sup_{0=t_0 \leq t_1 \leq \dots \leq t_N=1} \sum_{i=1}^N \ell_R(\gamma|_{[t_{i-1}, t_i]}) \\ &= \ell_R(\gamma). \end{aligned}$$

On the other hand, also the inequality

$$\ell_R(\gamma) \leq \inf_{\tilde{\gamma}} \ell_R(\tilde{\gamma}) = d(\gamma(0), \gamma(1)) \leq \ell(\gamma)$$

holds, where the infimum is taken over all piecewise smooth curves between $\gamma(0)$ and $\gamma(1)$. Combined, the two inequalities imply $\ell(\gamma) = \ell_R(\gamma)$.

Recall, that a smooth curve $\gamma : I \rightarrow M$ is called a *Riemannian geodesic*, if $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$, where ∇ denotes the Levi-Civita connection of (M, g) . A Riemannian geodesic is locally Riemannian length-minimizing, in the sense that for each $t_0 \in I$ there is a neighborhood I_0 such that for all $[a, b] \subset I_0$ we have $\ell_R(\gamma|_{[a, b]}) = d(\gamma(a), \gamma(b))$. On the other hand, if a curve $\tilde{\gamma}$ is Riemannian length-minimizing, i.e. $\ell_R(\tilde{\gamma}) = d(\tilde{\gamma}(0), \tilde{\gamma}(1))$, then $\tilde{\gamma}$ is a Riemannian geodesic. Proofs of the last two statements can be found for example in [Lee18].

As noted in [RM21] Remark 1.5, one can show, that for a smooth curve $\gamma : I \rightarrow M$ the notions of minimizing Riemannian geodesic and metric geodesic with respect to the induced metric d coincide. Indeed, let (M, g) be a Riemannian manifold, denote the induced metric as d , and the Levi-Civita connection as ∇ , and assume that $\gamma : I \rightarrow M$ is a piecewise smooth curve. Let $p := \gamma(0)$, $q := \gamma(1)$.

First, assume that γ is a minimizing Riemannian geodesic, i.e. $\ell_R(\gamma) = d(a, b)$. The curve being a Riemannian geodesic implies that the scalar $\sqrt{g(\dot{\gamma}, \dot{\gamma})} =: \lambda$ is constant, since ∇ is compatible with g and $\dot{\gamma}$ is parallel along γ . We have

$$d(p, q) = \ell_R(\gamma) = \int_0^1 \sqrt{g(\dot{\gamma}, \dot{\gamma})} = \lambda$$

and, since the restriction of any minimizing Riemannian geodesic is also minimizing,

$$d(\gamma(t), \gamma(t')) = \ell_R(\gamma|_{[t, t']}) = \int_t^{t'} \sqrt{g(\dot{\gamma}, \dot{\gamma})} dt = \lambda \cdot |t - t'|$$

for all $0 \leq t < t' \leq 1$. Thus γ is a metric geodesic with respect to d .

On the other hand, assume that γ is a metric geodesic with respect to d . In particular, γ is length-minimizing, implying $d(p, q) = \ell(\gamma) = \ell_R(\gamma)$, by the discussion above. But this means that γ is length-minimizing with respect to the Riemannian length and thus has to be a Riemannian geodesic.

6.3 The Concept of Geodesic Complexity

Many configuration spaces of systems in practice have the structure of metric spaces or even the structure of Riemannian manifolds, which can also be viewed as metric spaces as discussed before. For example, the space $(\mathbb{S}^1)^n$, which we have considered initially, resembles the configuration space of a planar robot arm with n joints. In this case, one can consider the problem of geodesic motion planning, which asks for a motion planning algorithm $s : X \times X \rightarrow X^I$, such that $s(x, x')$ is a geodesic for each input (x, x') . Since we have seen that geodesics are shortest possible paths, this could be a desirable feature for a motion planning in practical applications.

Definition 6.8. Let (X, d) be a metric space and let $GX \subseteq X^I$ be the subspace consisting of all continuous geodesics on X , which we will refer to as the *geodesic path space* of X . We define the *g-paths map* of X as

$$\bar{\pi}_{X,d} : GX \rightarrow X \times X, \quad \gamma \mapsto (\gamma(0), \gamma(1)).$$

We sometimes just write $\bar{\pi}_X$ or $\bar{\pi}$ instead of $\bar{\pi}_{X,d}$ if it does not cause any confusion over which space or metric is used. Note that the geodesic path space GX inherits a topology as a subspace of X^I .

As we have seen before in the case of the d-paths map, the g-paths map coincides with the restriction of the path fibration to the geodesic path space, i.e. $\bar{\pi}_{X,d} = \pi_X|_{GX}$. Again, as discussed in [RM21] Remark 1.10, the g-paths map fails to be a fibration in general:

Example 6.9. Consider $\mathbb{S}^1$ with the usual Riemannian manifold structure. Let $x, y \in \mathbb{S}^1$. As seen in section 6.2, the notion of geodesic with respect to the induced distance function on $\mathbb{S}^1$ is the same as the notion of a smooth minimizing geodesic with respect to the Riemannian metric. This means, if $x = -y$, there are exactly two geodesics between x and y , whereas if $x \neq -y$, there is only one geodesic joining them. In other words, the fiber $(\bar{\pi})^{-1}(x, y)$ consists of two geodesics for $x = -y$ and

one geodesic for $x \neq -y$. But since $\mathbb{S}^1$ is connected, if π_G would be a fibration, the fibers of π_G would have to be homotopy equivalent by proposition 3.19. As we have seen, this is not the case, so $\bar{\pi}$ cannot be a fibration.

The following definition is due to [RM21].

Definition 6.10. Let (X, d) be a geodesic metric space. The *geodesic complexity* $GC(X, d)$ is the smallest integer k , such that there exists a (possibly discontinuous) map $s : X \times X \rightarrow GX$ and a partition of $X \times X$ into k subsets

$$X \times X = U_1 \cup \dots \cup U_k, \quad U_i \cap U_j = \emptyset, i \neq j,$$

such that:

- i) Each U_i is locally compact.
- ii) s is a (possibly discontinuous) section of $\bar{\pi}_{X,d} : GX \rightarrow X \times X$, i.e. $\bar{\pi}_{X,d} \circ s = \text{id}_{X \times X}$.
- iii) $s|_{U_i} : U_i \rightarrow GX$ is continuous for all $i = 1, \dots, k$.

If no such number exists, set $GC(X, d) := \infty$.

We denote the sets U_i given above as *domains of continuity* for $\bar{\pi}_{X,d}$ and refer to the maps $s|_{U_i}$ as *local sections* of $\bar{\pi}_{X,d}$.

Note that in the definition given here, we used a partition into k subsets instead of $k + 1$ subsets, as originally used by the authors in [RM21]. We do this, in order for geodesic complexity to fit better to our definitions of topological complexity and directed topological complexity. Keep this in mind when comparing the propositions of this thesis with the results from [RM21]. Furthermore, note that the authors of [RM21] omit the requirement that the metric space is geodesic in their definition of geodesic complexity. However, they state a later point in the paper that the condition for the metric space to be geodesic is necessary for geodesic complexity to be finite. Otherwise, even a discontinuous section of $\bar{\pi}$ could not exist.

Example 6.11. Consider $\mathbb{R}^n$ with metric d induced by the standard norm. Similar to the arguments used in example 3.11 i), there is a continuous section s of π , that assigns to each pair $(a, b) \in \mathbb{R}^n \times \mathbb{R}^n$ the straight line segment between them. As we have seen in example 6.4, a straight line segment is a geodesic with respect to d , so s is indeed a continuous section of $\bar{\pi}$. Also, $\mathbb{R}^n$ is locally compact, hence $GC(\mathbb{R}^n, d) = 1$.

Two metric spaces $(X, d_X), (Y, d_Y)$ being homotopy-equivalent as topological spaces does not suffice for $GC(X, d_X)$ and $GC(Y, d_Y)$ to be the same: We will see later with proposition 6.15, that for some $k \in \mathbb{N}$ with $k > 1$ there exists a metric d_k on $\mathbb{R}^k$ with $GC(\mathbb{R}^k, d_k) \geq k > 1$. On the other hand, we know that $GC(\mathbb{R}^k, d) = 1$, where d denotes the standard metric,

thus $\text{GC}(\mathbb{R}^k, d_k) \neq \text{GC}(R^{k+1}, d)$.

However, as noted in the introduction, geodesic complexity is invariant under isometries.

Definition 6.12. A map $f : (X, d_X) \rightarrow (Y, d_Y)$ between metric spaces is called *distance-preserving*, if $d_Y(f(a), f(b)) = d_X(a, b)$ for all $a, b \in X$. A bijection g between metric spaces is called an *isometry*, if both g and g^{-1} are distance-preserving.

Note that a distance-preserving map $f : (X, d_X) \rightarrow (Y, d_Y)$ is automatically injective. Furthermore, it is continuous with regard to the respective metric topologies, since for a converging sequence $x_n \rightarrow x$, we have $d_Y(f(x), f(x_n)) = d_X(x, x_n)$ and thus $f(x_n) \rightarrow f(x)$. Here we used that continuity and sequential continuity coincide for metric spaces. In particular, this implies that every isometry is a homeomorphism. Moreover, f induces a continuous map $Gf : GX \rightarrow GY, \gamma \mapsto f \circ \gamma$. Continuity is shown similar to continuity of the map Pf in the directed case, and the property that geodesics get mapped to geodesics is easily verified. For this map we have

$$\begin{aligned} \pi_X \circ Gf(\gamma) &= \pi_X(f \circ \gamma) \\ &= (f(\gamma(0)), f(\gamma(1))) \\ &= (f \times f)(\gamma(0), \gamma(1)) \\ &= (f \times f) \circ \pi_Y(\gamma), \end{aligned}$$

i.e. the following diagram commutes

$$\begin{array}{ccc} GX & \xrightarrow{Gf} & GY \\ \pi_X \downarrow & & \downarrow \pi_Y \\ X \times X & \xrightarrow{f \times f} & Y \times Y \end{array}$$

Proposition 6.13. If $f : (X, d_X) \rightarrow (Y, d_Y)$ is an isometry, then $\text{GC}(X, d_X) = \text{GC}(Y, d_Y)$.

Proof. Let $\text{GC}(X, d_X) = k < \infty$, i.e. there is a section $s : X \times X \rightarrow GX$ of π_X and a partition of

$$X \times X = U_1 \cup \dots \cup U_k$$

into pairwise disjoint subsets, such that each U_i is locally compact and $s|_{U_i}$ is continuous for all i . Define

$$V_i := (f \times f)(U_i).$$

Since f is a bijection, those sets form a partition of $Y \times Y$. Now f is an isometry and thus a homeomorphism, hence each V_i is locally compact.

A section $r : Y \times Y \rightarrow GY$ of π_Y can then be obtained by setting

$$r := Gf \circ s \circ (f \times f)^{-1}.$$

Then for each i we have

$$r|_{V_i} = Gf \circ s|_{U_i} \circ (f \times f)^{-1}|_{V_i},$$

which is a composition of continuous maps. Hence $\text{GC}(Y, d_Y) \leq \text{GC}(X, d_X)$.

Similarly, one can show $\text{GC}(X, d_X) \leq \text{GC}(Y, d_Y)$.

This also shows that $\text{GC}(X, d_X) = \text{GC}(Y, d_Y) = \infty$, if either of them is infinite. For example, let $\text{GC}(Y, d_Y) = \infty$ and assume $\text{GC}(X, d_X) < \infty$. But then we know $\text{GC}(Y, d_Y) \leq \text{GC}(X, d_X)$, a contradiction. $\square$

6.4 Relations to Topological Complexity

In this chapter, we will discuss differences between geodesic complexity of a space (X, d) and the topological complexity of its underlying space X , equipped with the metric topology.

Of course, by proposition 4.11 iii), for a geodesic space (X, d) we have that $\text{TC}(X) \leq \text{GC}(X, d)$, since a section of $\bar{\pi}$ is just a special case of a section of π .

Recall, that a subspace $(K, d) \subseteq (X, d)$ is called *convex*, if for any $x, y \in K$, every geodesic from x to y is contained in K . For locally compact convex subspaces, one obtains the following result, which can be found in [RM21], theorem 2.2.

Proposition 6.14. Let (X, d) be a geodesic space and $K \subseteq X$ be a locally compact convex subspace. Then

$$\text{TC}(K) \leq \text{GC}(K, d) \leq \text{GC}(X, d).$$

Proof. The first inequality was already discussed above. In order to show the other inequality, let $k := \text{GC}(X, d)$. This means we have a partition of $X \times X$ into k locally compact and pairwise disjoint subsets U_i together with continuous sections $s_i : U_i \rightarrow GX$ of π_X . Consider

$$K_i := U_i \cap (K \times K).$$

Those sets then form a partition of $K \times K$ into pairwise disjoint and locally compact subsets, since products of locally compact spaces and intersections of locally compact subspaces of a Hausdorff space are again locally compact, see chapter 2.2.

Now K is convex, so for every i , the local section $s_i : U_i \rightarrow GX$ of π_X restricts to a local section $s_i|_{K_i} : K_i \rightarrow GK$ of $\bar{\pi}_K$. Hence we obtain

$$\text{GC}(K, d) \leq \text{GC}(X, d).$$

$\square$

As mentioned in [RM21], at first one could hope, that topological and geodesic complexity will actually coincide for a well-behaved class of metric spaces, for example

manifolds. Unfortunately, even when restricting our attention to closed Riemannian manifolds, this is not true in general, as one can deduce from the next proposition. The proof is beyond the scope of this thesis and can be found in [RM21] in section 2.

Proposition 6.15. *For any $k \in \mathbb{N}^+$, there exists a metric g on $\mathbb{S}^{k+3}$ with $\text{GC}(\mathbb{S}^{k+3}, g) - \text{TC}(\mathbb{S}^{k+3}) \geq k$. Furthermore, for every $k \in \mathbb{N}^+$ there exists a metric g' on $\mathbb{R}^k$, such that $\text{GC}(\mathbb{R}^k, g') \geq k$.*

Remark 6.16. Note that the proposition we stated here appears a little bit different than the theorem we referenced. This is due to two things: First, the author of [RM21] uses definitions for topological and geodesic complexity, which are 1 lower than the definitions we used in this thesis. Second, the theorem in the reference is stated in terms of a closed Riemannian manifold (M, g) with $\text{GC}(M, g) - \text{TC}(M) \geq k$, but this is proven by choosing $M = \mathbb{S}^{k+3}$ and constructing a non-standard Riemannian metric g on $\mathbb{S}^{k+3}$, for which the torus $(T^{k+2}, g) \subseteq (\mathbb{S}^{k+3}, g)$ is a convex subspace. Proposition 6.14 then yields

$$\text{GC}(\mathbb{S}^{k+3}, g) \geq \text{TC}(T^{k+2}) = k + 3,$$

where we used that the topological complexity of the n -torus is $\text{TC}(T^n) = \text{TC}((\mathbb{S}^1)^n) = n + 1$, see 4.43. Then, since $\text{TC}(\mathbb{S}^{k+3}) \leq 3$ by proposition 4.42, one obtains

$$\text{GC}(\mathbb{S}^{k+3}) - \text{TC}(\mathbb{S}^{k+3}) \geq k.$$

Similarly, one obtains a metric g' with

$$\text{GC}(\mathbb{R}^k) \geq \text{TC}(T^{k-1}) = k$$

Now this proposition has some implications. It shows, that even when restricting ourselves to spheres, which are relatively “nice” spaces, there exist metrics such that geodesic and topological complexity differ by an arbitrarily large amount. On the other hand, as noted in [RM21] chapter 8, while $\text{cat}(X) \leq \text{TC}(X) \leq \text{GC}(X, d)$ holds true as in proposition 4.19, an upper bound of the form

$$\text{GC}(X, d) \leq \text{cat}(X \times X)$$

cannot be true in general for all (geodesic) metric spaces (X, d) . This follows from the previous proposition, since we know that there exists a metric g on $\mathbb{R}^2$ with $\text{GC}(\mathbb{R}^2, g) > 1$, but $\text{cat}(\mathbb{R}^2 \times \mathbb{R}^2) = 1$ by contractibility of $\mathbb{R}^2$.

6.5 The Total Cut Locus and its Connection to Geodesic Complexity

In this chapter, we introduce the concept of the total cut locus of a metric space and explore its connection to geodesic complexity. We will see that for a certain

class of metric spaces, geodesic complexity depends only on the structure of the total cut locus. The definition is taken from [RM21].

Definition 6.17. Let (X, d) be a metric space. The *total cut locus* C_X of X is defined as the set of all pairs of points in X for which there is more than one geodesic joining them, i.e.

$$C_X \subseteq \{(x, y) \in X \times X \mid |(\bar{\pi}_X)^{-1}(x, y)| > 1\}.$$

The *cut locus* of a point $x \in X$ is defined as

$$C_X(x) := \{y \in X \mid (x, y) \in C\}.$$

Note that a geodesic space X is uniquely geodesic if and only if its total cut locus C_X is empty. Before moving to the main result of this chapter, we need some additional results from [BH99], chapter I, 3.8 Corollary, 3.12 Lemma and 3.13 Corollary.

Proposition 6.18. *A length space is proper if and only if it is complete and locally compact.*

Proposition 6.19. *Let (X, d) be a proper metric space and let $x, y \in X$. Suppose that there is a unique geodesic γ from x to y . Furthermore, for all $n \in \mathbb{N}$ let γ_n be a geodesic in X , such that $\gamma_n(0)$ converges to x and $\gamma_n(1)$ converges to y . Then γ_n converges uniformly to γ .*

Remark 6.20. Note that our definition of a geodesic coincides with the notion of a *linearly reparametrized geodesic* given in [BH99], hence the statement of proposition 6.19 at first glance appears a little bit different than the one stated in [BH99].

Remark 6.21. By proposition 3.3 we know that the compact-open topology on X^I is the same as the topology of uniform convergence whenever X is a metric space. If X is also a geodesic space, we can define a (possibly discontinuous) section $s : X \times X \rightarrow GX$ of $\bar{\pi}$ by assigning to $(x, y) \in X \times X$ some arbitrary geodesic between them. Proposition 6.19 then implies that at every point (x, y) for which the assigned geodesic is uniquely determined, this section must be continuous.

Corollary 6.22. *If a proper metric space X is uniquely geodesic, then there is a global continuous section of $\bar{\pi}_X$, implying $\text{GC}(X) = 1$.*

We have to make one additional observation, before we get to the main result.

Remark 6.23. Any geodesic space (X, d) is a length space. To see this, take a pair $(x, y) \in X \times X$. Then there is a geodesic γ joining those two points. Since γ is a geodesic, we have $\ell(\gamma) = d(x, y)$. As discussed before, geodesics are length-minimizing, so any other path between x and y must have at least length $\ell(\gamma)$, hence the

infimum of the length of rectifiable curves between x and y must be equal to $\ell(\gamma) = d(x, y)$.

Note that the converse does not always hold, as noted in [BH99] on page 35: Consider $\mathbb{R}^2 \setminus \{0\}$ with the metric induced by the standard metric on $\mathbb{R}^2$. This is a length space, but there is no geodesic between 1 and -1 .

However, if X is a complete and locally compact length space, then it is also geodesic by the Hopf-Rinow theorem, see proposition 6.7.

Now we are ready to move to the main result of this section. The proof is based upon [RM21], Theorem 3.3

Proposition 6.24. *Let (X, d) be a locally compact, geodesic and complete metric space. Then there is a continuous section of π_X over the complement of the total cut locus of X .*

In particular, if the total cut locus of X is empty, it follows that $\text{GC}(X, d) = 1$.

Proof. First note that by the previous remark, X being geodesic implies that X is a length space, hence it is also proper by proposition 6.18.

Let $C_X \subseteq X \times X$ denote the total cut locus of X . Consider the complement $(X \times X) \setminus C_X$ of the total cut locus. This set contains all pairs of points for which there is exactly one geodesic connecting them. As already discussed before, by proposition 6.19 there is a local section of π_X over $(X \times X) \setminus C_X$.

In particular, if $C = \emptyset$, then there is a global section of π_X over $X \times X$. $\square$

Thus, for a locally compact, geodesic, complete metric space, the main challenge of geodesic motion planning depends on the structure of the total cut locus.

Example 6.25. i) Consider $X = \mathbb{S}^1 \subseteq \mathbb{R}^2$. Then

$$C_X = \{(x, y) \in X \times X \mid x = -y\},$$

so there exists a continuous section of π over

$$X \setminus C_X = \{(x, y) \in X \times X \mid x \neq -y\}.$$

A continuous section of π over C_X can be constructed as in example 4.5 ii): Let U_2 and s_2 be defined as in the example. Then $C_X \subseteq U_2$, so a continuous section of π over C_X can be obtained by restricting s_2 to C_X .

ii) This example is based upon [RM21], Remark 3.4. Consider (I^2, d_1) where $d_1((x, y), (x', y')) = |x - x'| + |y - y'|$ is the ℓ_1 -metric. Let $(x, y), (x', y') \in I^2$ with $x \neq x'$ and $y \neq y'$. Consider the paths

$$\begin{aligned} \gamma_{x,x'} &: t \mapsto ((1-t)x + tx', y), \\ \gamma_{y,y'} &: t \mapsto (x', (1-t)y + ty'), \\ \tilde{\gamma}_{y,y'} &: t \mapsto (x, (1-t)y + ty'), \\ \tilde{\gamma}_{x,x'} &: t \mapsto ((1-t)x + tx', y'). \end{aligned}$$

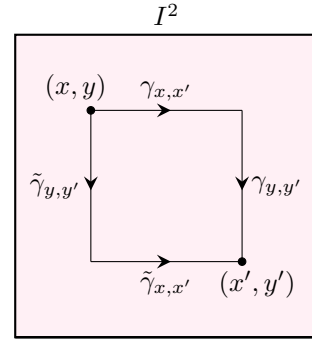


Figure 12: The paths $\gamma = \gamma_{x,x'} * \gamma_{y,y'}$ and $\tilde{\gamma} = \tilde{\gamma}_{y,y'} * \tilde{\gamma}_{x,x'}$ in I^2 .

Furthermore let $\gamma := \gamma_{x,x'} * \gamma_{y,y'}$ and $\tilde{\gamma} := \tilde{\gamma}_{y,y'} * \tilde{\gamma}_{x,x'}$, where $*$ denotes concatenation, as shown in fig. 12. Both paths are length-minimizing with respect to d_1 and can thus be reparametrized to be geodesics, as mentioned before. On the other hand, let $(a, b), (a', b')$ with $a = a'$ or $b = b'$. Without loss of generality, assume $a = a'$. In this case, there exists only one geodesic, which is the straight line segment between (a, b) and (a', b') , which has length $|b - b'| = d_1((a, b), (a', b'))$. This is due to the fact that any other path would either be longer with respect to the second component, or would have a deviation from a with respect to the first component, leading to a length of the path which is bigger than $|b - b'|$.

In summary, this tells us that the total cut locus of this space is

$$\{((x, y), (x', y')) \mid x \neq x' \text{ and } y \neq y'\},$$

which means there is a local section of π_{I^2} over the set

$$\{((x, y), (x', y')) \mid x = x' \text{ or } y = y'\}.$$

However, there is even a global section $s : I^2 \rightarrow GX$ of π_{I^2} given by

$$s((x, y), (x', y'))(t) := (1-t) \cdot (x, y) + t \cdot (x', y'),$$

which simply assigns to $((x, y), (x', y')) \in I^2$ the straight line segment between them (Continuity is again implied by corollary 3.6). To see that this indeed always yields a geodesic with respect to d_1 , consider a path γ of this form, i.e.

$$\begin{aligned} \gamma(t) &= (1-t) \cdot (x, y) + t \cdot (x', y') \\ &= ((1-t)x + tx', (1-t)y + ty') \end{aligned}$$

for some $(x, y), (x', y') \in I^2$. Then for $0 \leq t < t' \leq 1$

we have

$$\begin{aligned}
d_1(\gamma(t), \gamma(t')) &= |(1-t)x + tx' - (1-t')x - t'y'| \\
&\quad + |(1-t)y + ty' - (1-t')y - t'y'| \\
&= |t(x' - x) - t'(x' - x)| \\
&\quad + |t(y' - y) - t'(y' - y)| \\
&= d_1((x, y), (x', y')) \cdot |t' - t|.
\end{aligned}$$

Since there exists a global section of π_{I^2} , we obtain $GC(I^2, d_1) = 1$.

Part ii) of the previous example shows that the converse of proposition 6.24 does not hold: The space (I^2, d_1) is locally compact, geodesic and complete, has $GC(I^2, d_1) = 1$, but its total cut locus is nonempty.

As is usual in mathematics, given some proposition, we should ask ourselves, if the requirements are really necessary.

Of course, the condition that the space is geodesic cannot be omitted, since otherwise even a discontinuous section could not exist over the complement of the total cut locus.

Regarding local compactness, in [BH99] Chapter I, 3.14 Exercise, the authors mention a space which is complete and uniquely geodesic, but for which the geodesics do not vary continuously with their endpoints:

Example 6.26. For $n \in \mathbb{N}$ let Y_n be the subspace of $\mathbb{S}^2$ obtained as follows: Denote as N, S the north and south pole of $\mathbb{S}^2$ respectively. Consider two geodesic arcs, each going from S to N , which meet at an angle of $\pi/4$. Denote as $\tilde{Y}_n$ the area enclosed by those two arcs, and denote as D_n the open ball around N with radius $1/n$ with respect to the metric on $\mathbb{S}^2$. Then define $Y_n := \tilde{Y}_n \setminus D_n$ and equip it with the induced length metric. The boundary of Y_n then consists of three geodesic segments with respect to this metric, and two of those segments have length $\pi - 1/n$. Let α_n denote one of them.

We then define the space X as follows: Take the interval $[0, \pi]$, then for each $n > 2$, attach Y_n to $[0, \pi]$ by an isometry which maps the segment $\alpha_n \rightarrow [0, \pi - 1/n]$, see fig. 13. Then equip X with the unique length metric, that restricts to the given metrics on $[0, \pi]$ and on each Y_n .

Then the claim is, that X is a complete and uniquely geodesic space, but geodesic segments in X do not vary continuously with their endpoints. The property of X being uniquely geodesic follows, since each Y_n as well as the interval $[0, \pi]$ is uniquely geodesic. We will not discuss completeness here and only sketch an idea how one could prove, that geodesic segments do not vary continuously.

We proceed as follows: First, identify N with π and S

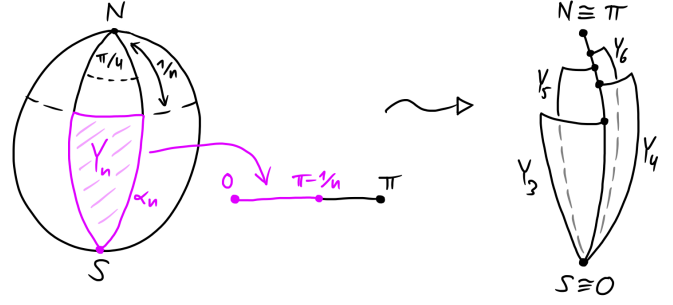


Figure 13: Construction of the space X .

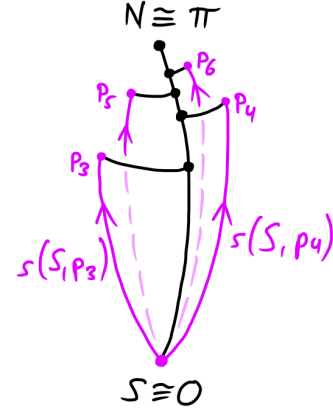


Figure 14: The converging sequence $(S, p_n) \rightarrow (S, N)$ together with the non-converging sequence $(s(S, p_n))_{n \geq 2}$.

with 0 on the interval $[0, \pi]$. Then for each n , let p_n be the point of $Y_n \subseteq X$ which has distance equal to $\pi - 1/n$ from S and maximum distance from $\pi - 1/n \in [0, \pi]$, i.e. the unique point lying on the boundary of Y_n at an intersection of two of the geodesic segments, and which is not attached to $[0, \pi]$.

Then for each n , the unique geodesic (with respect to the induced length metric on X) between S and p_n is given by the geodesic segment of length $\pi - 1/n$ which is not attached to $[0, 1/n]$, whereas the unique geodesic segment between S and N is given by the map

$$I \rightarrow [0, \pi] \subseteq X, \quad t \mapsto t \cdot \pi.$$

Furthermore, let $s : X \times X \rightarrow GX$ be the map, which assigns to each pair of points its unique geodesic segment in GX . Now consider the sequence $(S, p_n)_{n > 2}$ of points in $X \times X$. This sequence converges to (S, N) with respect to the product metric on $X \times X$ induced by the length metric on X . But $s(S, p_n)$ does not converge to $s(S, N)$ with respect to the compact-open topology on GX , which coincides with the topology of uniform convergence in this case, see proposition 3.3.

See fig. 14 for a visualization of this construction.

In other words, as also mentioned in [RM21] Remark 3.5, there is a space which is geodesic, complete and has an empty total cut locus, but for which there is no global continuous section of $\bar{\pi}$. It follows by proposition 6.24 that this space is not locally compact.

7 Concluding Remarks and Further Work

We have seen throughout the thesis, that there are several different kinds of upper and lower bounds for topological complexity, which have proven to be useful for computing the complexity of certain spaces. We have seen that $\text{TC}(X) \leq \overrightarrow{\text{TC}}(\vec{X})$ for strongly connected d-spaces $\vec{X}$ and $\text{TC}(Y) \leq \text{GC}(Y, d)$ for geodesic spaces (Y, d) , so the lower bounds for topological complexity immediately carry over to those cases, i.e.

$$\begin{aligned} \text{cat}(X) &\leq \overrightarrow{\text{TC}}(\vec{X}), & \text{zcl}(X; \mathbb{F}) &< \overrightarrow{\text{TC}}(\vec{X}), \\ \text{cat}(Y) &\leq \text{GC}(Y, d), & \text{zcl}(Y; \mathbb{F}) &< \text{GC}(Y, d). \end{aligned}$$

For upper bounds, the situation is more difficult and requires further research.

Consider the upper bound given by the LS-category of $X \times X$ as well as the dimensional upper bound for topological complexity. We have seen, that one can show

$$\text{TC}(X) \leq \text{cat}(X \times X) \leq 2 \cdot \dim + 1$$

by using that $\pi : X^I \rightarrow X \times X$ is a fibration, hence $\text{TC}(X) = \text{secat}(\pi) \leq \text{cat}(X \times X)$.

Unfortunately, it turned out that neither $\vec{\pi}$ nor $\bar{\pi}$ are fibrations, so this approach cannot be used directly for directed and geodesic complexity. However, the authors in [GFS20] mention in chapter 6 that

$$\overrightarrow{\text{TC}}(\vec{X}) \leq 2 \cdot \dim(X) + 1$$

may be shown for “nice” cubical complexes $\vec{X}$, but, as it seems for now, a proof has yet to be formulated. In general, this inequality does not hold for arbitrary directed spaces, as we have seen in example 5.13.

For geodesic complexity, the upper bound by $\text{cat}(X \times X)$ cannot be true in general, as is implied by proposition 6.15: There is a metric g on $\mathbb{R}^n$ with $\text{GC}(\mathbb{R}^n, g) \geq n$, but $\text{cat}(\mathbb{R}^n \times \mathbb{R}^n) = 1$, since $\mathbb{R}^n$ is contractible. Regarding dimensional bounds, a question is raised in [RM21] chapter 8. It is unclear, whether the inequality $\text{GC}(X, d) \leq \dim(X \times X)$ holds for sufficiently “nice” metric spaces. On the one hand, the author was not able to find a metric space a metric space for which $\text{GC}(X, d) \leq \dim(X \times X)$ could be proven, but on the other hand, the author mentions that there does not seem to be a reason, why the inequality should be true.

Furthermore, in [RM21] chapter 8, a second question is raised, if the categorical lower and upper bounds from proposition 4.19 carry over to the geodesic setting, when replacing $\text{cat}(X)$ with $\text{Gcat}(X, d)$, which is a geodesic version of LS-category. For a metric space (X, d) , it

is defined as the smallest integer k , for which there is a decomposition of X into k locally compact subsets E_i , such that each E_i is contractible in X via some homotopy $h : I \times E_i \rightarrow X$, with the additional property, that $t \mapsto h(t, x)$ is a geodesic in X for each $x \in E_i$. The proof for $\text{Gcat}(X, d) \leq \text{GC}(X, d)$ can be adapted from the undirected case, but the situation is unclear for the upper bound.

At this point, it should be noted, that while $\bar{\pi}$ fails to be a fibration in general, in some cases it features the structure of a level-wise stratified covering with certain properties. This then yields a lower bound for geodesic complexity, which proves useful for computations. If some specific additional requirements for $\bar{\pi}$ are met, this lower bound even turns into an equality. For details, see [RM21] chapter 3, as well as the subsequent chapters for explicit computations. Although we did not cover this topic here, as it would have been way beyond the scope of this thesis, this could be an interesting foundation for future work.

Lastly, let us return one more time to the setting of directed spaces, and in particular to the notion of a regular space, see definition 5.16. We have seen in remark 5.19, that each space X with finite topological complexity can be considered regular. Indeed, as mentioned by the authors of [GFS20], all directed spaces they have considered in their paper are regular, and it is unclear, if there even exist d-spaces which are not regular.

8 Appendix

Lemma 8.1. *The map*

$$H : I \times PX \rightarrow PX, \quad (t, \gamma) \mapsto H_t(\gamma) := \alpha_{\gamma, t}$$

as defined in proposition 5.34 is continuous.

Proof. Define

$$Z := \{(\gamma_1, \gamma_2, \gamma_3) \in (PX)^3 \mid \bar{\pi}(\gamma_2) = (\gamma_1(1), \gamma_3(0))\},$$

i.e. for $(\gamma_1, \gamma_2, \gamma_3) \in Z$ we have $\gamma_1(1) = \gamma_2(0)$ and $\gamma_2(1) = \gamma_3(0)$.

Then concatenation of three paths $C : Z \rightarrow PX$ given by

$$(\gamma_1, \gamma_2, \gamma_3) \mapsto (\gamma_1 * \gamma_2) * \gamma_3$$

is a continuous operation, as already discussed in example 3.8. Note that

$$C(\gamma_1, \gamma_2, \gamma_3)(\tau) = \begin{cases} \gamma_1(4\tau), & \tau \in [0, \frac{1}{4}], \\ \gamma_2(4\tau - 1), & \tau \in [\frac{1}{4}, \frac{1}{2}], \\ \gamma_3(2\tau - 1), & \tau \in [\frac{1}{2}, 1]. \end{cases}$$

Next consider for $t \in I$ the map $\varphi_t : I \rightarrow I$ given by

$$\varphi_t(\tau) := \begin{cases} \frac{\tau}{2t}, & \tau \in [0, \frac{t}{2}], \\ \frac{2\tau + 2 - 3t}{8(1-t)}, & \tau \in [\frac{t}{2}, 1 - \frac{t}{2}], \\ \frac{\tau + t - 1}{t}, & \tau \in [1 - \frac{t}{2}, 1]. \end{cases}$$

We have that the map $I \rightarrow I^I, t \mapsto \varphi_t$ is continuous, since $(\tau, t) \mapsto \varphi_t(\tau)$ is continuous.

Denote as $\rho : I \times I \times PX \rightarrow PX$ the restriction map from example 3.8 iii), which is continuous. Then for $t, \tau \in I$ and $\gamma \in PX$ we have

$$\begin{aligned} & C(\rho(0, t/2, \gamma), s(\gamma(t/2), \gamma(1 - t/2)), \rho(1 - t/2, 1, \gamma))(\tau) \\ &= \begin{cases} \gamma(2 \cdot t \cdot \tau), & \tau \in [0, \frac{1}{4}], \\ s(\gamma(t/2), \gamma(1 - t/2))(4\tau - 1), & \tau \in [\frac{1}{4}, \frac{1}{2}], \\ \gamma(\tau \cdot t - t + 1), & \tau \in [\frac{1}{2}, 1]. \end{cases} \end{aligned}$$

and thus

$$\begin{aligned} & C(\rho(0, t/2, \gamma), s(\gamma(t/2), \gamma(1 - t/2)), \rho(1 - t/2, 1, \gamma))(\varphi_t(\tau)) \\ &= \begin{cases} \gamma(\tau), & \tau \in [0, \frac{t}{2}], \\ s(\gamma(t/2), \gamma(1 - t/2))\left(\frac{\tau - t/2}{1 - t}\right), & \tau \in [\frac{t}{2}, 1 - \frac{t}{2}], \\ \gamma(\tau), & \tau \in [1 - \frac{t}{2}, 1]. \end{cases} \\ &= \alpha_{\gamma, t}(\tau) \\ &= H(\gamma, t)(\tau). \end{aligned}$$

Let $J : PX \rightarrow Z$ be the map given by

$$\gamma \mapsto (\rho(0, t/2, \gamma), s(\gamma(t/2), \gamma(1 - t/2)), \rho(1 - t/2, 1, \gamma)).$$

But this means, we can write H as the composition

$$\begin{array}{ccc}
 I \times PX & \xrightarrow{(t, \gamma) \mapsto (\varphi_t, J(\gamma))} & I^I \times Z \\
 & \searrow H & \downarrow \text{id} \times C \\
 & & I^I \times PX \\
 & & \downarrow (\varphi, \gamma) \mapsto \gamma \circ \varphi \\
 & & PX
 \end{array}$$

All of the individual maps are continuous: Continuity of C, ρ and $(t \mapsto \varphi_t)$ was discussed before, s is continuous by assumption and continuity of the precomposition map $(\gamma, \varphi_t) \mapsto \gamma \circ \varphi_t$ was discussed in example 3.8 ii). $\square$

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